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LEE et al.

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Oct. 2, 2018 (KR) 10-2018-0117819

(57) **ABSTRACT**

A refrigerator may include a first tray assembly forming a part of an ice making cell, a second tray assembly forming another part of the ice making cell, a heater provided adjacent to at least one of the first or second tray assemblies, a driver connected to the second tray assembly, and a control unit to control the heater and the driver. The driving unit includes a cam capable of being connected to the second tray assembly, and a path in which a lever moves is formed inside the cam.

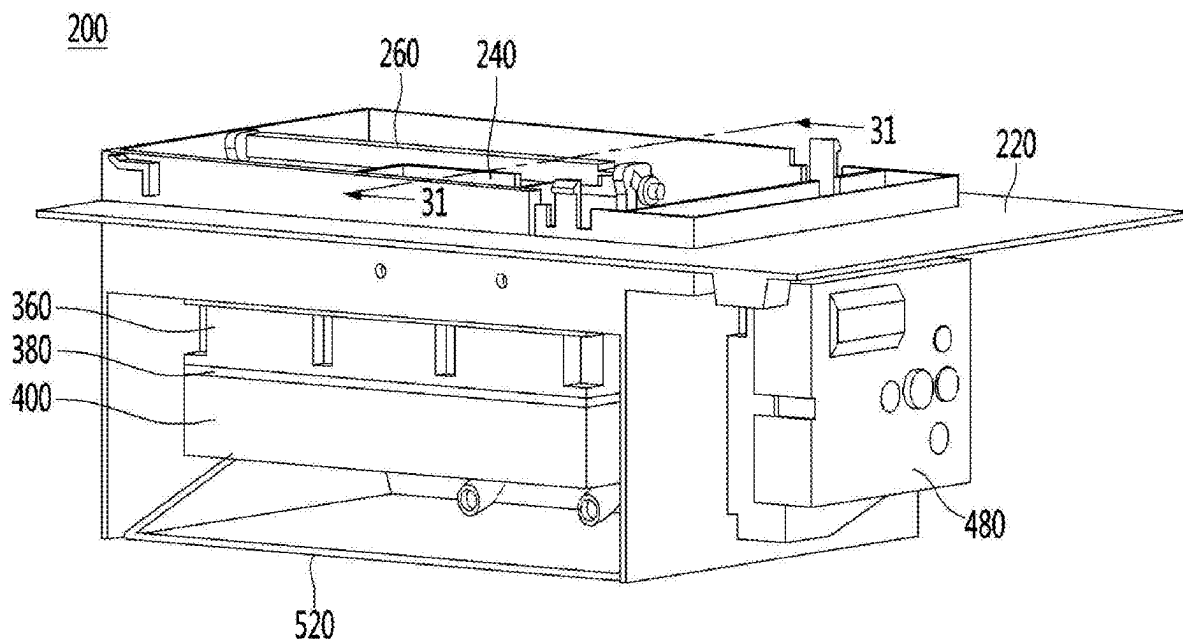


FIG. 1

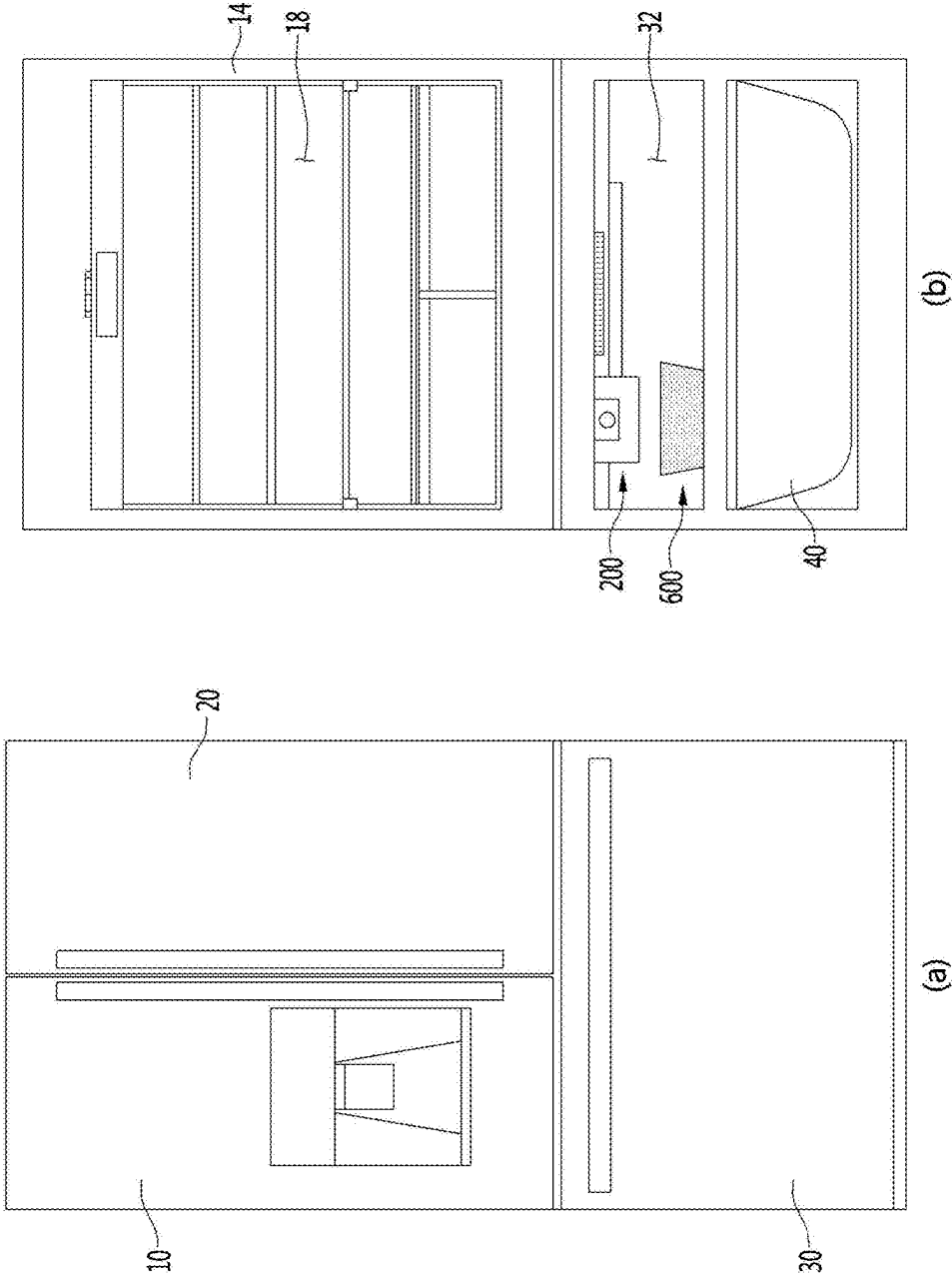


FIG. 2

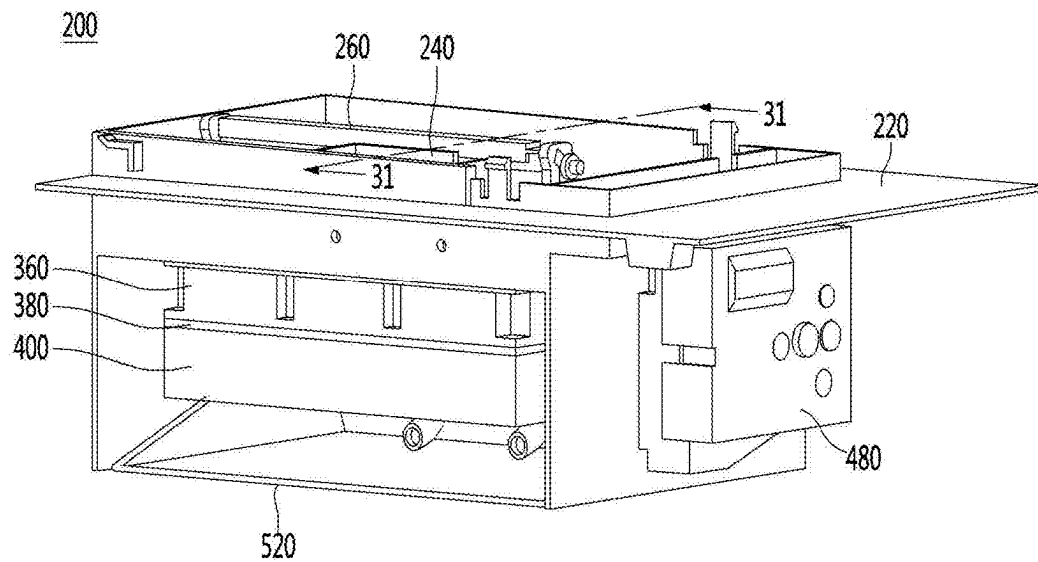


FIG. 3

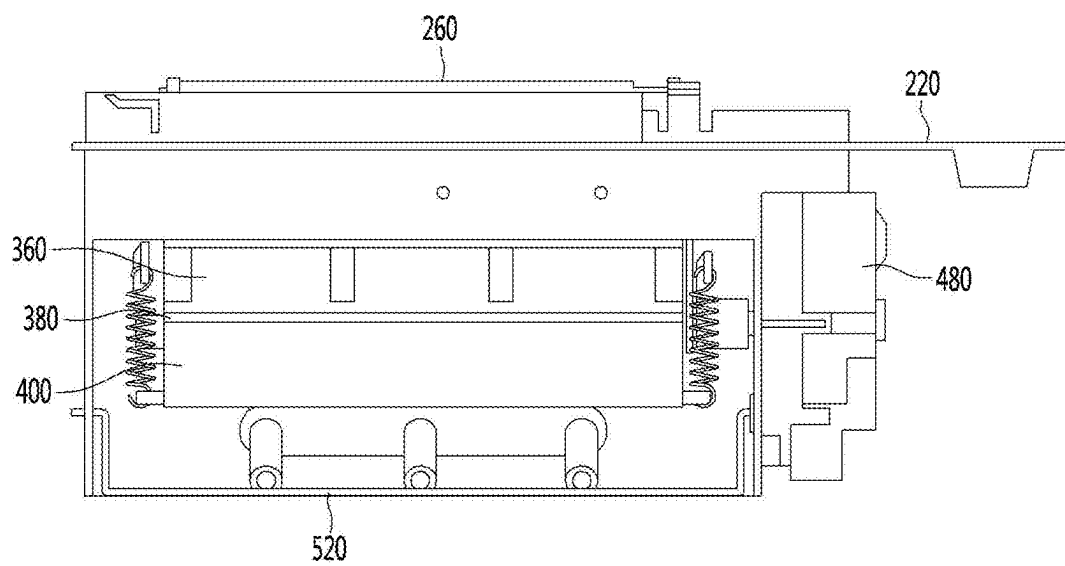


FIG. 4

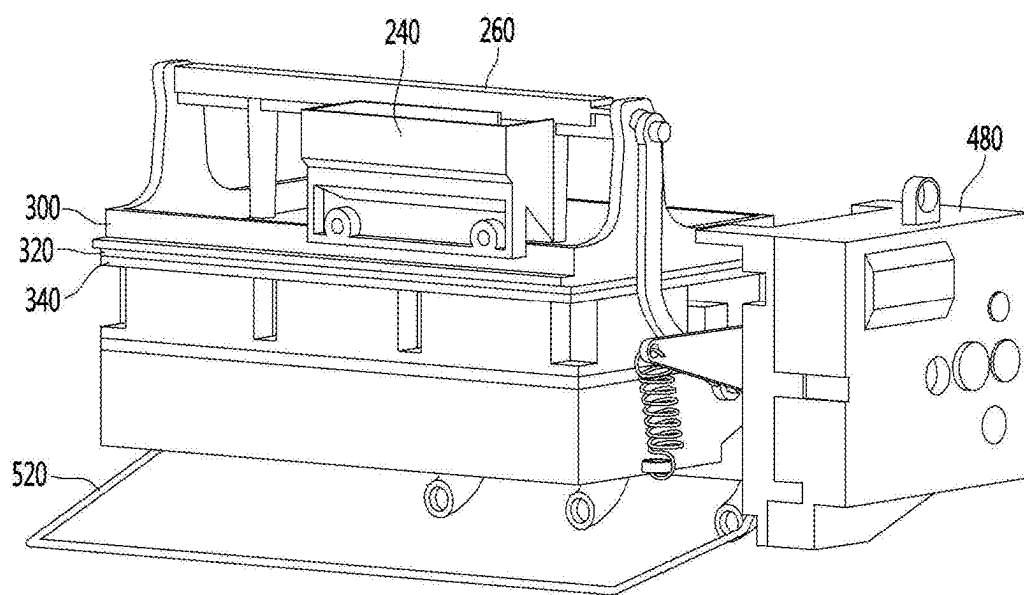


FIG. 5

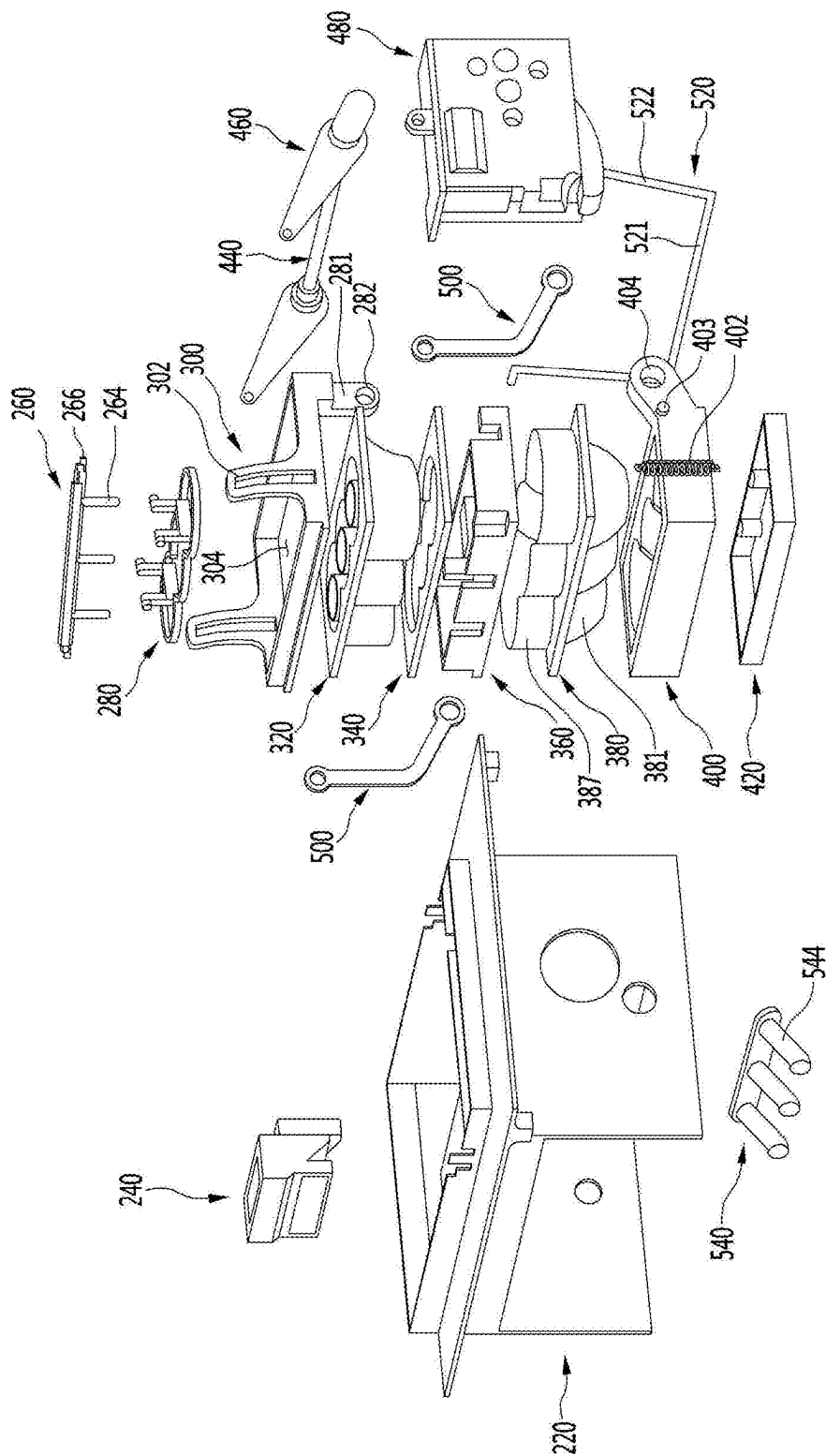


FIG. 6

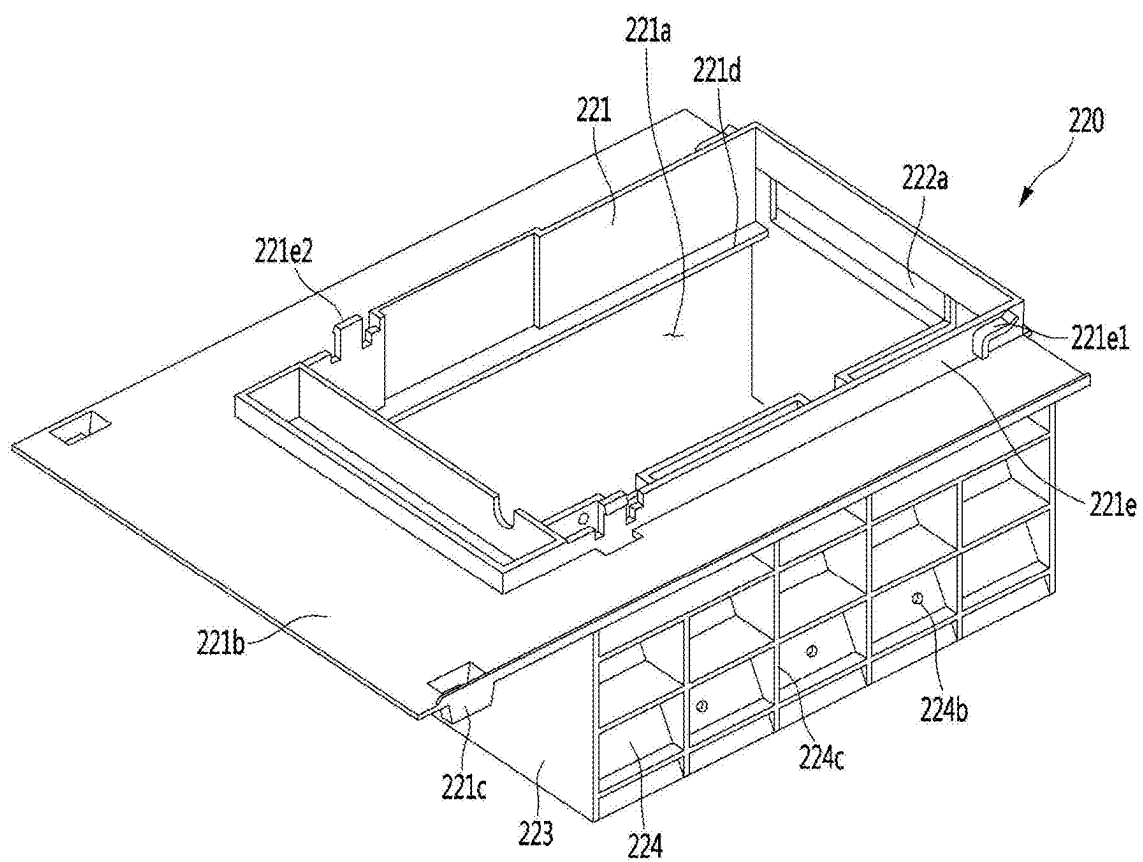


FIG. 7

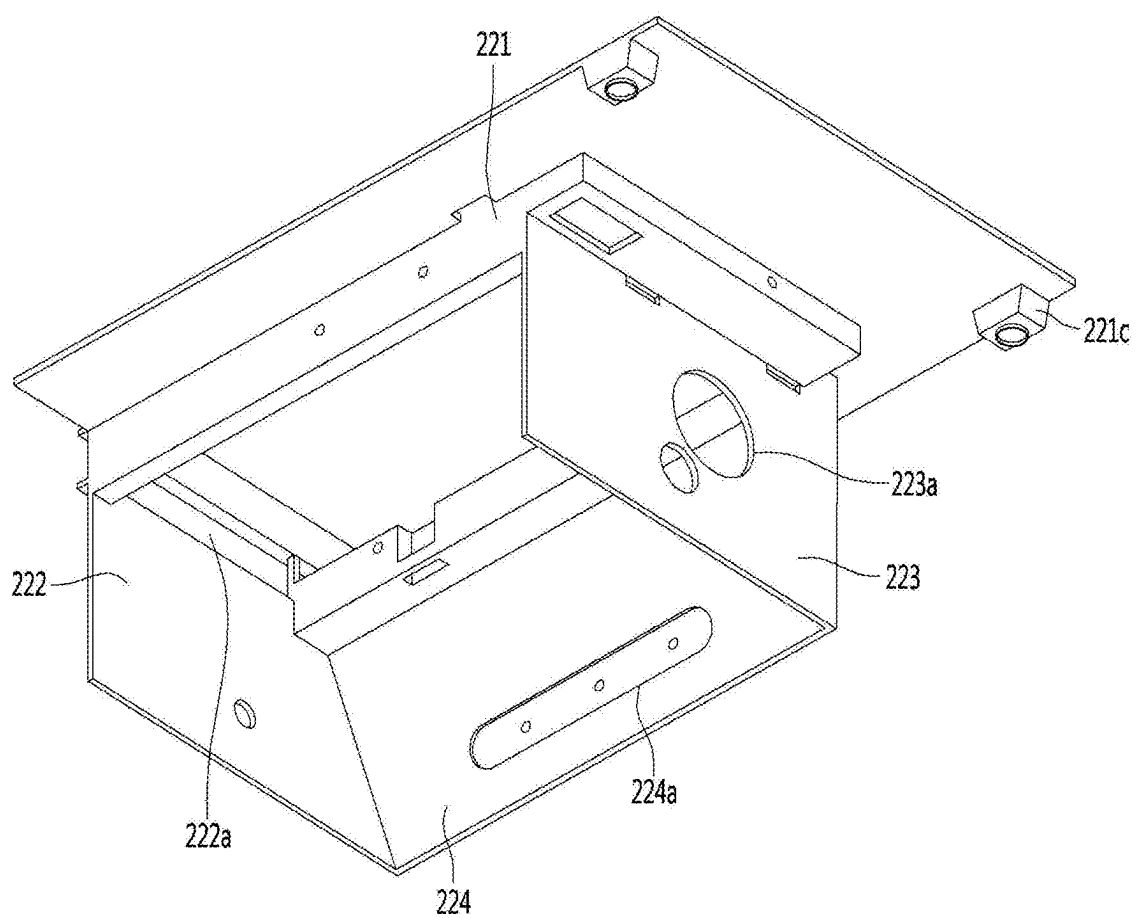


FIG. 8

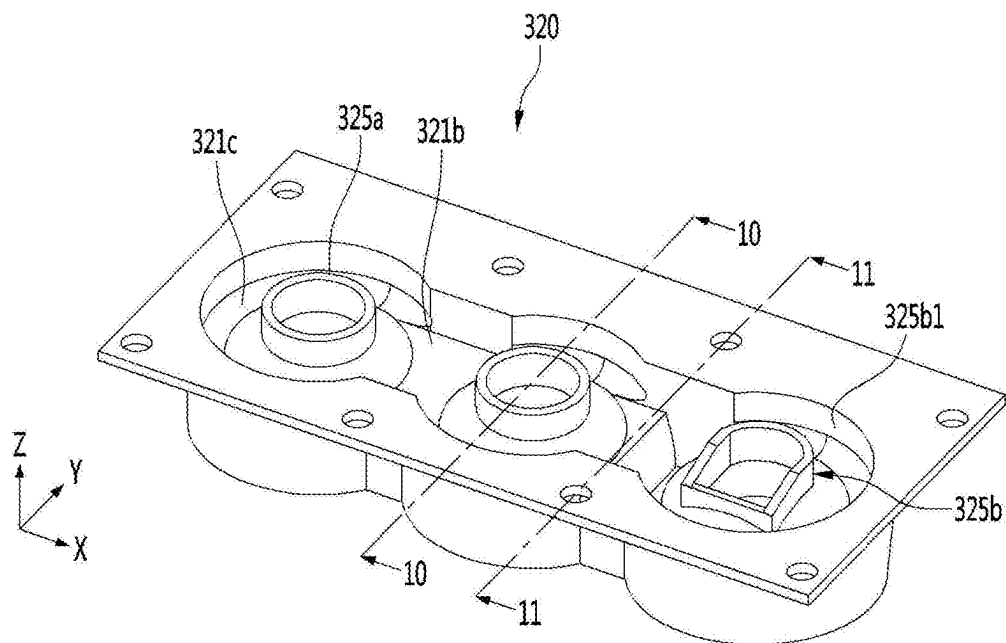


FIG. 9

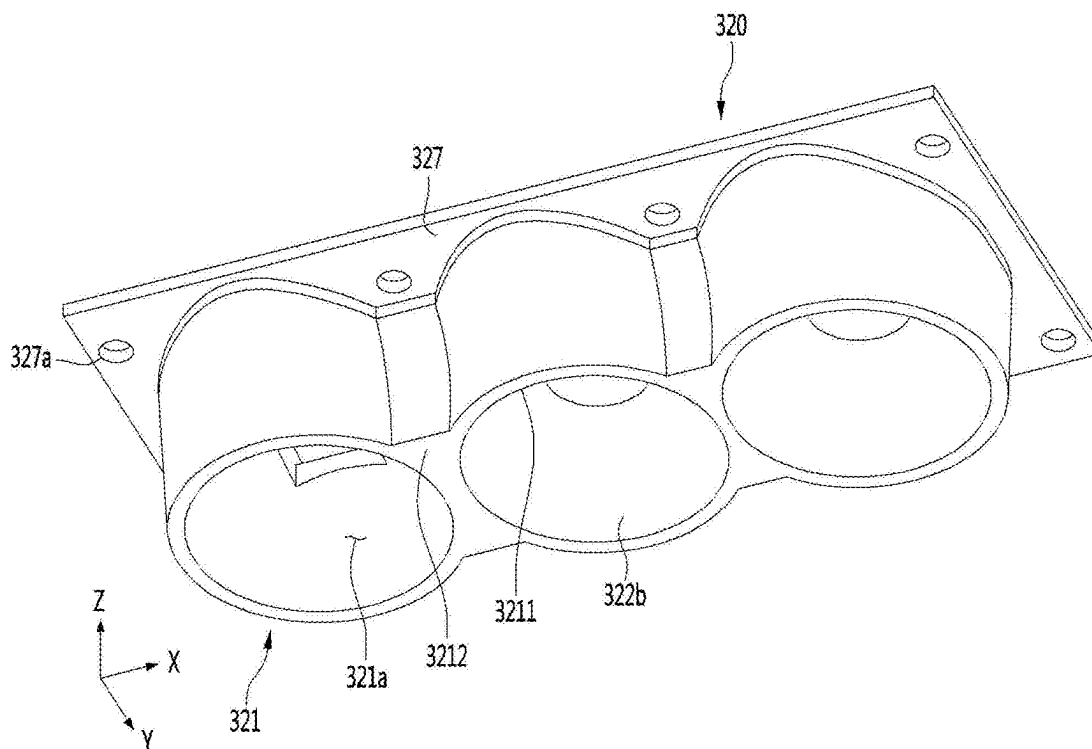


FIG. 10

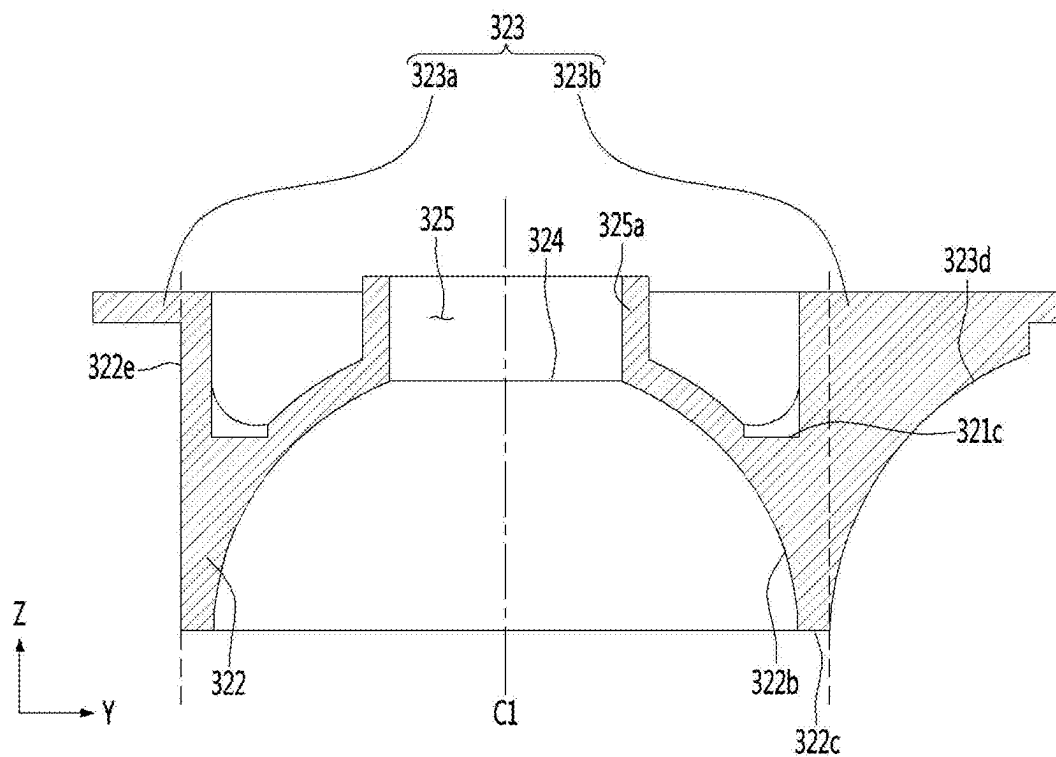


FIG. 11

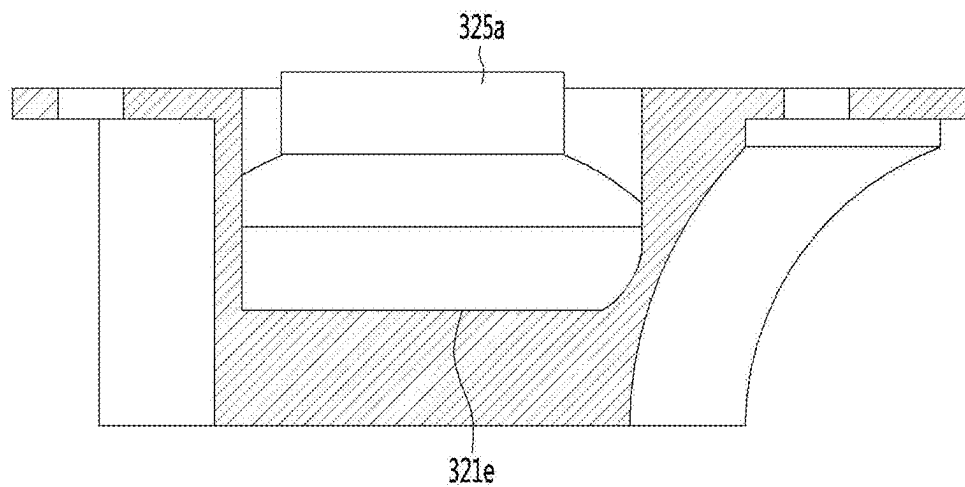


FIG. 12

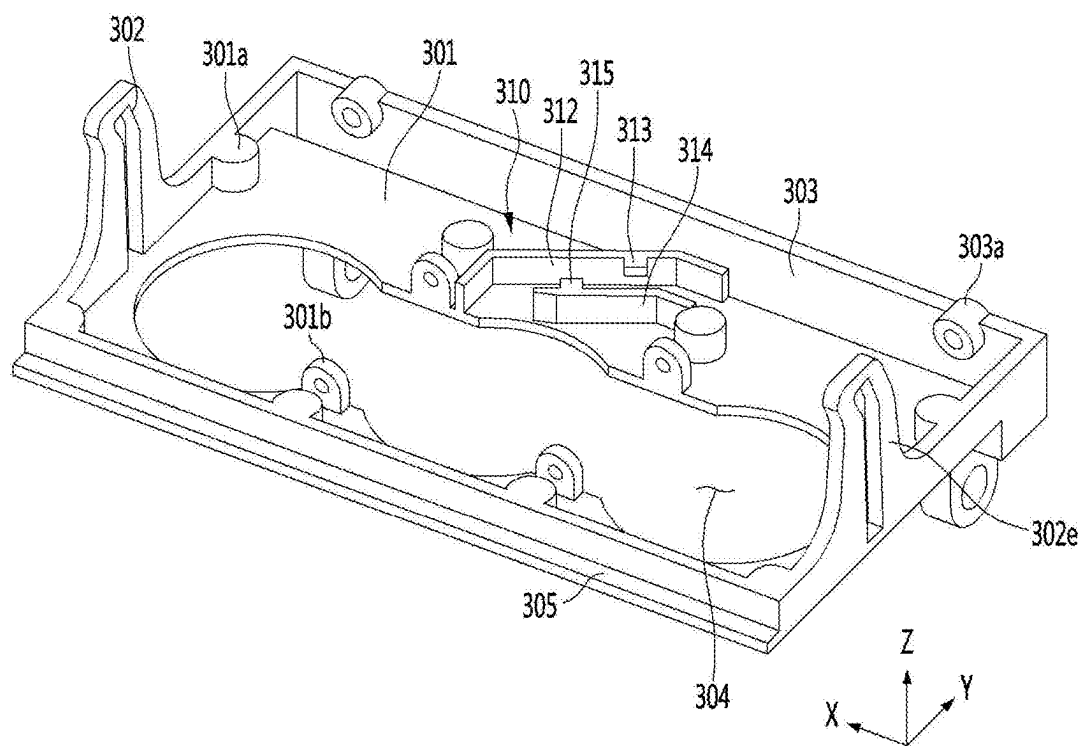


FIG. 13

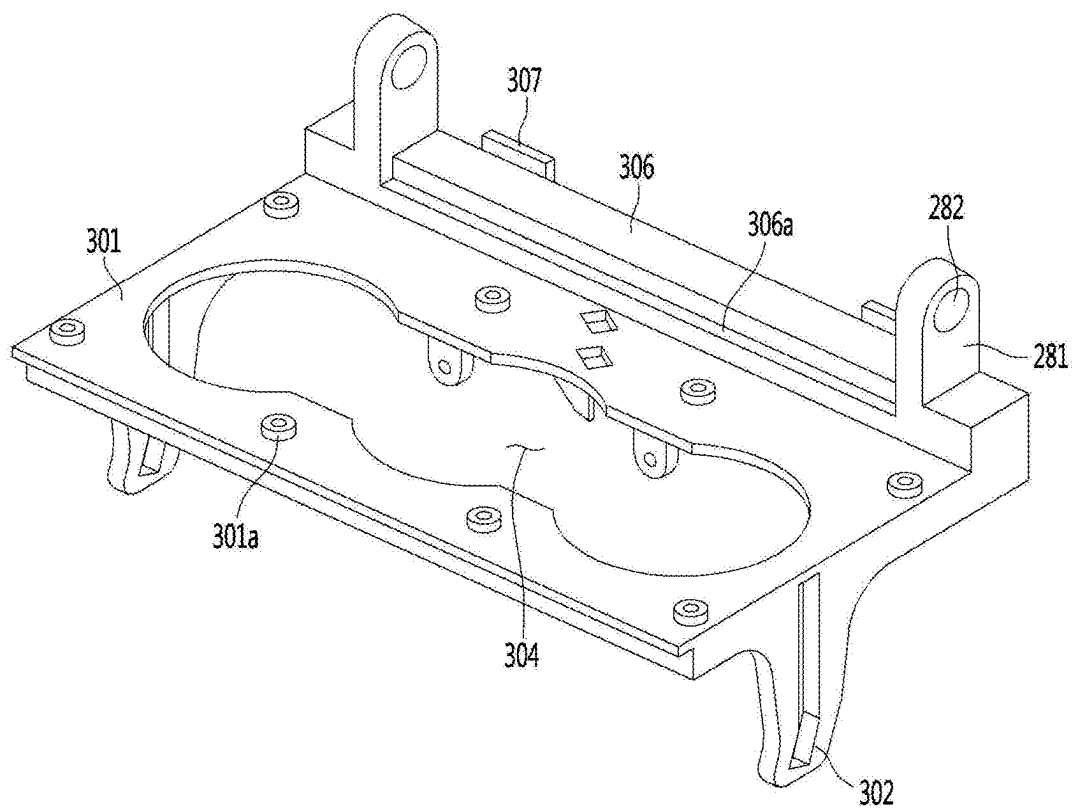


FIG. 14

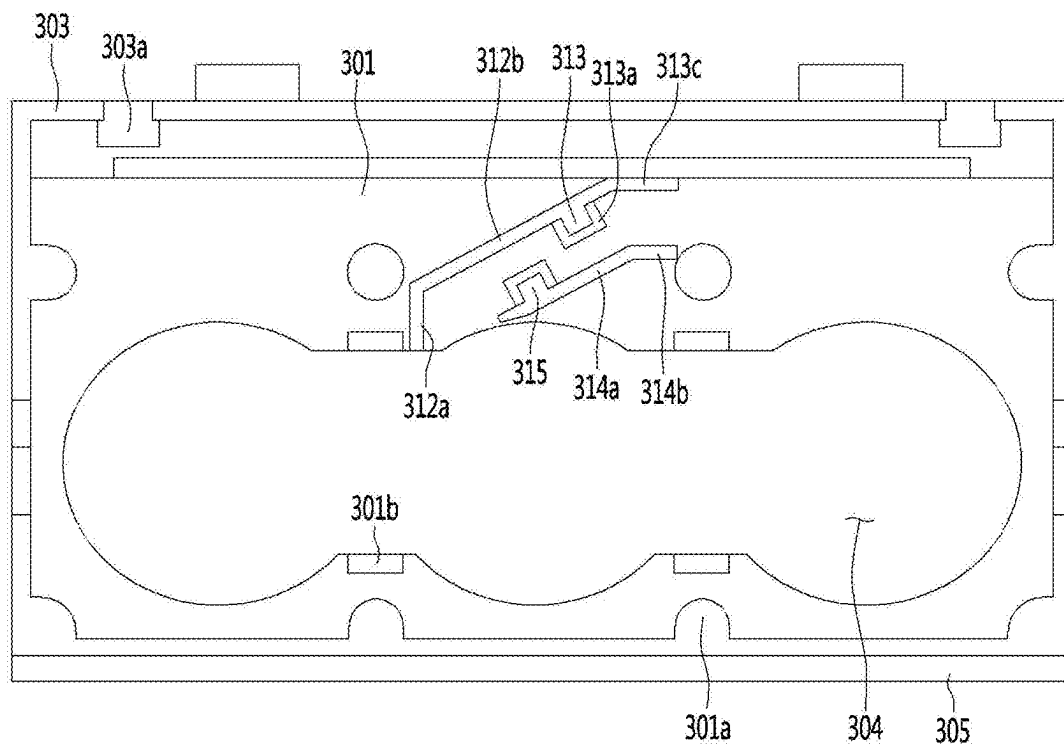


FIG. 15

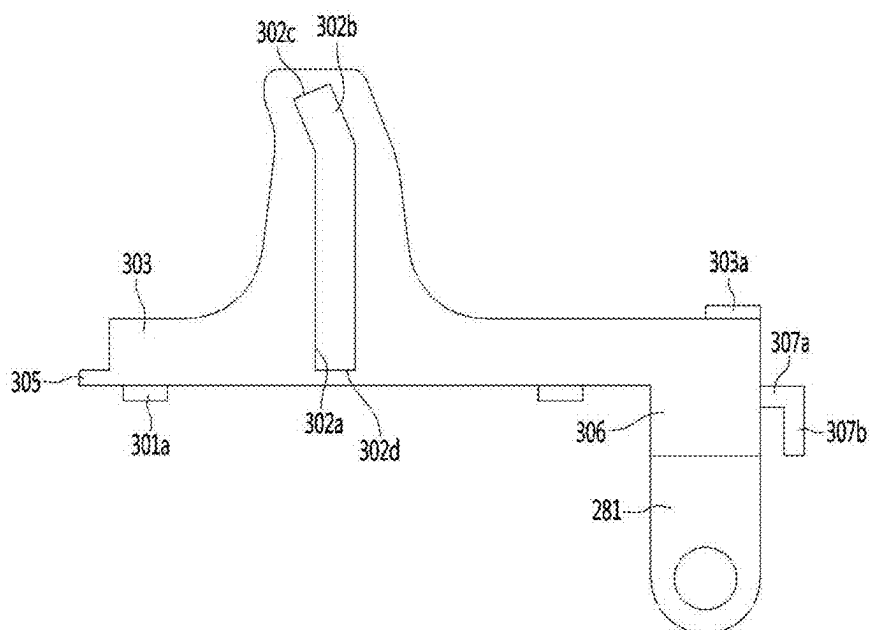


FIG. 16

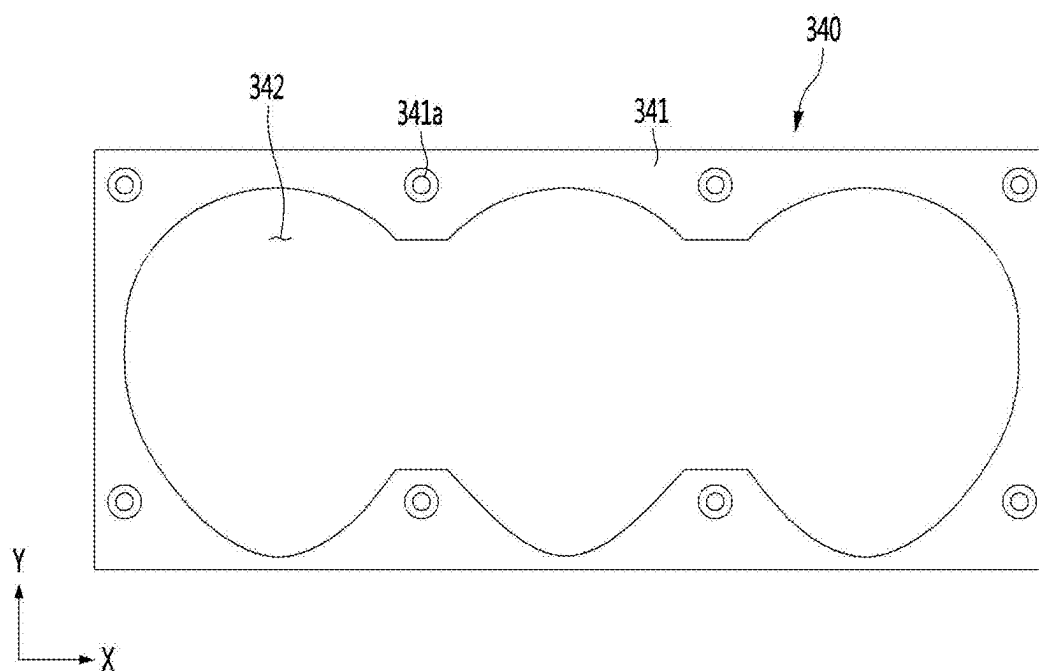
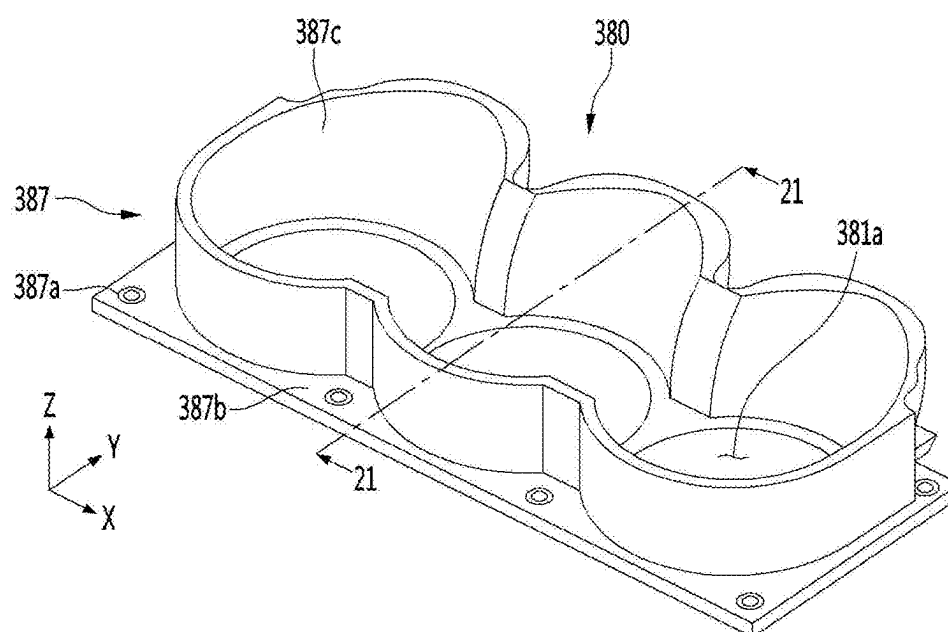


FIG. 17



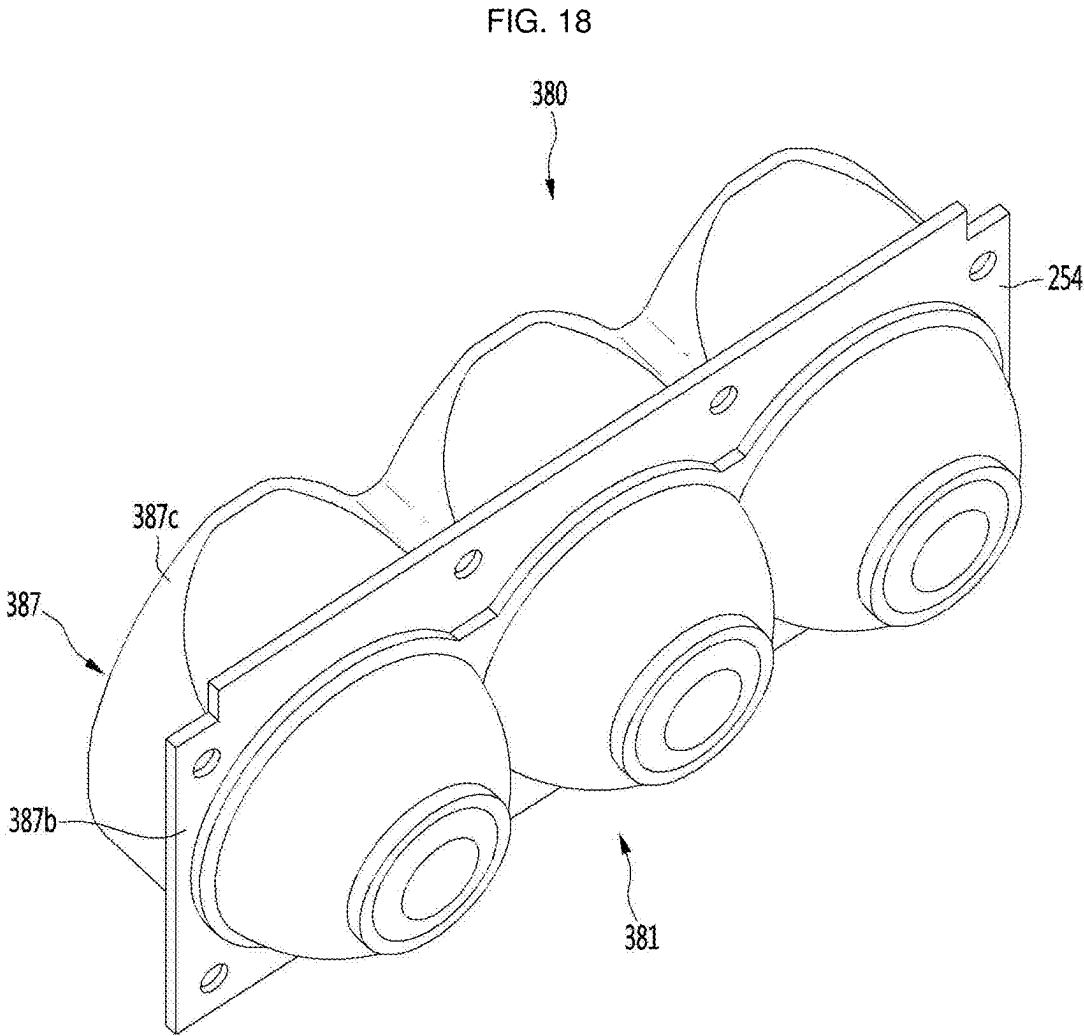


FIG. 19

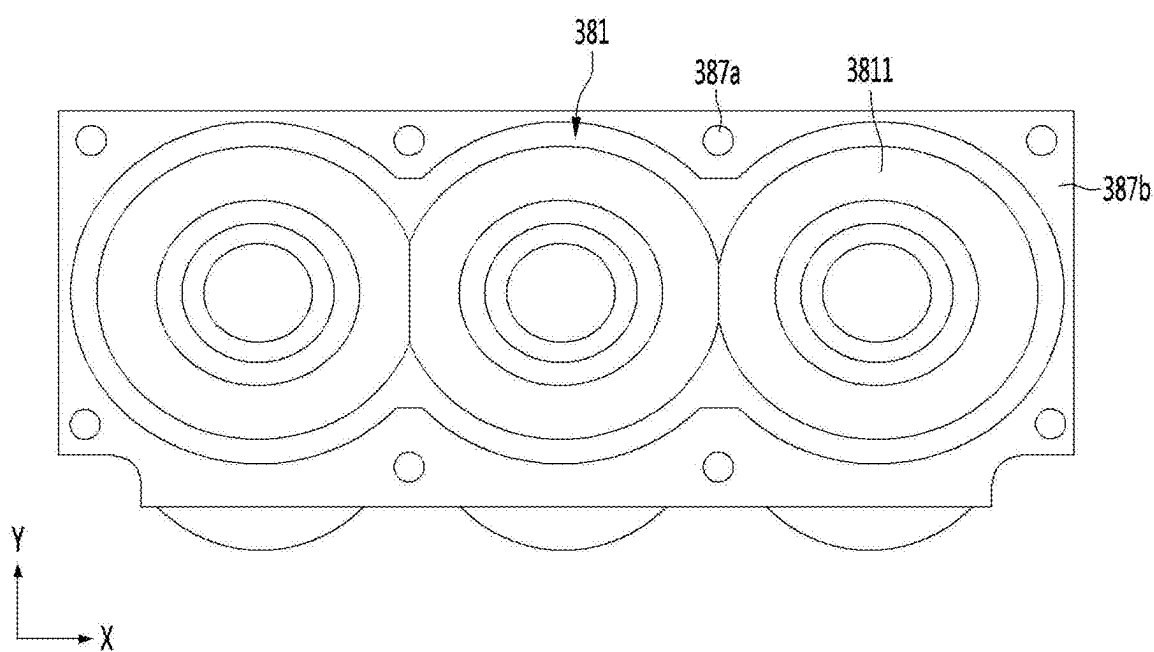


FIG. 20

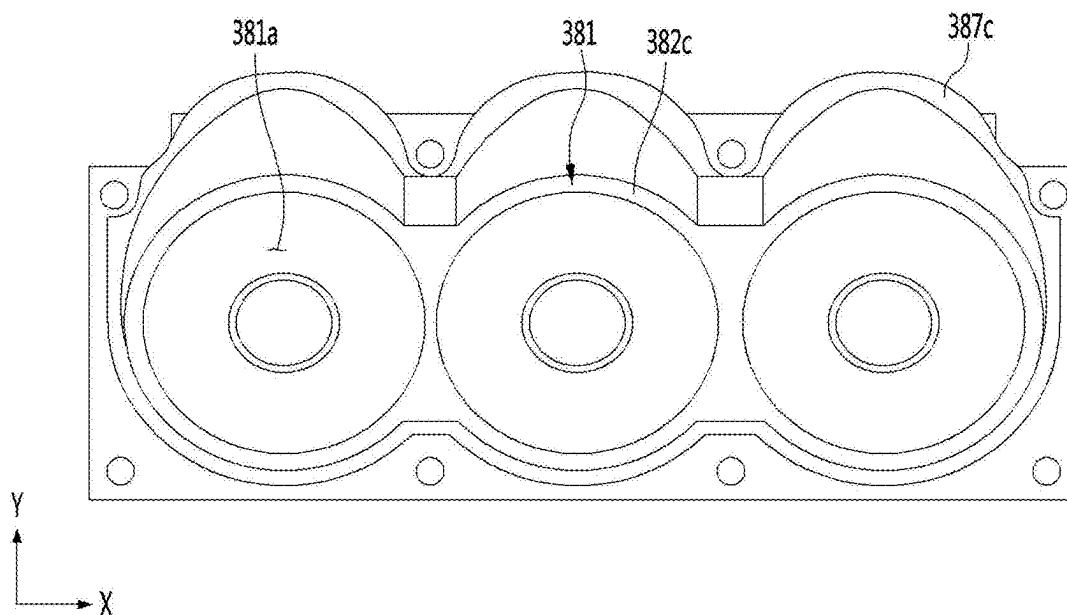


FIG. 21

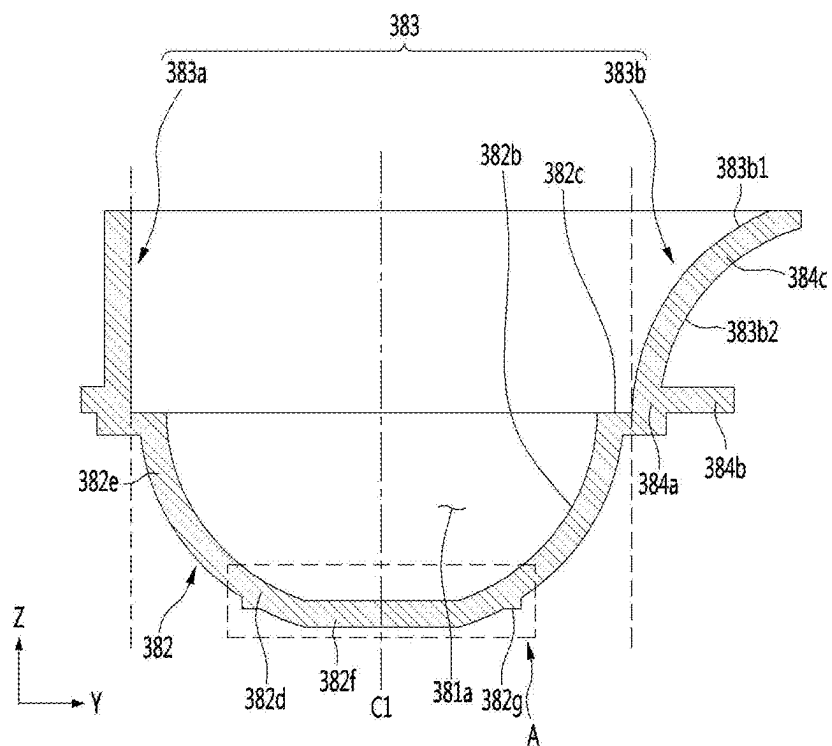


FIG. 22

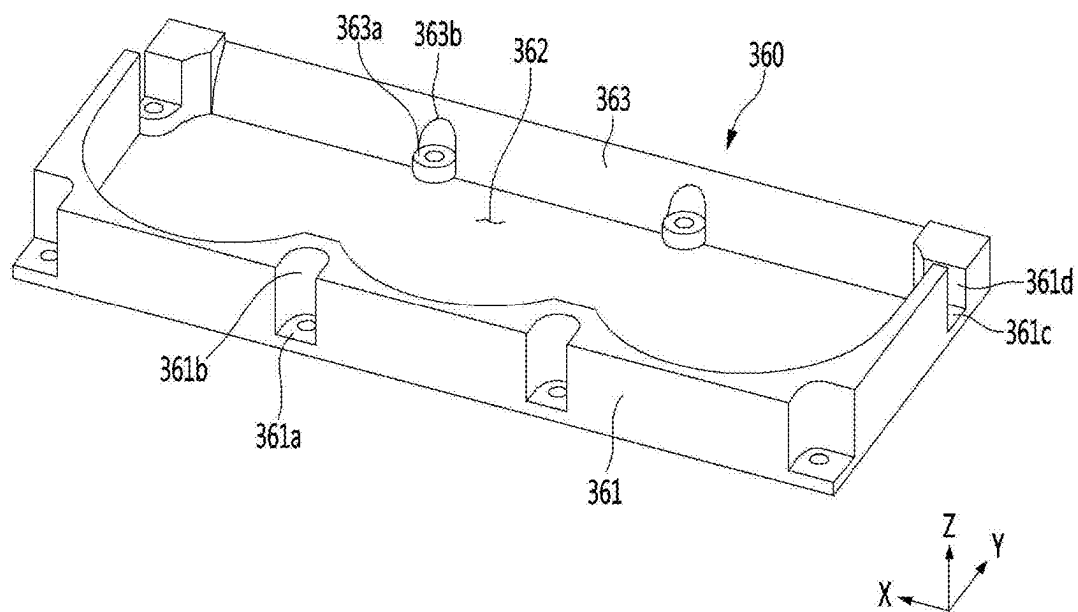


FIG. 23

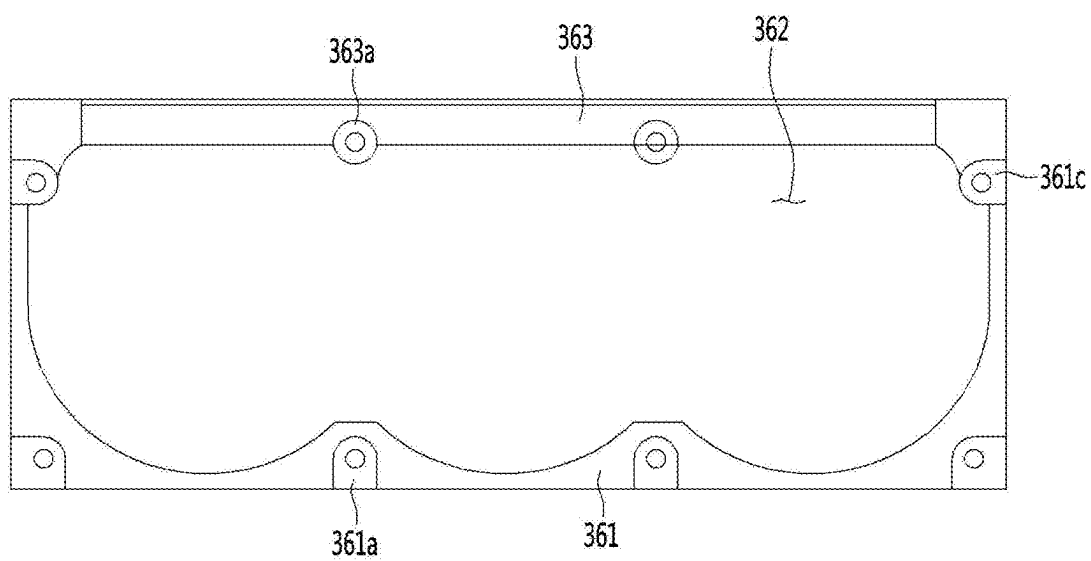


FIG. 24

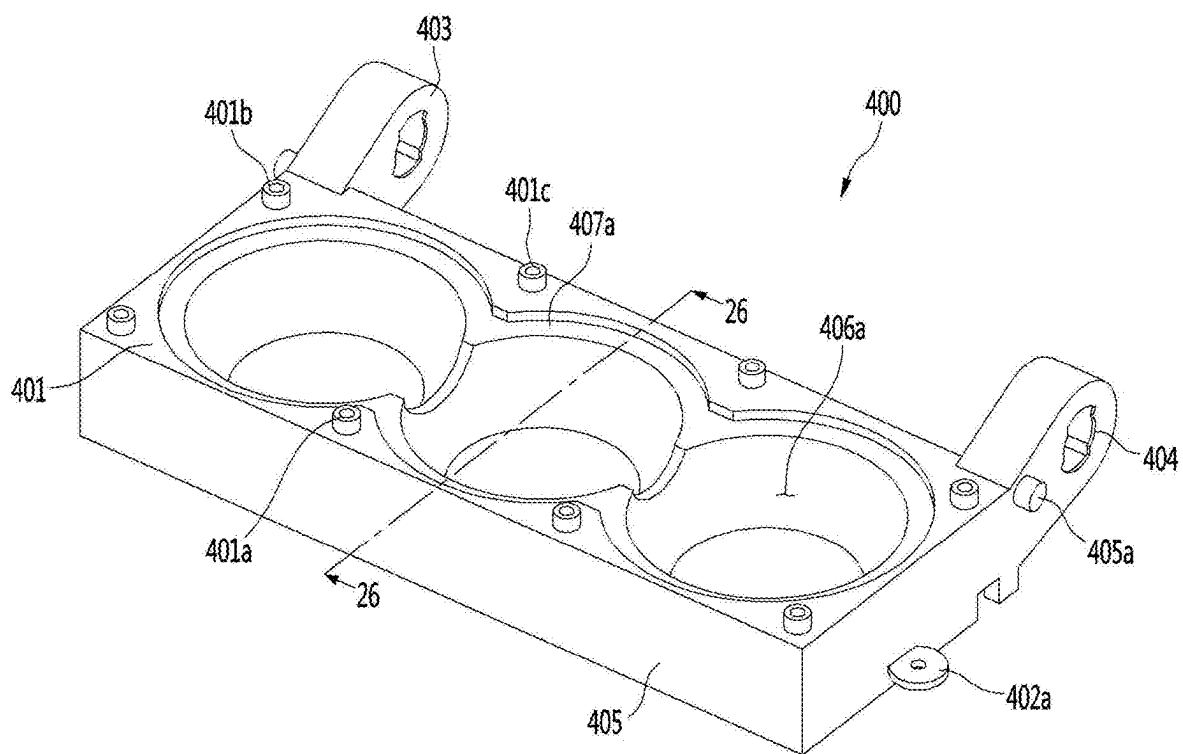


FIG. 25

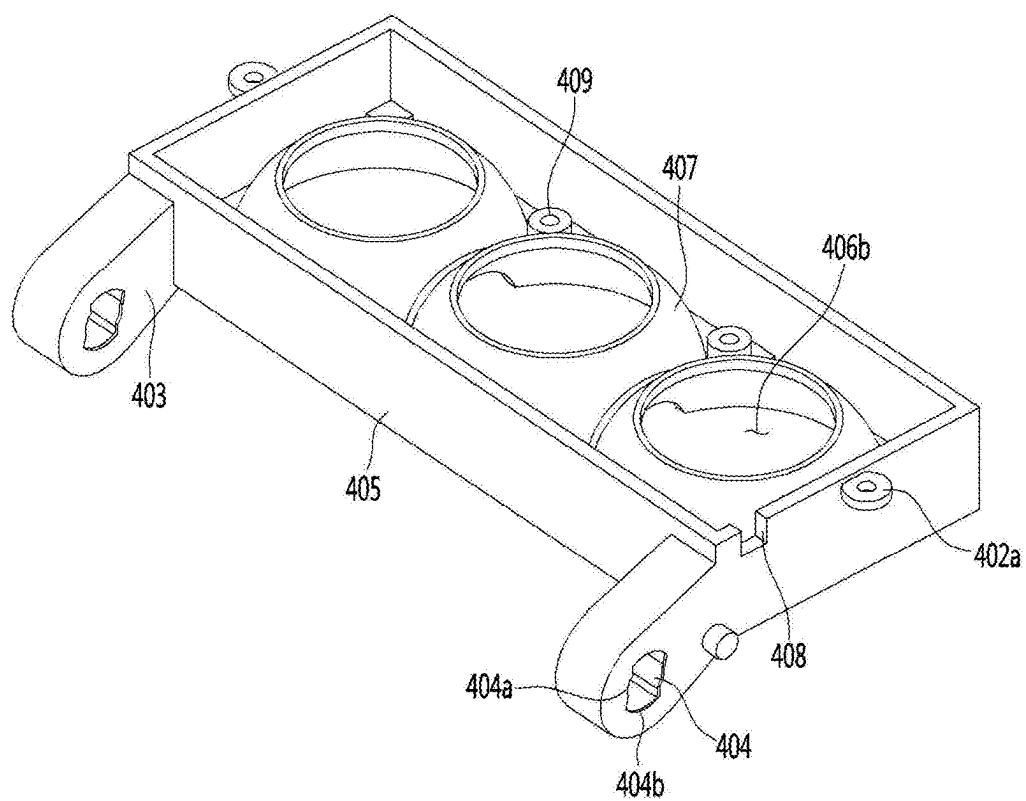


FIG. 27

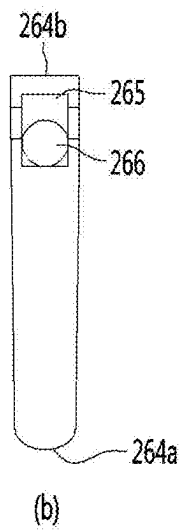
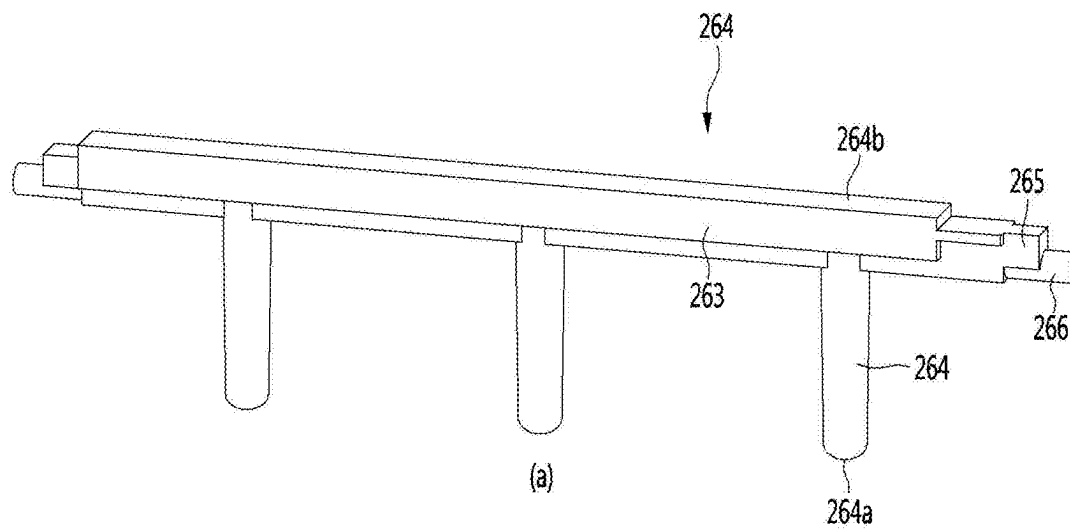


FIG. 28

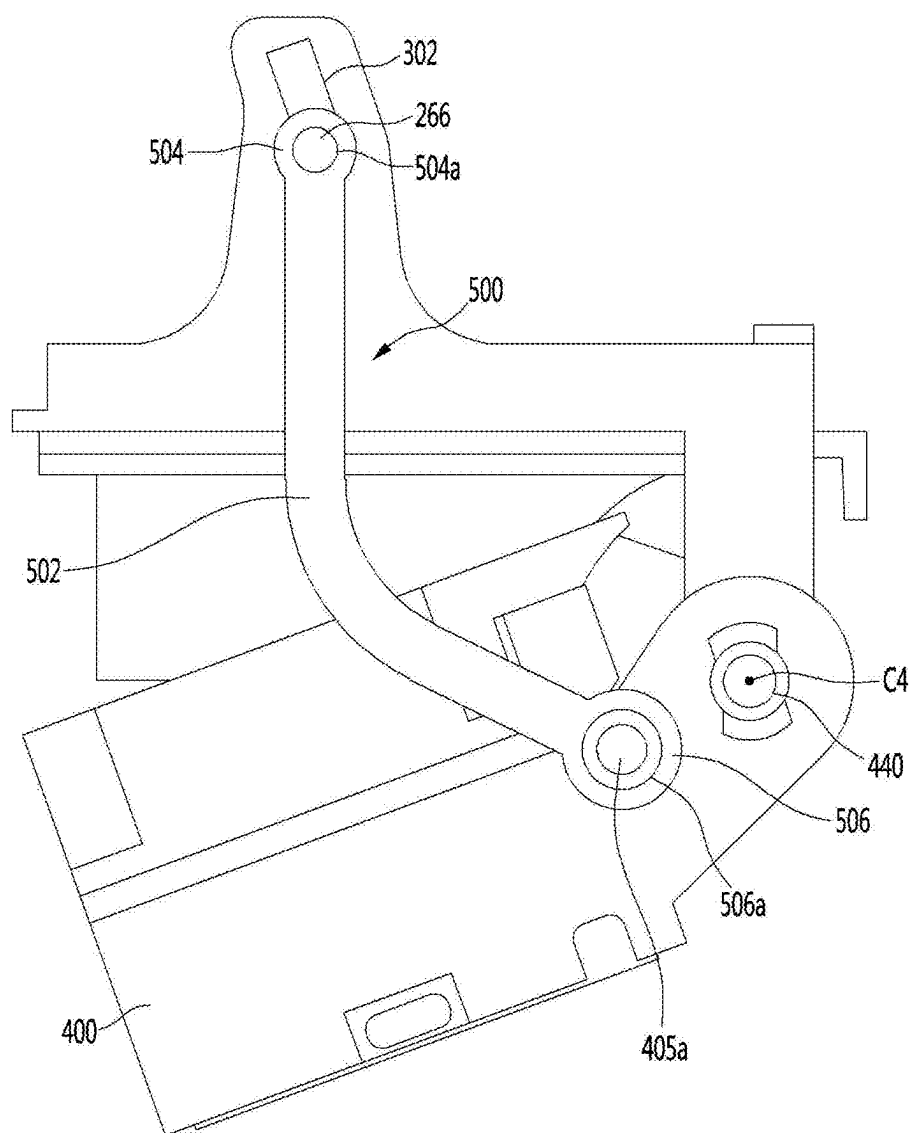


FIG. 29

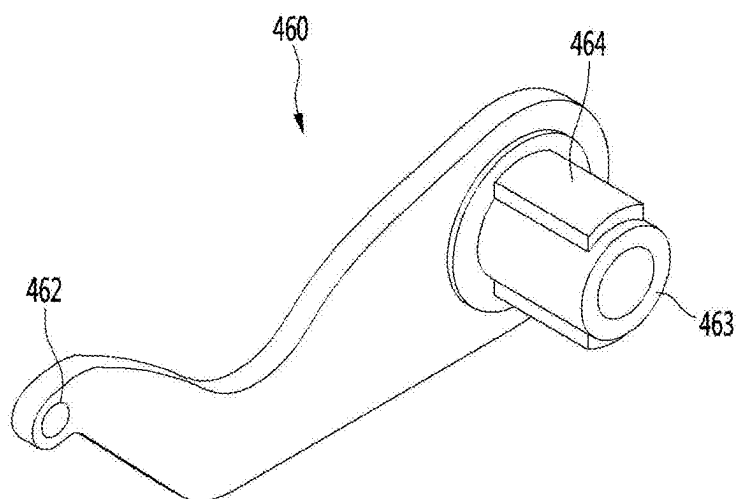


FIG. 30

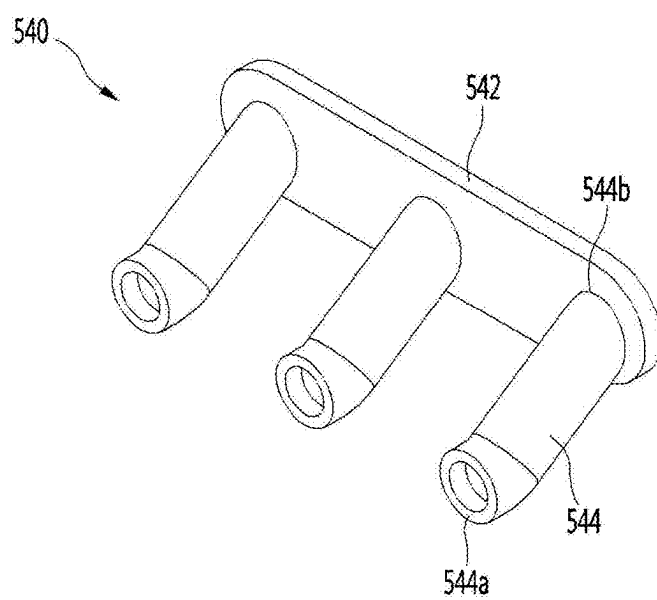


FIG. 31

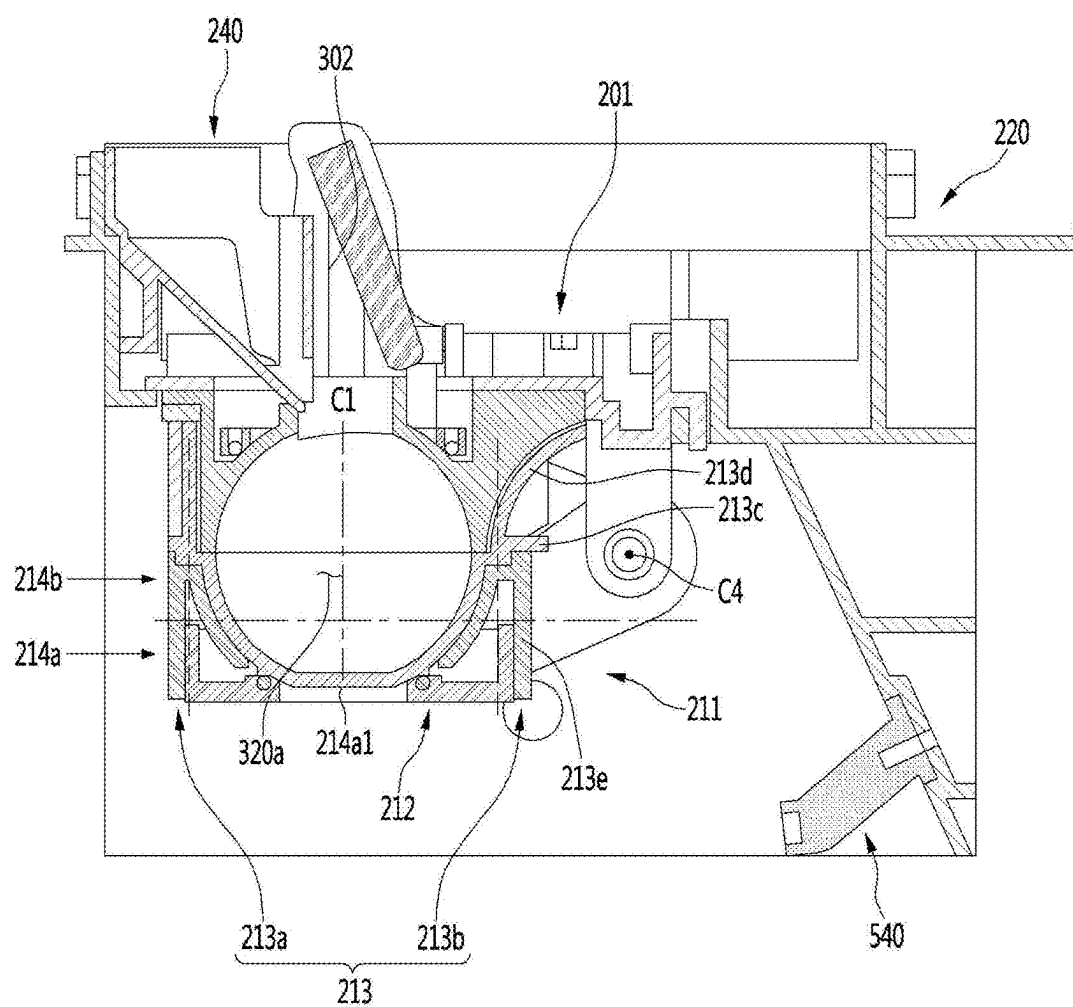


FIG. 32

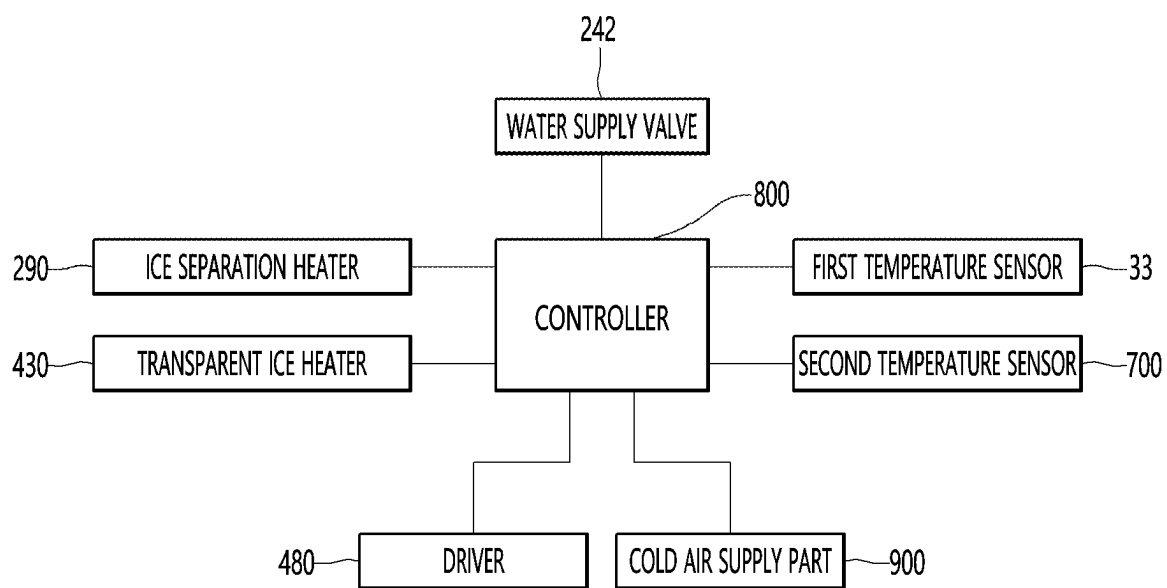


FIG. 33

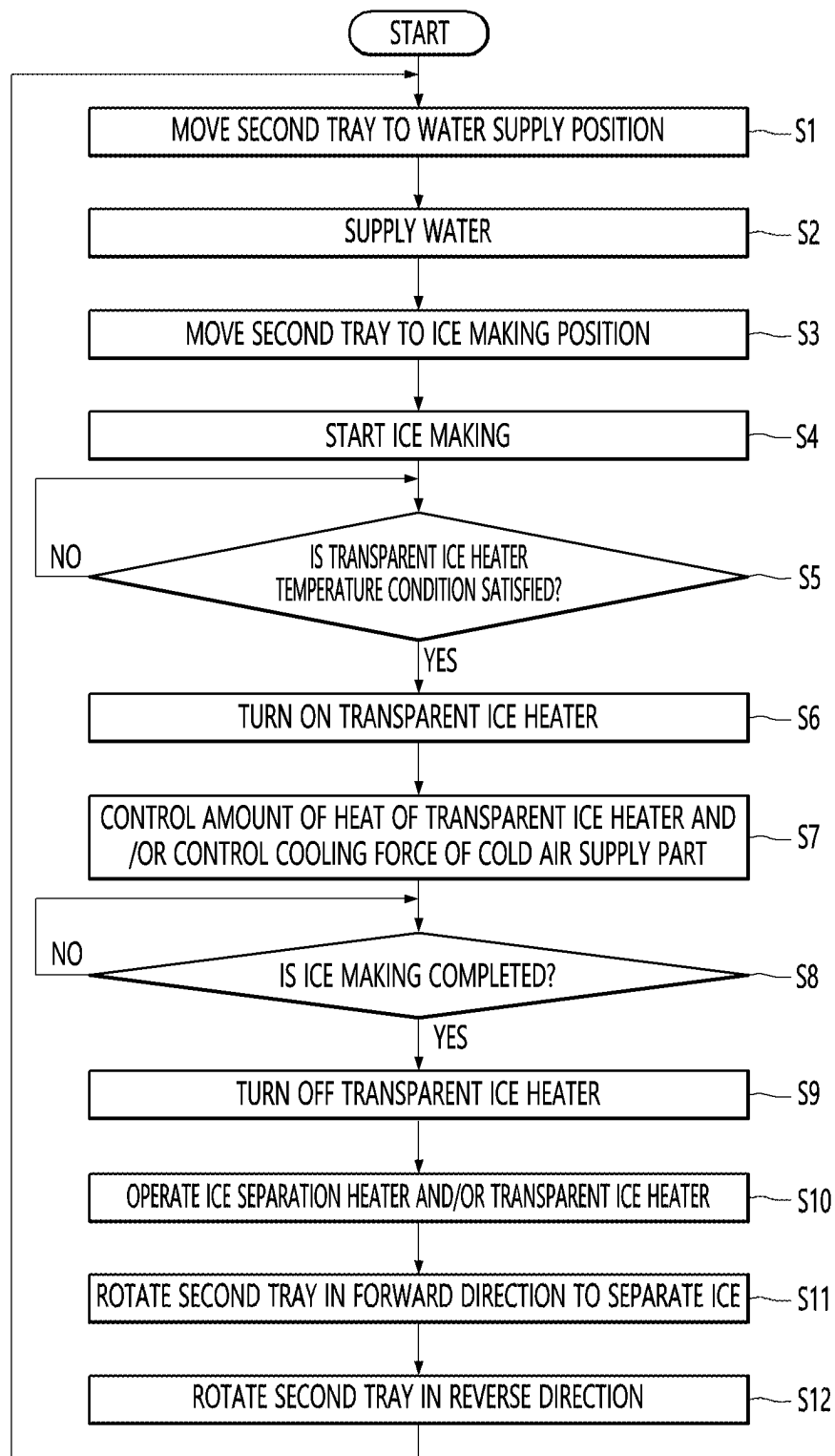


FIG. 34

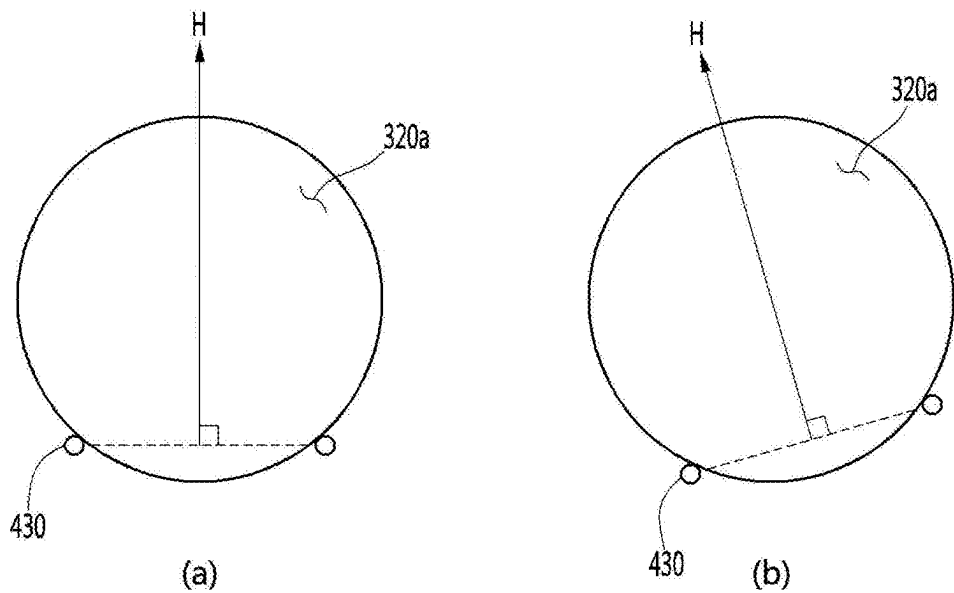


FIG. 35

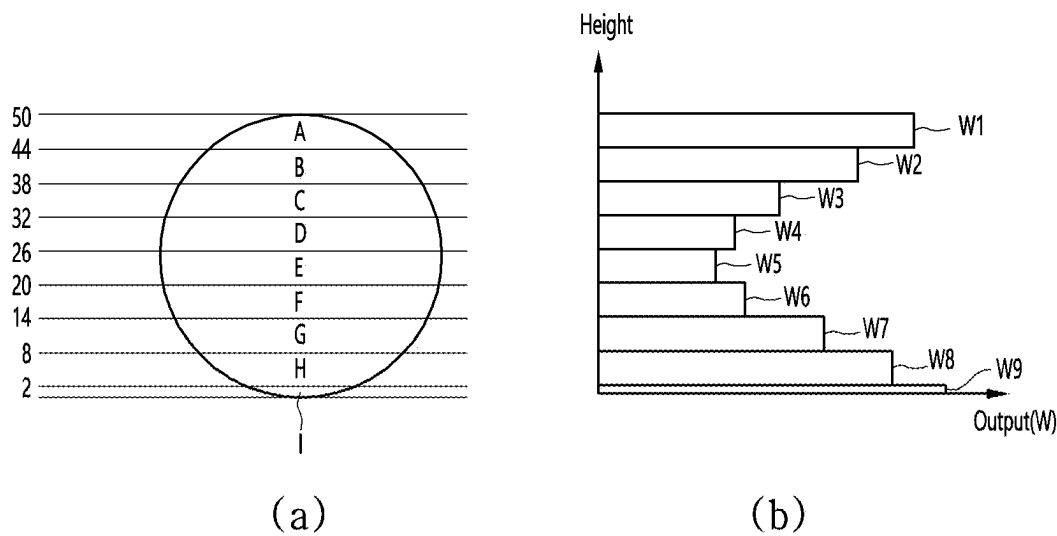


FIG. 36

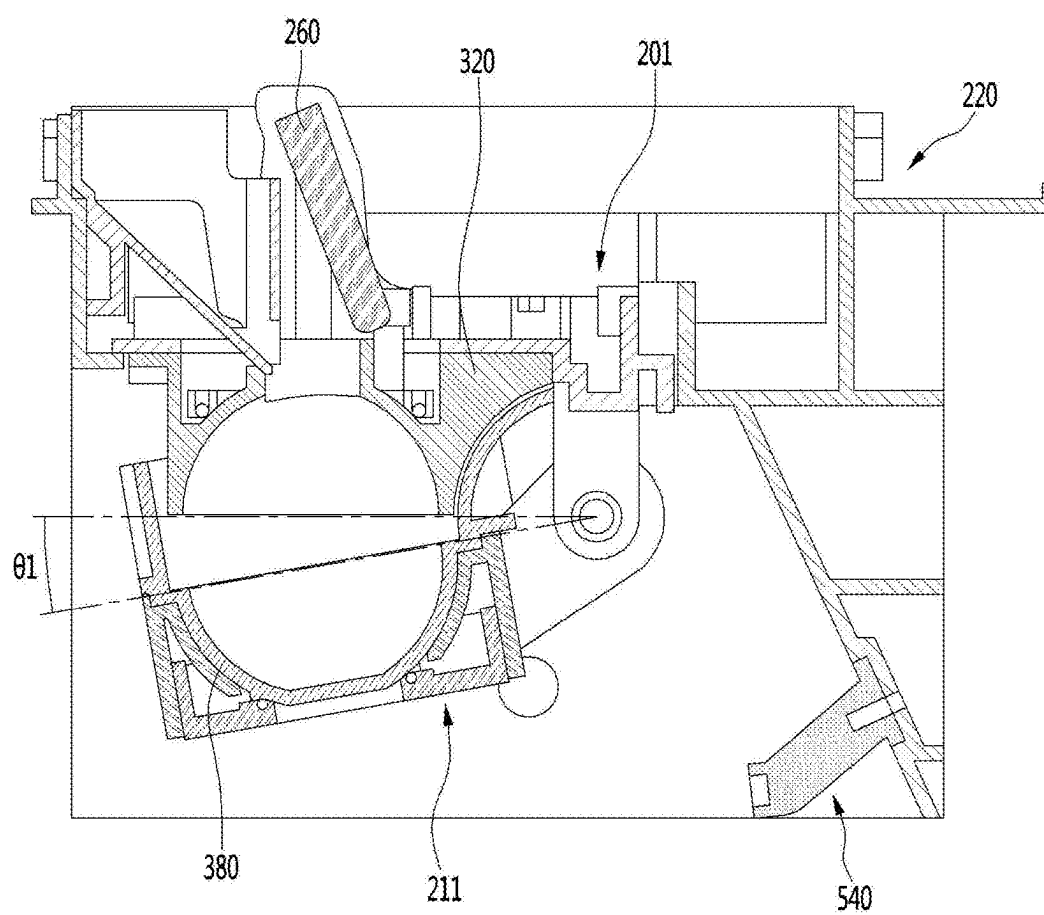


FIG. 37

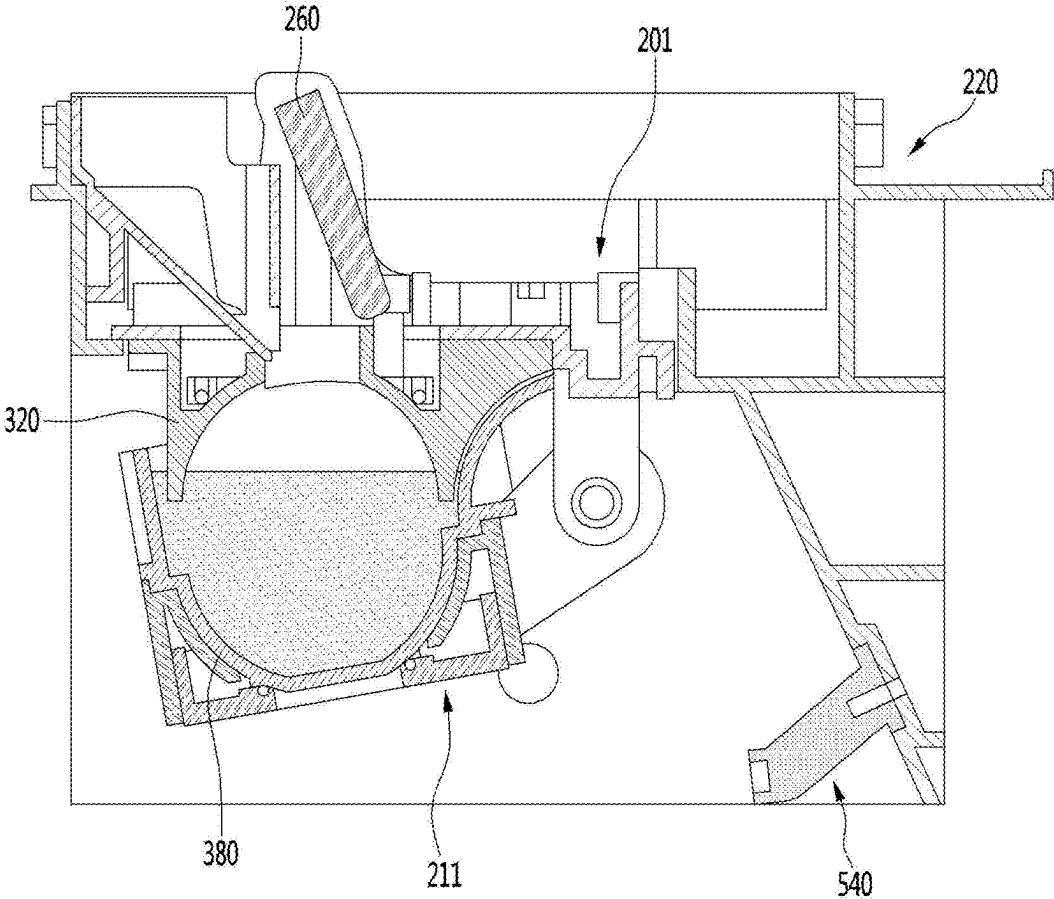


FIG. 38

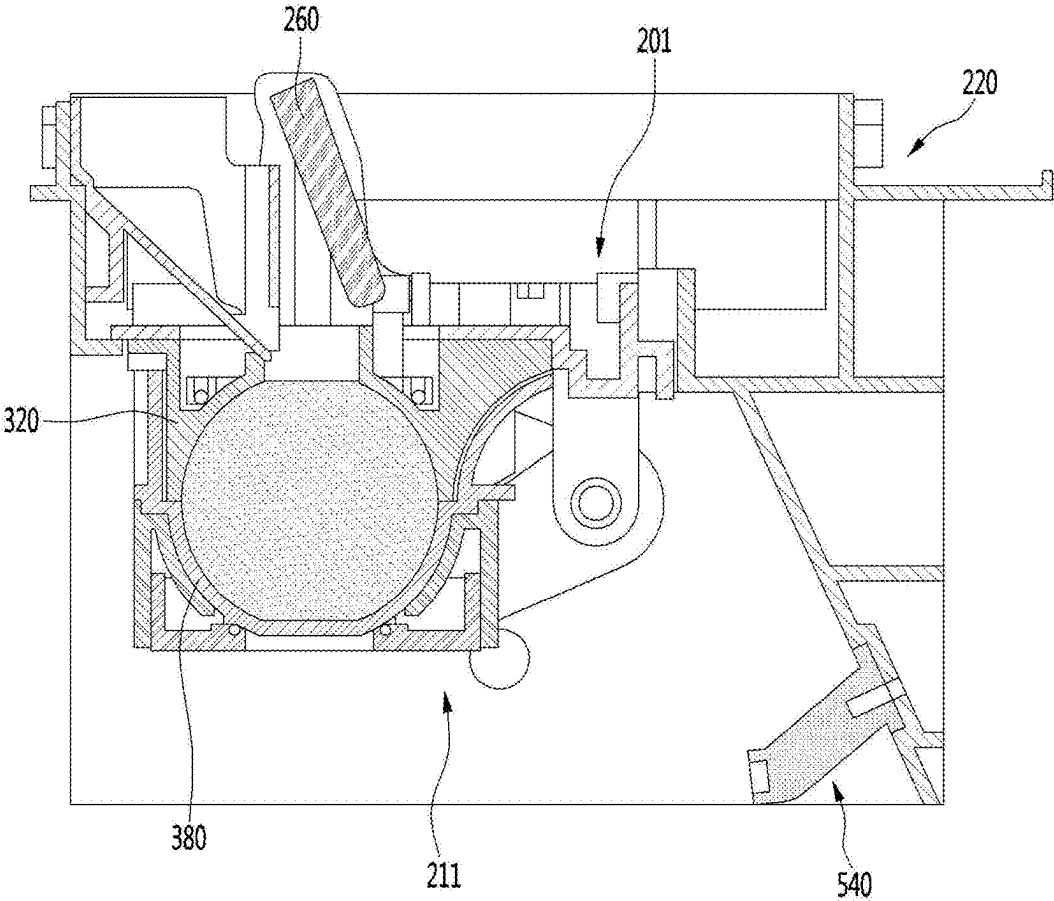


FIG. 39

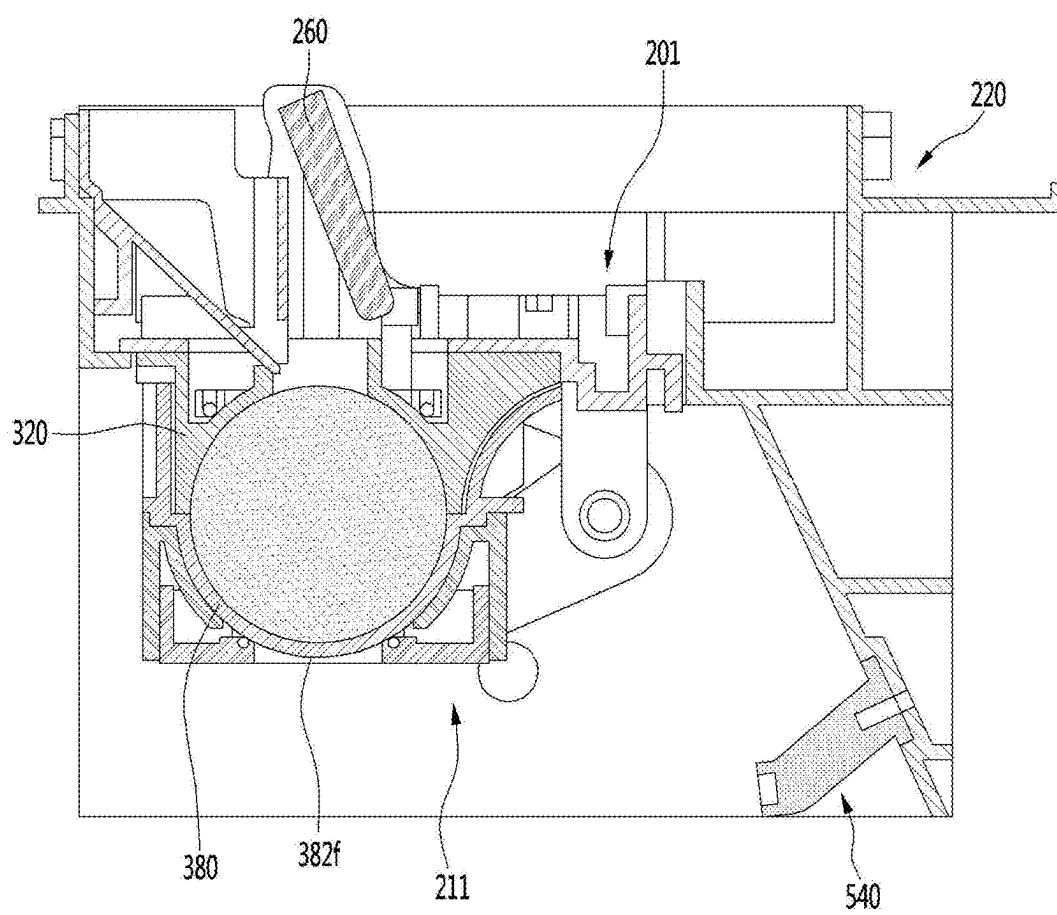


FIG. 40

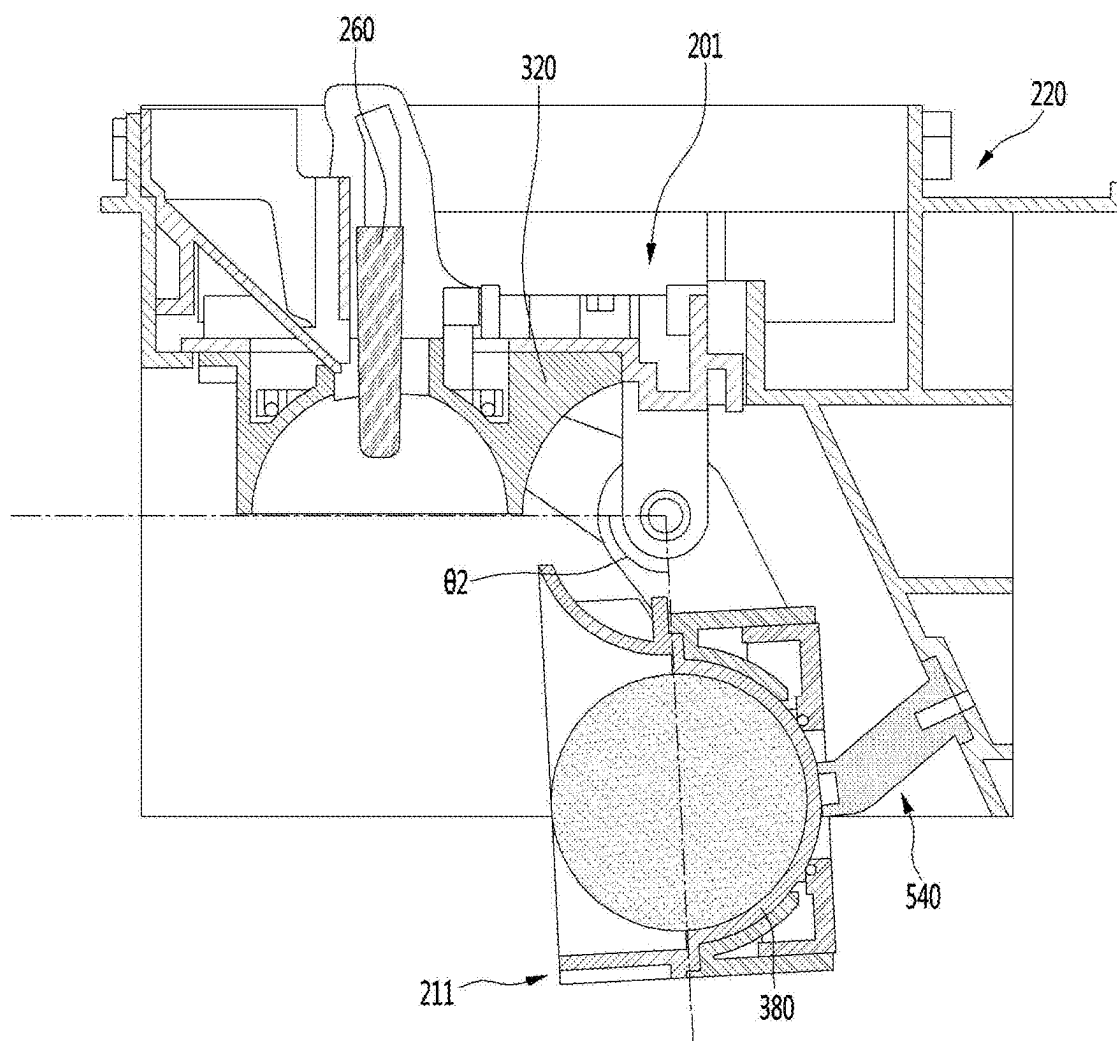


FIG. 41

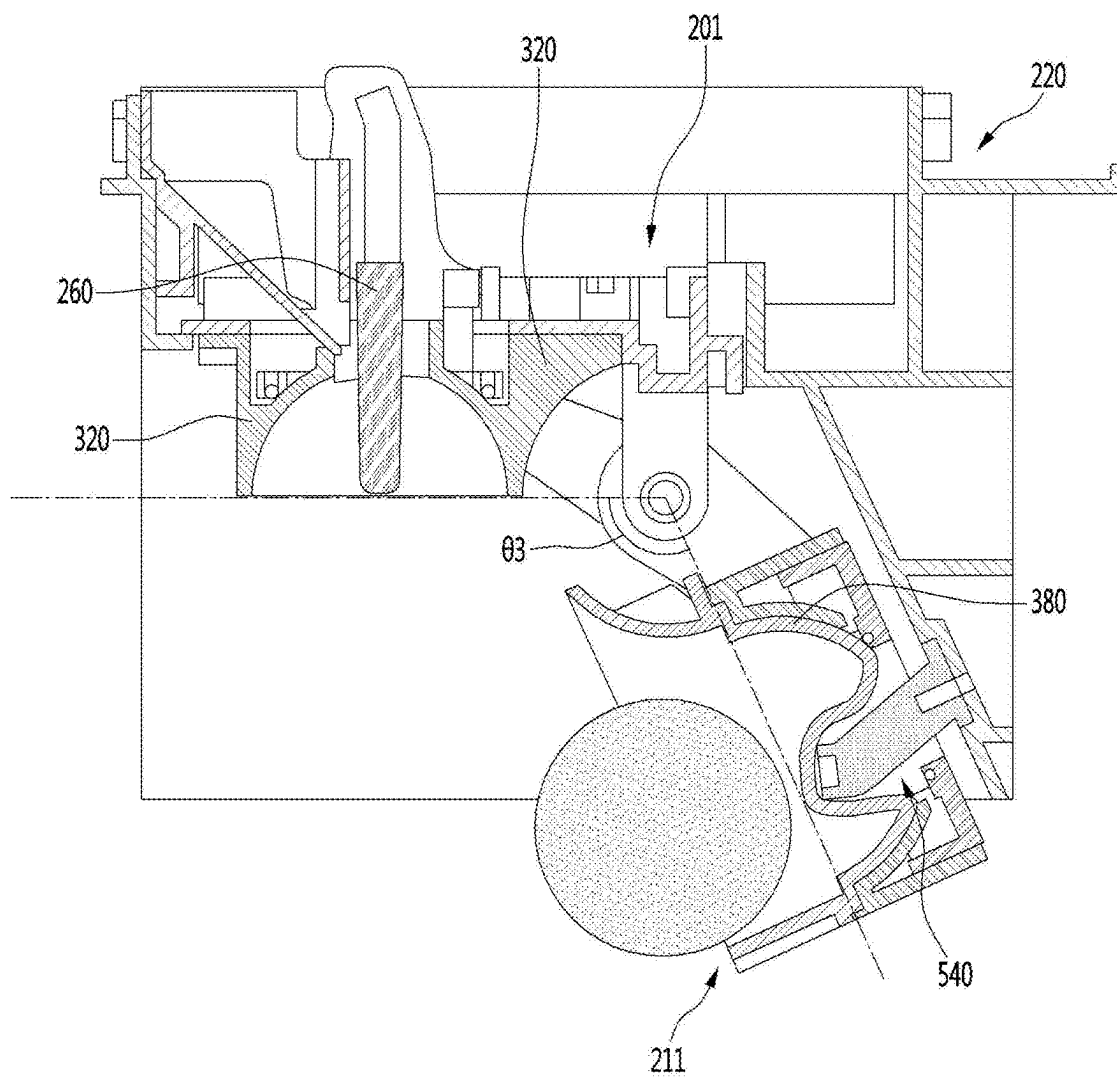


FIG. 42

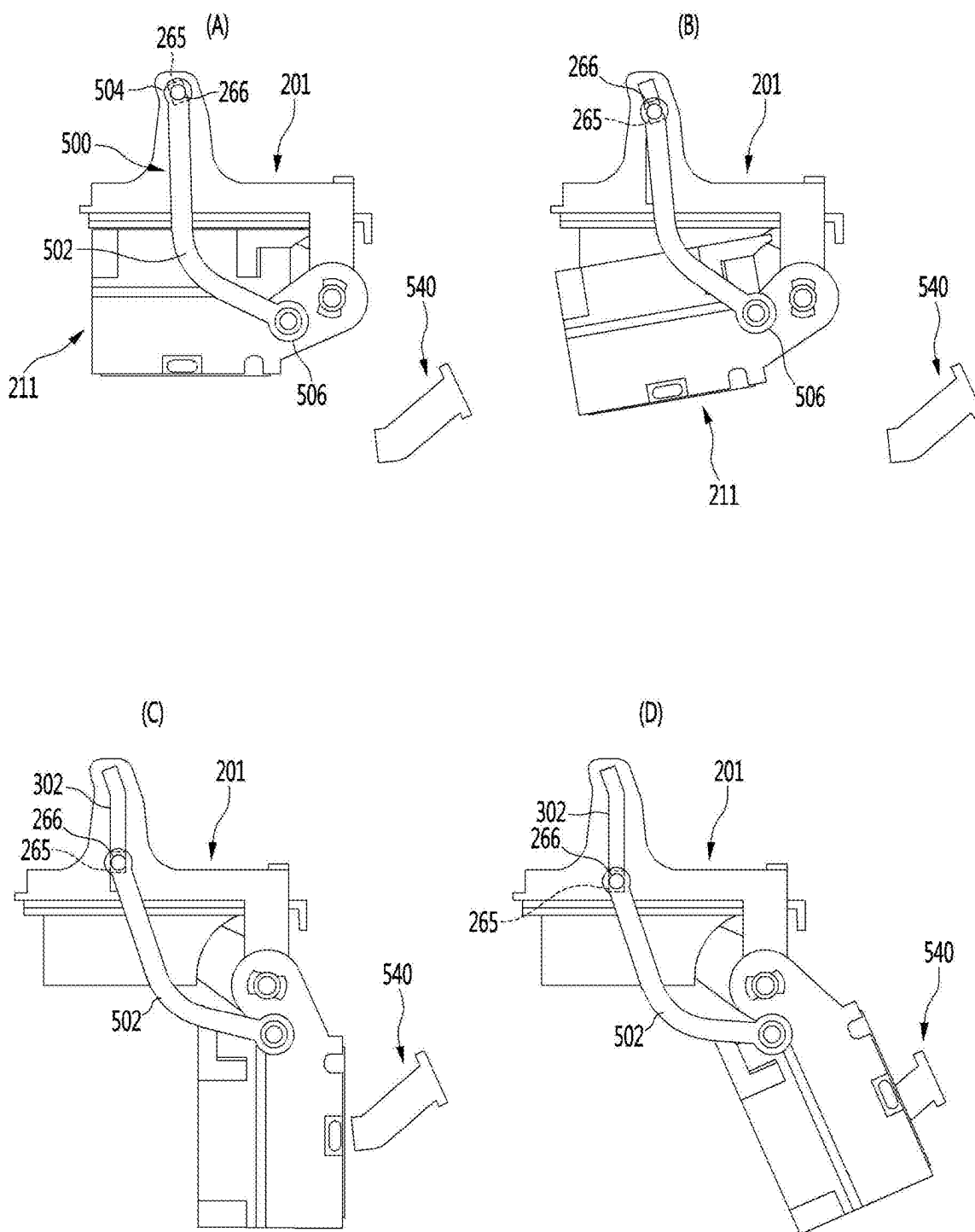


FIG. 43

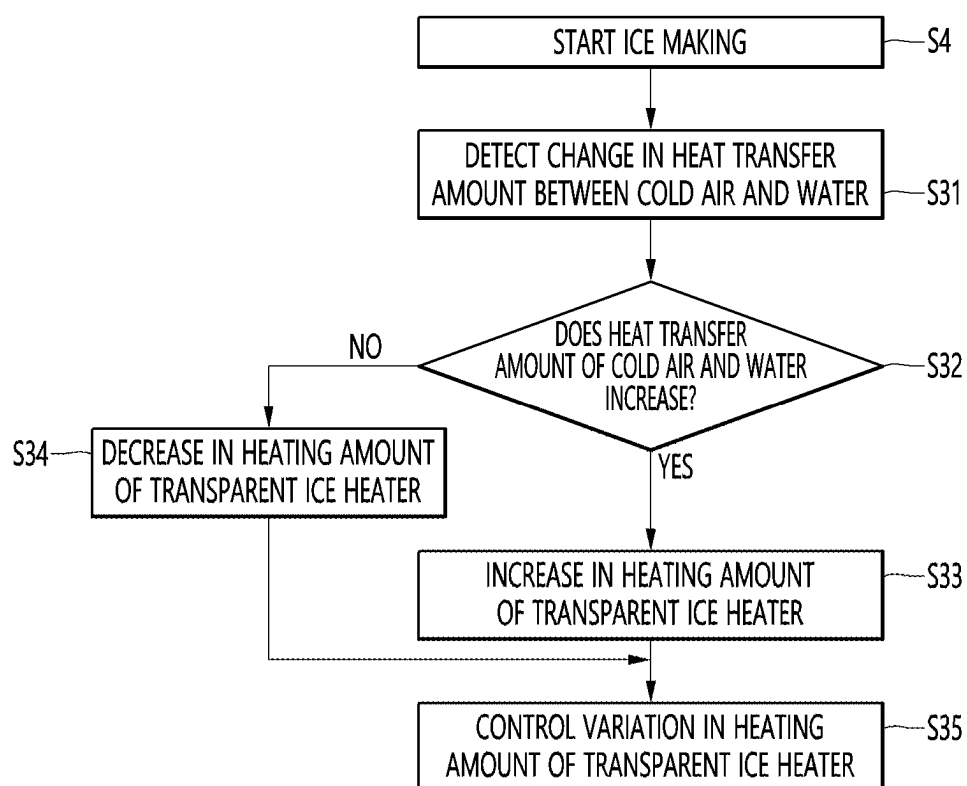


FIG. 44

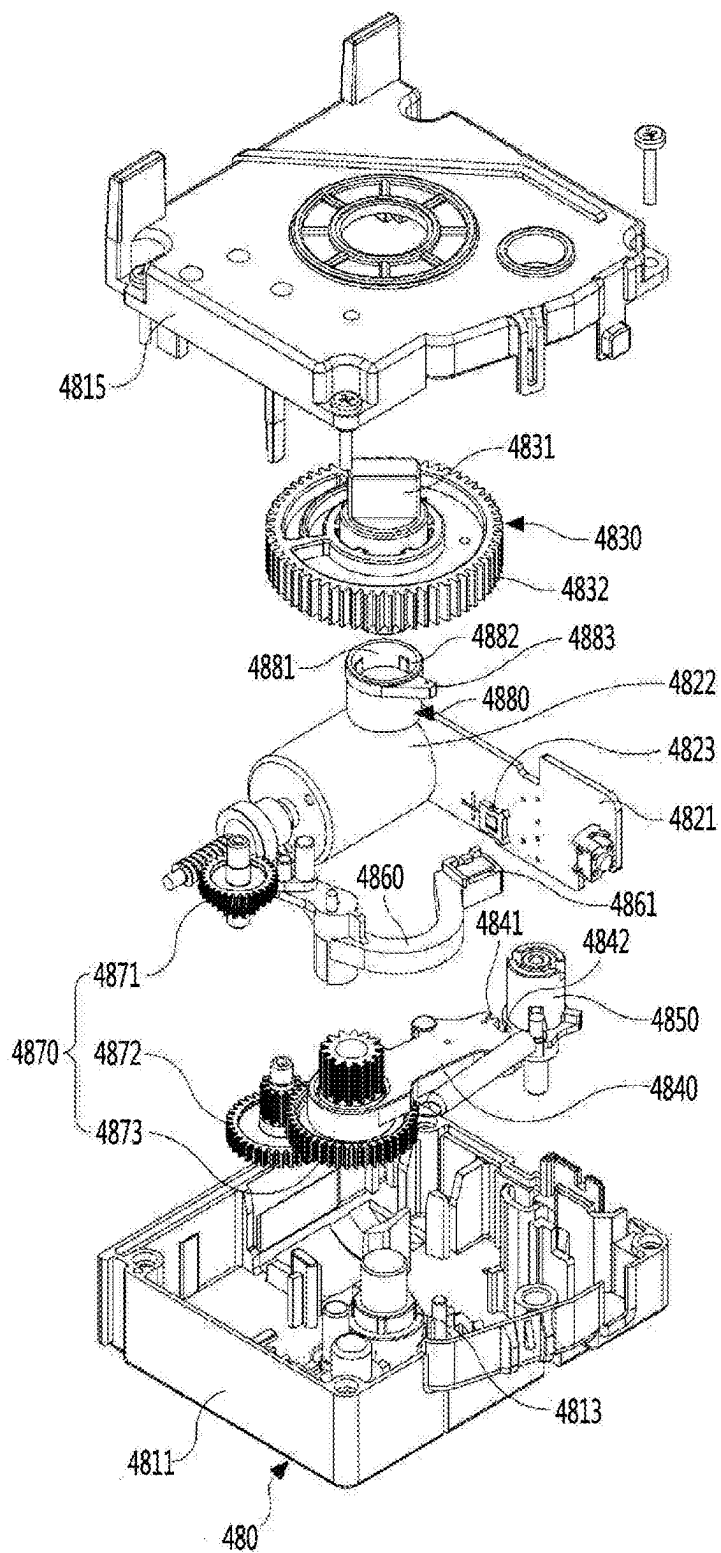


FIG. 45

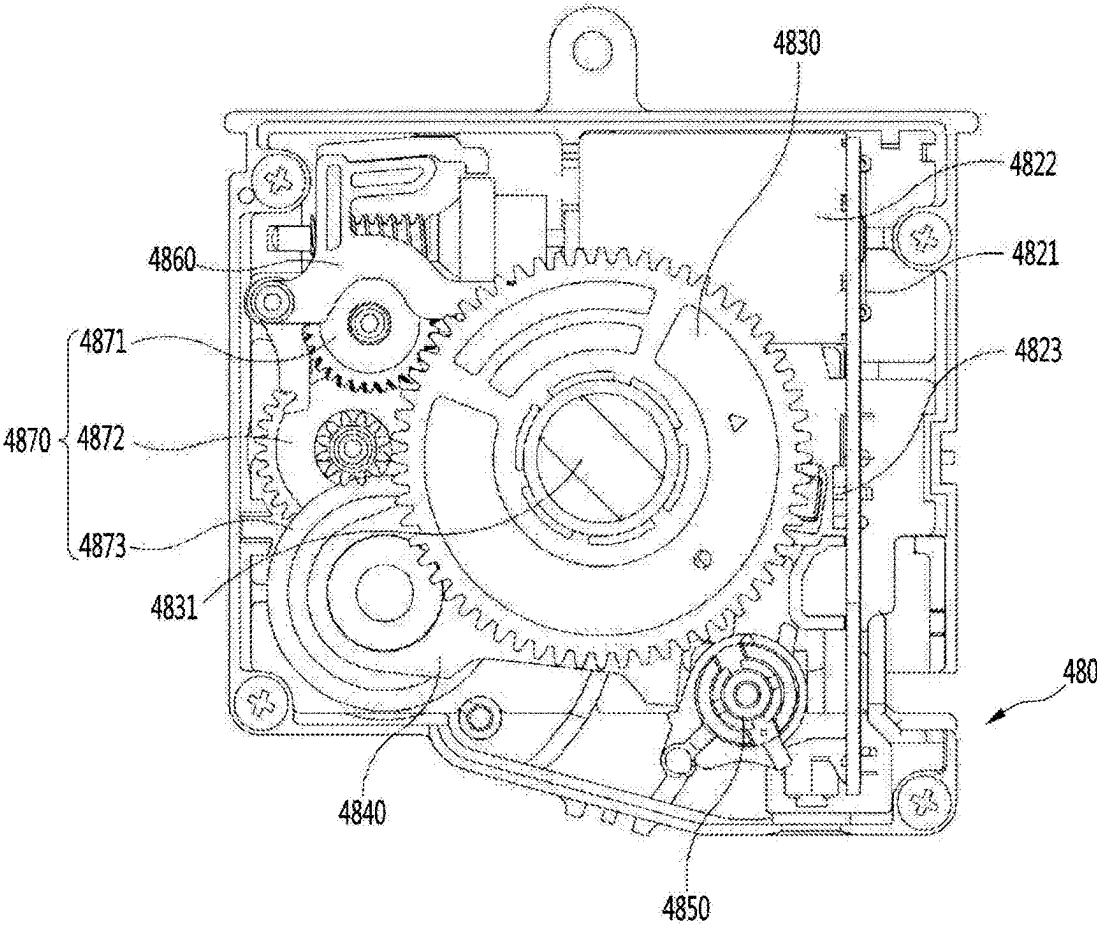


FIG. 46

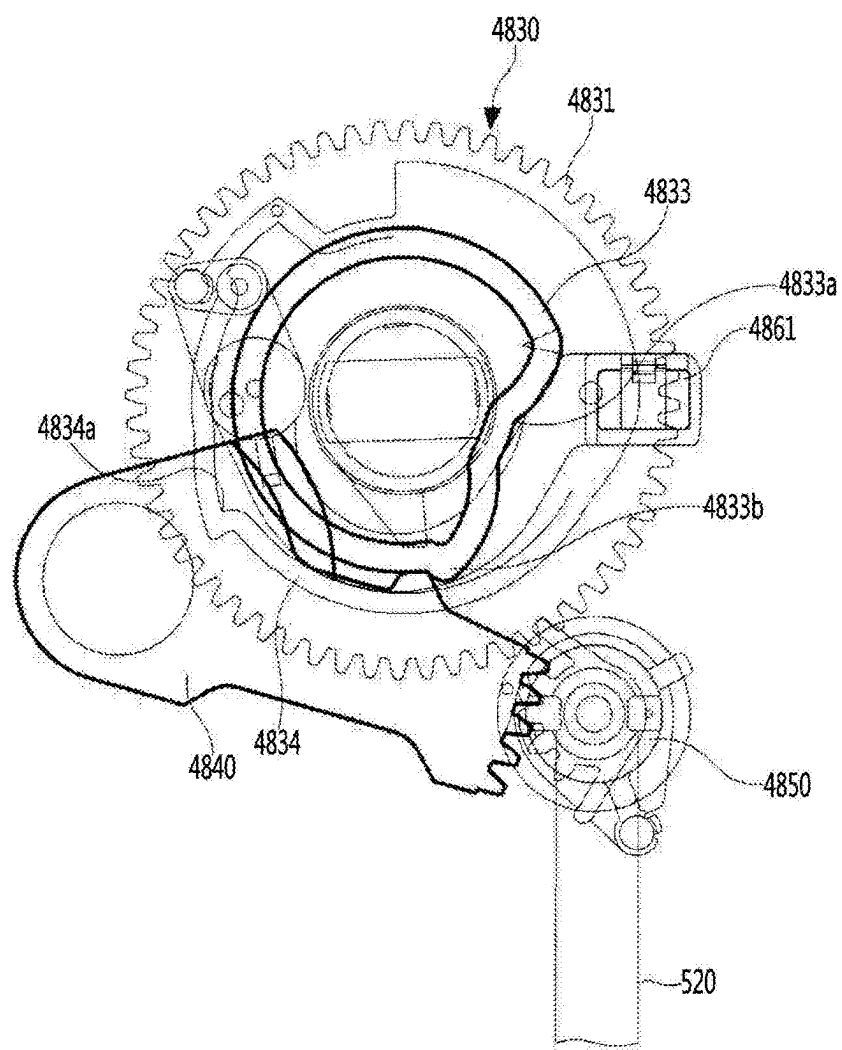
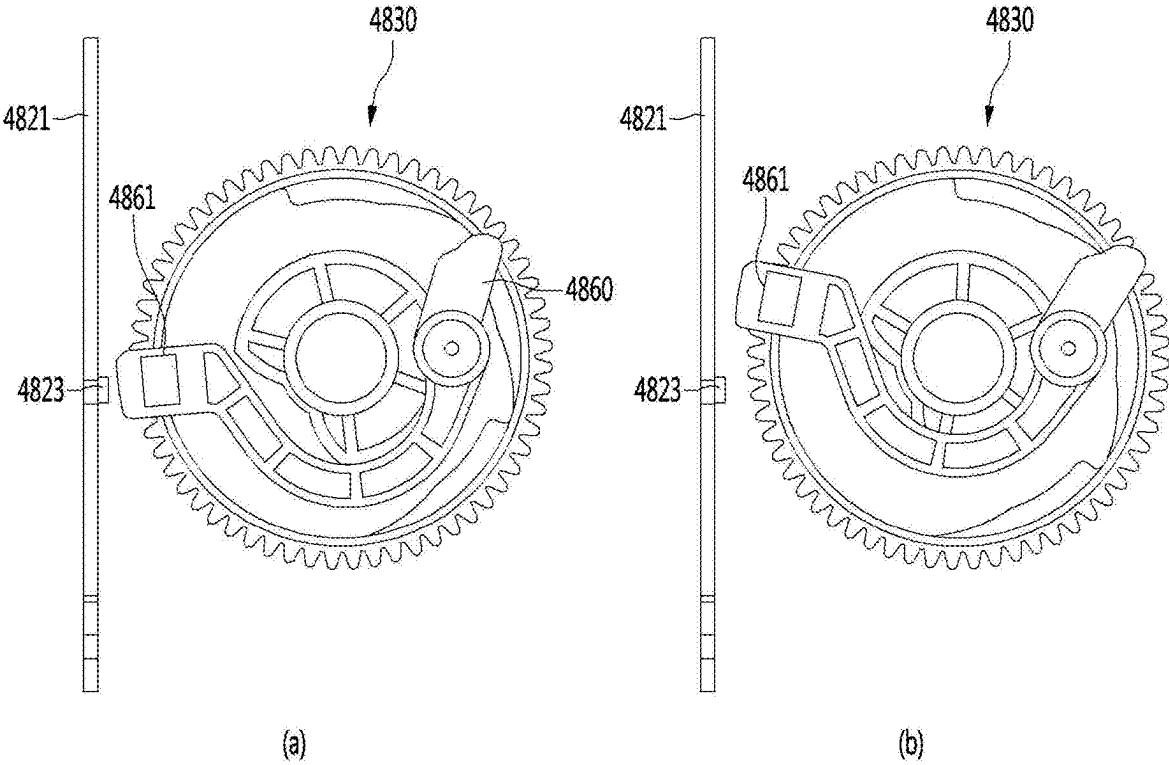


FIG. 47



REFRIGERATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/409,336, filed Jan. 10, 2024, which is a continuation of U.S. patent application Ser. No. 17/282,324, filed Apr. 1, 2021, now U.S. Pat. No. 11,898,785, which is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2019/012881, filed Oct. 1, 2019, which claims priority to Korean Patent Application Nos. 10-2018-0117785, 10-2018-0117819, 10-2018-0117821, 10-2018-0117822, all filed Oct. 2, 2018, 10-2018-0142117, filed Nov. 16, 2018, and 10-2019-0081712, filed Jul. 6, 2019, whose entire disclosures are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This specification relates to a refrigerator.

BACKGROUND ART

[0003] In general, refrigerators are home appliances for storing food at a low temperature in a storage space that is covered by a door. The refrigerator may cool the inside of the storage space by using cold air to store the stored food in a refrigerated or frozen state. Generally, an ice maker for making ice is provided in the refrigerator. The ice maker makes ice by cooling water after accommodating the water supplied from a water supply source or a water tank into a tray. Also, the ice maker separates the made ice from the ice tray in a heating manner or twisting manner. As described above, the ice maker through which water is automatically supplied, and the ice automatically separated may be opened upward so that the made ice is pumped up. As described above, the ice made in the ice maker may have at least one flat surface such as crescent or cubic shape.

[0004] When the ice has a spherical shape, it is more convenient to use the ice, and also, it is possible to provide different feeling of use to a user. Also, even when the made ice is stored, a contact area between the ice cubes may be minimized to minimize a mat of the ice cubes.

[0005] An ice maker is disclosed in Korean Registration No. 10-1850918 (hereinafter, referred to as a “prior art document 1”) that is a prior art document.

[0006] The ice maker disclosed in the prior art document 1 includes an upper tray in which a plurality of upper cells, each of which has a hemispherical shape, are arranged, and which includes a pair of link guide parts extending upward from both side ends thereof, a lower tray in which a plurality of upper cells, each of which has a hemispherical shape and which is rotatably connected to the upper tray, a rotation shaft connected to rear ends of the lower tray and the upper tray to allow the lower tray to rotate with respect to the upper tray, a pair of links having one end connected to the lower tray and the other end connected to the link guide part, and an upper ejecting pin assembly connected to each of the pair of links in a state in which both ends thereof are inserted into the link guide part and elevated together with the upper ejecting pin assembly.

[0007] In the prior art document 1, although the spherical ice is made by the hemispherical upper cell and the hemispherical lower cell, since the ice is made at the same time

in the upper and lower cells, bubbles containing water are not completely discharged but are dispersed in the water to make opaque ice.

[0008] An ice maker is disclosed in Japanese Patent Laid-Open No. 9-269172 (hereinafter, referred to as a “prior art document 2”) that is a prior art document.

[0009] The ice maker disclosed in the prior art document 2 includes an ice making plate and a heater for heating a lower portion of water supplied to the ice making plate. In the case of the ice maker disclosed in the prior art document 2, water on one surface and a bottom surface of an ice making block is heated by the heater in an ice making process. Thus, when solidification proceeds on the surface of the water, and also, convection occurs in the water to make transparent ice. When growth of the transparent ice proceeds to reduce a volume of the water within the ice making block, the solidification rate is gradually increased, and thus, sufficient convection suitable for the solidification rate may not occur. Thus, in the case of the prior art document 2, when about $\frac{2}{3}$ of water is solidified, a heating amount of heater increases to suppress an increase in the solidification rate. However, the prior art document 2 discloses a feature in which when the volume of water is simply reduced, only the heating amount of heater increases and does not disclose a structure and a heater control logic for making ice having high transparency without reducing the ice making rate.

DISCLOSURE

Technical Problem

[0010] Embodiments provide a refrigerator capable of making ice having uniform transparency by reducing transfer of heat, which is transferred to one tray adjacent to an operating heater, to an ice making cell provided by the other tray in an ice making process.

[0011] Embodiments also provide a refrigerator in which transparency per unit height is uniform even while transparent ice is made.

[0012] Embodiments also provide a refrigerator in which water of an ice making cell is prevented from leaking between tray assemblies at an ice making position.

Technical Solution

[0013] A refrigerator according to one aspect may include a first tray assembly defining one portion of an ice making cell and a second tray assembly defining the other portion of the ice making cell. The tray assembly may be defined as a tray. The tray assembly may be defined as a tray and a tray case surrounding the tray. The first tray assembly may include a first tray, and the second tray assembly may include a second tray.

[0014] The refrigerator may further include a heater disposed adjacent to at least one of the first tray assembly or the second tray assembly. Any one tray assembly of the first and second tray assemblies may be closer to the ice separation heater than the other tray assembly. The heater may be disposed on the one tray assembly.

[0015] The refrigerator may further include a driver connected to the second tray assembly. The second tray assembly may be in contact with the first tray assembly in an ice making process and be spaced apart from at least a portion of the first tray assembly in an ice separation process by the

driver. The refrigerator may further include a controller configured to control the heater and the driver.

[0016] The controller may control a cooler so that the cold air is supplied to the ice making cell after the second tray assembly moves to an ice making position when the water is completely supplied to the ice making cell. The controller may control the second tray assembly so that the second tray assembly moves in a reverse direction after moving to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell.

[0017] The controller may control the second tray assembly so that the supply of the water starts after the second tray assembly moves to a water supply position in the reverse direction when the ice is completely separated. The controller may control the heater to be turned on so that ice is easily separated from the tray assemblies before the second tray assembly moves in the forward direction to an ice separation position.

[0018] An additional heater may be disposed on the other tray assembly. An amount of heat of the additional heater may be less than that of the heater in at least a section in which the cooler supplies cold.

[0019] The refrigerator may further include a cam. The cam may have a path in which a lever moves therein. The cam may be directly or indirectly connected to the second tray assembly.

[0020] The controller may control the driver so that a position of the second tray is determined according to a movement position (linear/rotational movement) of the driver. The controller may control the driver so that a position of the cam is determined according to a movement position (linear/rotational movement) of the driver. A gear may be disposed on an outer circumferential surface of the cam. A rotation shaft may be disposed at a central portion of the cam. After the ice making in the ice making cell is completed, the controller may control the cam to move in the first direction (or forward direction) until the second tray is moved to the ice making position.

[0021] The refrigerator may further include a pusher provided with a first edge, on which a surface configured to press the ice or the tray assembly is formed, a bar extending from the first edge, and a second edge disposed at an end of the bar so that the ice is easily separated from the tray assemblies.

[0022] The controller may control at least one of the pusher or the second tray assembly to move so as to change a relative position between the pusher and the second tray assembly. In the ice separation process, the controller may control the cam to be stopped after additionally moving in the first direction after the second tray assembly moves to the ice separation position so that pressing force applied to the ice in the second tray (or the second tray assembly) increases.

[0023] In the ice separation process, the controller may control the cam to be stopped after additionally moving in the first direction after the second tray assembly moves to the ice separation position so that a decrease in pressing force applied by the pusher to the ice in the second tray (or the second tray assembly) due to deformation of the second tray (or the second tray assembly) is reduced.

[0024] The refrigerator may further include a bracket to which the pusher is fixed. In the ice separation process, the controller may control the cam to be stopped after addition-

ally moving in the first direction after the second tray assembly moves to the ice separation position so that a decrease in pressing force applied by the pusher to the ice in the second tray (or the second tray assembly) due to deformation of the bracket is reduced. The controller may control the second tray (or the second tray assembly) and the cam to rotatably move.

[0025] The controller may control the second tray (or the second tray assembly) and the cam to rotatably move, and the ice separation position may be a position at which a rotation angle of the cam is greater than 90 degrees based on the ice making position. The rotation angle of the cam may be greater than 90 degrees and less than 180 degrees. The rotation angle of the cam may be greater than 90 degrees and less than 150 degrees. The rotation angle of the cam may be greater than 90 degrees and less than 140 degrees.

[0026] The controller may control the cam to move in a second direction (reverse direction) until the second tray (or the second tray assembly) moves to the water supply position after the ice completely separated. The controller may control the cam to be stopped after additionally moving in the second direction after the second tray (or the second tray assembly) moves to the water supply position. The second direction may be a direction opposite to a direction of gravity. In consideration of the inertia of the tray (tray assembly) and the motor, it may be preferable that the cam additionally rotates in the direction opposite to the direction of gravity.

[0027] The controller may control the second tray (or the second tray assembly) and the cam to rotatably move, and the water supply position may be a position before at least a portion of the ice making cell formed by the second tray assembly reaches a horizontal reference line passing through a center of a rotation shaft of the driver.

[0028] At the ice making position, the rotation angle of the cam may be set to zero. The controller may control the second tray (or the second tray assembly) and the cam to rotatably move, and at the water supply position, the rotation angle of the cam may be greater than zero. The rotation angle of the cam may be greater than 0 degrees and less than 20 degrees. The rotation angle of the cam may be greater than 5 degrees and less than 15 degrees.

[0029] The controller may control the cam to move in the second direction (reverse direction) until the second tray (or the second tray assembly) moves to the ice making position after water is completely supplied to the ice making cell.

[0030] In the ice making process, the controller may control the cam to additionally move in the second direction after the second tray (or the second tray assembly) moves to the ice making position so that coupling force between the first and second trays increases. The controller may control the second tray (or the second tray assembly) and the cam to rotatably move, and the ice making position may be a position at which at least a portion of the ice making cell formed by the second tray assembly reaches a horizontal reference line passing through a center of a rotation shaft of the driver.

[0031] The controller may control the second tray (or the second tray assembly) and the cam to rotatably move, and at the ice making position, the position of the cam may be greater than negative (−) 30 degrees and less than 0 degree. The rotation angle of the cam may be greater than negative (−) 25 degrees and less than negative (−) 5 degrees. The

rotation angle of the cam may be greater than negative (−) 20 degrees and less than negative (−) 10 degrees.

Advantageous Effects

[0032] According to the embodiments, since the heater is turned on in at least a portion of the sections while the cooler supplies cold air, the ice making rate may decrease by the heat of the heater so that the bubbles dissolved in the water inside the ice making cell move toward the liquid water from the portion at which the ice is made, thereby making the transparent ice.

[0033] In addition, according to the embodiments, one or more of the cooling power of the cooler and the heating amount of heater may be controlled to vary according to the mass per unit height of water in the ice making cell to make the ice having the uniform transparency as a whole regardless of the shape of the ice making cell.

[0034] In addition, according to this embodiment, as the coupling force between the first and second tray assemblies increases, the water may be prevented from leaking between the first and second tray assemblies.

[0035] In addition, according to this embodiment, the pressing force of the pusher may increase during the ice separation process to easily separate the ice from the tray.

[0036] Also, the heating amount of transparent ice heater and/or the cooling power of the cooler may vary in response to the change in the heat transfer amount between the water in the ice making cell and the cold air in the storage chamber, thereby making the ice having the uniform transparency as a whole.

DESCRIPTION OF DRAWINGS

[0037] FIGS. 1 (a) to (b) are front views of a refrigerator according to an embodiment of the present invention.

[0038] FIG. 2 is a perspective view of an ice maker according to an embodiment.

[0039] FIG. 3 is a front view of the ice maker of FIG. 2.

[0040] FIG. 4 is a perspective view illustrating a state in which a bracket is removed from the ice maker of FIG. 3.

[0041] FIG. 5 is an exploded perspective view of the ice maker according to an embodiment.

[0042] FIGS. 6 and 7 are perspective views of the bracket according to an embodiment.

[0043] FIG. 8 is a perspective view of a first tray when viewed from an upper side.

[0044] FIG. 9 is a perspective view of the first tray when viewed from a lower side.

[0045] FIG. 10 is a cross-sectional view taken along line 10-10 of FIG. 8.

[0046] FIG. 11 is a cross-sectional view taken along line 11-11 of FIG. 8.

[0047] FIG. 12 is a perspective view of a first tray cover.

[0048] FIG. 13 is a bottom perspective view of the first tray cover.

[0049] FIG. 14 is a plan view of the first tray cover.

[0050] FIG. 15 is a side view of a first tray case.

[0051] FIG. 16 is a plan view of a first tray supporter.

[0052] FIG. 17 is a perspective view of a second tray when viewed from an upper side according to an embodiment of the present invention.

[0053] FIG. 18 is a perspective view of the second tray when viewed from a lower side.

[0054] FIG. 19 is a bottom view of the second tray.

[0055] FIG. 20 is a plan view of the second tray.

[0056] FIG. 21 is a cutaway cross-sectional view taken along line 21-21 of FIG. 17.

[0057] FIG. 22 is a perspective view of a second tray cover.

[0058] FIG. 23 is a plan view of the second tray cover.

[0059] FIG. 24 is a perspective view illustrating an upper portion of a second tray supporter.

[0060] FIG. 25 is a perspective view illustrating a lower portion of the second tray supporter.

[0061] FIG. 26 is a cutaway cross-sectional view taken along line 26-26 of FIG. 24.

[0062] FIGS. 27 (a) to (b) are views of a first pusher according to an embodiment.

[0063] FIG. 28 is a view illustrating a state in which the first pusher is connected to a second tray assembly by a pusher link.

[0064] FIG. 29 is a perspective view illustrating a rotation arm of the present invention.

[0065] FIG. 30 is a perspective view of a second pusher according to an embodiment of the present invention.

[0066] FIG. 31 is a cutaway cross-sectional view taken along line 31-31 of FIG. 2.

[0067] FIG. 32 is a control block diagram of the refrigerator according to an embodiment of the present invention.

[0068] FIG. 33 is a flowchart for explaining a process of making ice in the ice maker according to an embodiment.

[0069] FIGS. 34 (a) to (b) are views for explaining a height reference depending on a relative position of the transparent heater with respect to the ice making cell.

[0070] FIGS. 35 (a) to (b) are views for explaining an output of the transparent heater per unit height of water within the ice making cell.

[0071] FIG. 36 is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at a water supply position.

[0072] FIG. 37 is a view illustrating a state in which supply of water is completed in FIG. 36.

[0073] FIG. 38 is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly at an ice making position.

[0074] FIG. 39 is a view illustrating a state in which a pressing part of the second tray is deformed in a state in which ice making is completed.

[0075] FIG. 40 is a cross-sectional view illustrating a position relationship between a first tray assembly and a second tray assembly in an ice separation process.

[0076] FIG. 41 is a cross-sectional view illustrating a position relationship between the first tray assembly and the second tray assembly at the ice separation position.

[0077] FIGS. 42 (A) to (D) are views illustrating an operation of the pusher link when the second tray assembly moves from the ice making position to the ice separation position.

[0078] FIG. 43 is a view for explaining a method for controlling a refrigerator when a heat transfer amount between cold air and water vary in an ice making process.

[0079] FIG. 44 is an exploded perspective view of a driver according to an embodiment of the present invention.

[0080] FIG. 45 is a plan view illustrating an internal configuration of the driver.

[0081] FIG. 46 is a view illustrating a cam and an operation lever of the driver.

[0082] FIGS. 47 (a) to (b) are views illustrating a position relationship between a hall sensor and a magnet depending on rotation of the cam.

MODE FOR INVENTION

[0083] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It should be noted that when components in the drawings are designated by reference numerals, the same components have the same reference numerals as far as possible even though the components are illustrated in different drawings. Further, in description of embodiments of the present disclosure, when it is determined that detailed descriptions of well-known configurations or functions disturb understanding of the embodiments of the present disclosure, the detailed descriptions will be omitted.

[0084] Also, in the description of the embodiments of the present disclosure, the terms such as first, second, A, B, (a) and (b) may be used. Each of the terms is merely used to distinguish the corresponding component from other components, and does not delimit an essence, an order or a sequence of the corresponding component. It should be understood that when one component is “connected”, “coupled” or “joined” to another component, the former may be directly connected or jointed to the latter or may be “connected”, coupled” or “joined” to the latter with a third component interposed therebetween.

[0085] The refrigerator according to an embodiment may include a tray assembly defining a portion of an ice making cell that is a space in which water is phase-changed into ice, a cooler supplying cold air to the ice making cell, a water supply part supplying water to the ice making cell, and a controller. The refrigerator may further include a temperature sensor detecting a temperature of water or ice of the ice making cell. The refrigerator may further include a heater disposed adjacent to the tray assembly. The refrigerator may further include a driver to move the tray assembly. The refrigerator may further include a storage chamber in which food is stored in addition to the ice making cell. The refrigerator may further include a cooler supplying cold to the storage chamber. The refrigerator may further include a temperature sensor sensing a temperature in the storage chamber. The controller may control at least one of the water supply part or the cooler. The controller may control at least one of the heater or the driver.

[0086] The controller may control the cooler so that cold is supplied to the ice making cell after moving the tray assembly to an ice making position. The controller may control the second tray assembly so that the second tray assembly moves to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell. The controller may control the tray assembly so that the supply of the water supply part after the second tray assembly moves to the water supply position in the reverse direction when the ice is completely separated. The controller may control the tray assembly so as to move to the ice making position after the water supply is completed.

[0087] According to an embodiment, the storage chamber may be defined as a space that is controlled to a predetermined temperature by the cooler. An outer case may be defined as a wall that divides the storage chamber and an external space of the storage chamber (i.e., an external space

of the refrigerator). An insulation material may be disposed between the outer case and the storage chamber. An inner case may be disposed between the insulation material and the storage chamber.

[0088] According to an embodiment, the ice making cell may be disposed in the storage chamber and may be defined as a space in which water is phase-changed into ice. A circumference of the ice making cell refers to an outer surface of the ice making cell irrespective of the shape of the ice making cell. In another aspect, an outer circumferential surface of the ice making cell may refer to an inner surface of the wall defining the ice making cell. A center of the ice making cell refers to a center of gravity or volume of the ice making cell. The center may pass through a symmetry line of the ice making cell.

[0089] According to an embodiment, the tray may be defined as a wall partitioning the ice making cell from the inside of the storage chamber. The tray may be defined as a wall defining at least a portion of the ice making cell. The tray may be configured to surround the whole or a portion of the ice making cell. The tray may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The tray may be provided in plurality. The plurality of trays may contact each other. For example, the tray disposed at the lower portion may include a plurality of trays. The tray disposed at the upper portion may include a plurality of trays. The refrigerator may include at least one tray disposed under the ice making cell. The refrigerator may further include a tray disposed above the ice making cell. The first portion and the second portion may have a structure inconsideration of a degree of heat transfer of the tray, a degree of cold transfer of the tray, a degree of deformation resistance of the tray, a recovery degree of the tray, a degree of supercooling of the tray, a degree of attachment between the tray and ice solidified in the tray, and coupling force between one tray and the other tray of the plurality of trays.

[0090] According to an embodiment, the tray case may be disposed between the tray and the storage chamber. That is, the tray case may be disposed so that at least a portion thereof surrounds the tray. The tray case may be provided in plurality. The plurality of tray cases may contact each other. The tray case may contact the tray to support at least a portion of the tray. The tray case may be configured to connect components except for the tray (e.g., a heater, a sensor, a power transmission member, etc.). The tray case may be directly coupled to the component or coupled to the component via a medium therebetween. For example, if the wall defining the ice making cell is provided as a thin film, and a structure surrounding the thin film is provided, the thin film may be defined as a tray, and the structure may be defined as a tray case. For another example, if a portion of the wall defining the ice making cell is provided as a thin film, and a structure includes a first portion defining the other portion of the wall defining the ice making cell and a second part surrounding the thin film, the thin film and the first portion of the structure are defined as trays, and the second portion of the structure is defined as a tray case.

[0091] According to an embodiment, the tray assembly may be defined to include at least the tray. According to an embodiment, the tray assembly may further include the tray case.

[0092] According to an embodiment, the refrigerator may include at least one tray assembly connected to the driver to move. The driver is configured to move the tray assembly in at least one axial direction of the X, Y, or Z axis or to rotate about the axis of at least one of the X, Y, or Z axis. The embodiment may include a refrigerator having the remaining configuration except for the driver and the power transmission member connecting the driver to the tray assembly in the contents described in the detailed description. According to an embodiment, the tray assembly may move in a first direction.

[0093] According to an embodiment, the cooler may be defined as a part configured to cool the storage chamber including at least one of an evaporator or a thermoelectric element.

[0094] According to an embodiment, the refrigerator may include at least one tray assembly in which the heater is disposed. The heater may be disposed in the vicinity of the tray assembly to heat the ice making cell defined by the tray assembly in which the heater is disposed. The heater may include a heater to be turned on in at least partial section while the cooler supplies cold so that bubbles dissolved in the water within the ice making cell moves from a portion, at which the ice is made, toward the water that is in a liquid state to make transparent ice. The heater may include a heater (hereinafter referred to as an "ice separation heater") controlled to be turned on in at least a section after the ice making is completed so that ice is easily separated from the tray assembly. The refrigerator may include a plurality of transparent ice heaters. The refrigerator may include a plurality of ice separation heaters. The refrigerator may include a transparent ice heater and an ice separation heater. In this case, the controller may control the ice separation heater so that a heating amount of ice separation heater is greater than that of transparent ice heater.

[0095] According to an embodiment, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. The tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion.

[0096] For example, the first region may be defined in the first portion of the tray assembly. The first and second regions may be defined in the first portion of the tray assembly. Each of the first and second regions may be a portion of the one tray assembly. The first and second regions may be disposed to contact each other. The first region may be a lower portion of the ice making cell defined by the tray assembly. The second region may be an upper portion of an ice making cell defined by the tray assembly. The refrigerator may include an additional tray assembly. One of the first and second regions may include a region contacting the additional tray assembly. When the additional tray assembly is disposed in a lower portion of the first region, the additional tray assembly may contact the lower portion of the first region. When the additional tray assembly is disposed in an upper portion of the second region, the additional tray assembly and the upper portion of the second region may contact each other.

[0097] For another example, the tray assembly may be provided in plurality contacting each other. The first region may be disposed in a first tray assembly of the plurality of tray assemblies, and the second region may be disposed in a second tray assembly. The first region may be the first tray

assembly. The second region may be the second tray assembly. The first and second regions may be disposed to contact each other. At least a portion of the first tray assembly may be disposed under the ice making cell defined by the first and second tray assemblies. At least a portion of the second tray assembly may be disposed above the ice making cell defined by the first and second tray assemblies.

[0098] The first region may be a region closer to the heater than the second region. The first region may be a region in which the heater is disposed. The second region may be a region closer to a heat absorbing part (i.e., a coolant pipe or a heat absorbing part of a thermoelectric module) of the cooler than the first region. The second region may be a region closer to the through-hole supplying cold to the ice making cell than the first region. To allow the cooler to supply the cold through the through-hole, an additional through-hole may be defined in another component. The second region may be a region closer to the additional through-hole than the first region. The heater may be a transparent ice heater. The heat insulation degree of the second region with respect to the cold may be less than that of the first region.

[0099] The heater may be disposed in one of the first and second tray assemblies of the refrigerator. For example, when the heater is not disposed on the other one, the controller may control the heater to be turned on in at least a section of the cooler to supply the cold air. For another example, when the additional heater is disposed on the other one, the controller may control the heater so that the heating amount of heater is greater than that of additional heater in at least a section of the cooler to supply the cold air. The heater may be a transparent ice heater.

[0100] The embodiment may include a refrigerator having a configuration excluding the transparent ice heater in the contents described in the detailed description.

[0101] The embodiment may include a pusher including a first edge having a surface pressing the ice or at least one surface of the tray assembly so that the ice is easily separated from the tray assembly. The pusher may include a bar extending from the first edge and a second edge disposed at an end of the bar. The controller may control the pusher so that a position of the pusher is changed by moving at least one of the pusher or the tray assembly. The pusher may be defined as a penetrating type pusher, a non-penetrating type pusher, a movable pusher, or a fixed pusher according to a view point.

[0102] A through-hole through which the pusher moves may be defined in the tray assembly, and the pusher may be configured to directly press the ice in the tray assembly. The pusher may be defined as a penetrating type pusher.

[0103] The tray assembly may be provided with a pressing part to be pressed by the pusher, the pusher may be configured to apply a pressure to one surface of the tray assembly. The pusher may be defined as a non-penetrating type pusher.

[0104] The controller may control the pusher to move so that the first edge of the pusher is disposed between a first point outside the ice making cell and a second point inside the ice making cell.

[0105] The pusher may be defined as a movable pusher. The pusher may be connected to a driver, the rotation shaft of the driver, or the tray assembly that is connected to the driver and is movable. The controller may control the pusher to move at least one of the tray assemblies so that the first edge of the pusher is disposed between the first point outside

the ice making cell and the second point inside the ice making cell. The controller may control at least one of the tray assemblies to move to the pusher. Alternatively, the controller may control a relative position of the pusher and the tray assembly so that the pusher further presses the pressing part after contacting the pressing part at the first point outside the ice making cell. The pusher may be coupled to a fixed end. The pusher may be defined as a fixed pusher.

[0106] According to an embodiment, the ice making cell may be cooled by the cooler cooling the storage chamber. For example, the storage chamber in which the ice making cell is disposed may be a freezing compartment which is controlled at a temperature lower than 0 degree, and the ice making cell may be cooled by the cooler cooling the freezing compartment.

[0107] The freezing compartment may be divided into a plurality of regions, and the ice making cell may be disposed in one region of the plurality of regions.

[0108] According to an embodiment, the ice making cell may be cooled by a cooler other than the cooler cooling the storage chamber. For example, the storage chamber in which the ice making cell is disposed is a refrigerating compartment which is controlled to a temperature higher than 0 degree, and the ice making cell may be cooled by a cooler other than the cooler cooling the refrigerating compartment. That is, the refrigerator may include a refrigerating compartment and a freezing compartment, the ice making cell may be disposed inside the refrigerating compartment, and the ice maker cell may be cooled by the cooler that cools the freezing compartment.

[0109] The ice making cell may be disposed in a door that opens and closes the storage chamber.

[0110] According to an embodiment, the ice making cell is not disposed inside the storage chamber and may be cooled by the cooler. For example, the entire storage chamber defined inside the outer case may be the ice making cell. According to an embodiment, a degree of heat transfer indicates a degree of heat transfer from a high-temperature object to a low-temperature object and is defined as a value determined by a shape including a thickness of the object, a material of the object, and the like. In terms of the material of the object, a high degree of the heat transfer of the object may represent that thermal conductivity of the object is high. The thermal conductivity may be a unique material property of the object. Even when the material of the object is the same, the degree of heat transfer may vary depending on the shape of the object.

[0111] The degree of heat transfer may vary depending on the shape of the object. The degree of heat transfer from a point A to a point B may be influenced by a length of a path through which heat is transferred from the point A to the point B (hereinafter, referred to as a "heat transfer path"). The more the heat transfer path from the point A to the point B increases, the more the degree of heat transfer from the point A to the point B may decrease. The more the heat transfer path from the point A to the point B, the more the degree of heat transfer from the point A to the point B may increase.

[0112] The degree of heat transfer from the point A to the point B may be influenced by a thickness of the path through which heat is transferred from the point A to the point B. The more the thickness in a path direction in which heat is transferred from the point A to the point B decreases, the

more the degree of heat transfer from the point A to the point B may decrease. The greater the thickness in the path direction from which the heat from point A to point B is transferred, the more the degree of heat transfer from point A to point B.

[0113] According to an embodiment, a degree of cold transfer indicates a degree of heat transfer from a low-temperature object to a high-temperature object and is defined as a value determined by a shape including a thickness of the object, a material of the object, and the like. The degree of cold transfer is a term defined in consideration of a direction in which cold air flows and may be regarded as the same concept as the degree of heat transfer. The same concept as the degree of heat transfer will be omitted.

[0114] According to an embodiment, a degree of supercooling is a degree of supercooling of a liquid and may be defined as a value determined by a material of the liquid, a material or shape of a container containing the liquid, an external factor applied to the liquid during a solidification process of the liquid, and the like. An increase in frequency at which the liquid is supercooled may be seen as an increase in degree of the supercooling. The lowering of the temperature at which the liquid is maintained in the supercooled state may be seen as an increase in degree of the supercooling. Here, the supercooling refers to a state in which the liquid exists in the liquid phase without solidification even at a temperature below a freezing point of the liquid. The supercooled liquid has a characteristic in which the solidification rapidly occurs from a time point at which the supercooling is terminated. If it is desired to maintain a rate at which the liquid is solidified, it is advantageous to be designed so that the supercooling phenomenon is reduced.

[0115] According to an embodiment, a degree of deformation resistance represents a degree to which an object resists deformation due to external force applied to the object and is a value determined by a shape including a thickness of the object, a material of the object, and the like. For example, the external force may include a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. In another example, the external force may include a pressure on the ice or a portion of the tray assembly by the pusher for separating the ice from the tray assembly. For another example, when coupled between the tray assemblies, it may include a pressure applied by the coupling.

[0116] In terms of the material of the object, a high degree of the deformation resistance of the object may represent that rigidity of the object is high. The thermal conductivity may be a unique material property of the object. Even when the material of the object is the same, the degree of deformation resistance may vary depending on the shape of the object. The degree of deformation resistance may be affected by a deformation resistance reinforcement part extending in a direction in which the external force is applied. The more the rigidity of the deformation resistant resistance reinforcement part increases, the more the degree of deformation resistance may increase. The more the height of the extending deformation resistance reinforcement part increase, the more the degree of deformation resistance may increase.

[0117] According to an embodiment, a degree of restoration indicates a degree to which an object deformed by the external force is restored to a shape of the object before the external force is applied after the external force is removed and is defined as a value determined by a shape including a

thickness of the object, a material of the object, and the like. For example, the external force may include a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. In another example, the external force may include a pressure on the ice or a portion of the tray assembly by the pusher for separating the ice from the tray assembly. For another example, when coupled between the tray assemblies, it may include a pressure applied by the coupling force.

[0118] In view of the material of the object, a high degree of the restoration of the object may represent that an elastic modulus of the object is high. The elastic modulus may be a material property unique to the object. Even when the material of the object is the same, the degree of restoration may vary depending on the shape of the object. The degree of restoration may be affected by an elastic resistance reinforcement part extending in a direction in which the external force is applied. The more the elastic modulus of the elastic resistance reinforcement part increases, the more the degree of restoration may increase.

[0119] According to an embodiment, the coupling force represents a degree of coupling between the plurality of tray assemblies and is defined as a value determined by a shape including a thickness of the tray assembly, a material of the tray assembly, magnitude of the force that couples the trays to each other, and the like.

[0120] According to an embodiment, a degree of attachment indicates a degree to which the ice and the container are attached to each other in a process of making ice from water contained in the container and is defined as a value determined by a shape including a thickness of the container, a material of the container, a time elapsed after the ice is made in the container, and the like.

[0121] The refrigerator according to an embodiment includes a first tray assembly defining a portion of an ice making cell that is a space in which water is phase-changed into ice by cold, a second tray assembly defining the other portion of the ice making cell, a cooler supplying cold to the ice making cell, a water supply part supplying water to the ice making cell, and a controller. The refrigerator may further include a storage chamber in addition to the ice making cell. The storage chamber may include a space for storing food. The ice making cell may be disposed in the storage chamber. The refrigerator may further include a first temperature sensor sensing a temperature in the storage chamber. The refrigerator may further include a second temperature sensor sensing a temperature of water or ice of the ice making cell. The second tray assembly may contact the first tray assembly in the ice making process and may be connected to the driver to be spaced apart from the first tray assembly in the ice making process. The refrigerator may further include a heater disposed adjacent to at least one of the first tray assembly or the second tray assembly.

[0122] The controller may control at least one of the heater or the driver. The controller may control the cooler so that the cold is supplied to the ice making cell after the second tray assembly moves to an ice making position when the water is completely supplied to the ice making cell. The controller may control the second tray assembly so that the second tray assembly moves in a reverse direction after moving to an ice separation position in a forward direction so as to take out the ice in the ice making cell when the ice is completely made in the ice making cell. The controller may control the second tray assembly so that the supply of

the water supply part after the second tray assembly moves to the water supply position in the reverse direction when the ice is completely separated.

[0123] Transparent ice will be described. Bubbles are dissolved in water, and the ice solidified with the bubbles may have low transparency due to the bubbles. Therefore, in the process of water solidification, when the bubble is guided to move from a freezing portion in the ice making cell to another portion that is not yet frozen, the transparency of the ice may increase.

[0124] A through-hole defined in the tray assembly may affect the making of the transparent ice. The through-hole defined in one side of the tray assembly may affect the making of the transparent ice. In the process of making ice, if the bubbles move to the outside of the ice making cell from the frozen portion of the ice making cell, the transparency of the ice may increase. The through-hole may be defined in one side of the tray assembly to guide the bubbles so as to move out of the ice making cell. Since the bubbles have lower density than the liquid, the through-hole (hereinafter, referred to as an “air exhaust hole”) for guiding the bubbles to escape to the outside of the ice making cell may be defined in the upper portion of the tray assembly.

[0125] The position of the cooler and the heater may affect the making of the transparent ice. The position of the cooler and the heater may affect an ice making direction, which is a direction in which ice is made inside the ice making cell.

[0126] In the ice making process, when bubbles move or are collected from a region in which water is first solidified in the ice making cell to another predetermined region in a liquid state, the transparency of the made ice may increase. The direction in which the bubbles move or are collected may be similar to the ice making direction. The predetermined region may be a region in which water is to be solidified lately in the ice making cell.

[0127] The predetermined region may be a region in which the cold supplied by the cooler reaches the ice making cell late. For example, in the ice making process, the through-hole through which the cooler supplies the cold to the ice making cell may be defined closer to the upper portion than the lower part of the ice making cell so as to move or collect the bubbles to the lower portion of the ice making cell. For another example, a heat absorbing part of the cooler (that is, a refrigerant pipe of the evaporator or a heat absorbing part of the thermoelectric element) may be disposed closer to the upper portion than the lower portion of the ice making cell. According to an embodiment, the upper and lower portions of the ice making cell may be defined as an upper region and a lower region based on a height of the ice making cell.

[0128] The predetermined region may be a region in which the heater is disposed. For example, in the ice making process, the heater may be disposed closer to the lower portion than the upper portion of the ice making cell so as to move or collect the bubbles in the water to the lower portion of the ice making cell.

[0129] The predetermined region may be a region closer to an outer circumferential surface of the ice making cell than to a center of the ice making cell. However, the vicinity of the center is not excluded. If the predetermined region is near the center of the ice making cell, an opaque portion due to the bubbles moved or collected near the center may be easily visible to the user, and the opaque portion may remain until most of the ice until the ice is melted. Also, it may be

difficult to arrange the heater inside the ice making cell containing water. In contrast, when the predetermined region is defined in or near the outer circumferential surface of the ice making cell, water may be solidified from one side of the outer circumferential surface of the ice making cell toward the other side of the outer circumferential surface of the ice making cell, thereby solving the above limitation. The transparent ice heater may be disposed on or near the outer circumferential surface of the ice making cell. The heater may be disposed at or near the tray assembly.

[0130] The predetermined region may be a position closer to the lower portion of the ice making cell than the upper portion of the ice making cell. However, the upper portion is also not excluded. In the ice making process, since liquid water having greater density than ice drops, it may be advantageous that the predetermined region is defined in the lower portion of the ice making cell.

[0131] At least one of the degree of deformation resistance, the degree of restoration, and the coupling force between the plurality of tray assemblies may affect the making of the transparent ice. At least one of the degree of deformation resistance, the degree of restoration, and the coupling force between the plurality of tray assemblies may affect the ice making direction that is a direction in which ice is made in the ice making cell. As described above, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0132] To make the transparent ice, it may be advantageous for the refrigerator to be configured so that the direction in which ice is made in the ice making cell is constant. This is because the more the ice making direction is constant, the more the bubbles in the water are moved or collected in a predetermined region within the ice making cell. It may be advantageous for the deformation of the portion to be greater than the deformation of the other portion so as to induce the ice to be made in the direction of the other portion in a portion of the tray assembly. The ice tends to be grown as the ice is expanded toward a portion at which the degree of deformation resistance is low. To start the ice making again after removing the made ice, the deformed portion has to be restored again to make ice having the same shape repeatedly. Therefore, it may be advantageous that the portion having the low degree of the deformation resistance has a high degree of the restoration than the portion having a high degree of the deformation resistance.

[0133] The degree of deformation resistance of the tray with respect to the external force may be less than that of the tray case with respect to the external force, or the rigidity of the tray may be less than that of the tray case. The tray assembly allows the tray to be deformed by the external force, while the tray case surrounding the tray is configured to reduce the deformation. For example, the tray assembly may be configured so that at least a portion of the tray is surrounded by the tray case. In this case, when a pressure is applied to the tray assembly while the water inside the ice making cell is solidified and expanded, at least a portion of the tray may be allowed to be deformed, and the other part of the tray may be supported by the tray case to restrict the deformation. In addition, when the external force is

removed, the degree of restoration of the tray may be greater than that of the tray case, or the elastic modulus of the tray may be greater than that of the tray case. Such a configuration may be configured so that the deformed tray is easily restored.

[0134] The degree of deformation resistance of the tray with respect to the external force may be greater than that of the gasket of the refrigerator with respect to the external force, or the rigidity of the tray may be greater than that of the gasket. When the degree of deformation resistance of the tray is low, there may be a limitation that the tray is excessively deformed as the water in the ice making cell defined by the tray is solidified and expanded. Such a deformation of the tray may make it difficult to make the desired type of ice. In addition, the degree of restoration of the tray when the external force is removed may be configured to be less than that of the refrigerator gasket with respect to the external force, or the elastic modulus of the tray is less than that of the gasket.

[0135] The deformation resistance of the tray case with respect to the external force may be less than that of the refrigerator case with respect to the external force, or the rigidity of the tray case may be less than that of the refrigerator case. In general, the case of the refrigerator may be made of a metal material including steel. In addition, when the external force is removed, the degree of restoration of the tray case may be greater than that of the refrigerator case with respect to the external force, or the elastic modulus of the tray case is greater than that of the refrigerator case.

[0136] The relationship between the transparent ice and the degree of deformation resistance is as follows.

[0137] The second region may have different degree of deformation resistance in a direction along the outer circumferential surface of the ice making cell. The degree of deformation resistance of the portion of the second region may be greater than that of the another of the second region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0138] The first and second regions defined to contact each other may have different degree of deformation resistances in the direction along the outer circumferential surface of the ice making cell. The degree of deformation resistance of one portion of the second region may be greater than that of one portion of the first region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0139] In this case, as the water is solidified, a volume is expanded to apply a pressure to the tray assembly, which induces ice to be made in the other direction of the second region or in one direction of the first region. The degree of deformation resistance may be a degree that resists to deformation due to the external force. The external force may a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. The external force may be force in a vertical direction (Z-axis direction) of the pressure. The external force may be force acting in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0140] For example, in the thickness of the tray assembly in the direction of the outer circumferential surface of the ice

making cell from the center of the ice making cell, one portion of the second region may be thicker than the other of the second region or thicker than one portion of the first region. One portion of the second region may be a portion at which the tray case is not surrounded. The other portion of the second region may be a portion surrounded by the tray case. One portion of the first region may be a portion at which the tray case is not surrounded. One portion of the second region may be a portion defining the uppermost portion of the ice making cell in the second region. The second region may include a tray and a tray case locally surrounding the tray. As described above, when at least a portion of the second region is thicker than the other part, the degree of deformation resistance of the second region may be improved with respect to an external force. A minimum value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. A maximum value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the second region may be greater than that of the thickness of the other portion of the second region or greater than that of one portion of the first region. The uniformity of the thickness of one portion of the second region may be less than that of the thickness of the other portion of the second region or less than that of one of the thickness of the first region.

[0141] For another example, one portion of the second region may include a first surface defining a portion of the ice making cell and a deformation resistance reinforcement part extending from the first surface in a vertical direction away from the ice making cell defined by the other of the second region. One portion of the second region may include a first surface defining a portion of the ice making cell and a deformation resistance reinforcement part extending from the first surface in a vertical direction away from the ice making cell defined by the first region. As described above, when at least a portion of the second region includes the deformation resistance reinforcement part, the degree of deformation resistance of the second region may be improved with respect to the external force.

[0142] For another example, one portion of the second region may further include a support surface connected to a fixed end of the refrigerator (e.g., the bracket, the storage chamber wall, etc.) disposed in a direction away from the ice making cell defined by the other of the second region from the first surface. One portion of the second region may further include a support surface connected to a fixed end of the refrigerator (e.g., the bracket, the storage chamber wall, etc.) disposed in a direction away from the ice making cell defined by the first region from the first surface. As described above, when at least a portion of the second region includes a support surface connected to the fixed end, the degree of deformation resistance of the second region may be improved with respect to the external force.

[0143] For another example, the tray assembly may include a first portion defining at least a portion of the ice making cell and a second portion extending from a prede-

termined point of the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the first region. At least a portion of the second portion may include an additional deformation resistant resistance reinforcement part. At least a portion of the second portion may further include a support surface connected to the fixed end. As described above, when at least a portion of the second region further includes the second portion, it may be advantageous to improve the degree of deformation resistance of the second region with respect to the external force. This is because the additional deformation resistance reinforcement part is disposed at in the second portion, or the second portion is additionally supported by the fixed end.

[0144] For another example, one portion of the second region may include a first through-hole. As described above, when the first through-hole is defined, the ice solidified in the ice making cell of the second region is expanded to the outside of the ice making cell through the first through-hole, and thus, the pressure applied to the second region may be reduced. In particular, when water is excessively supplied to the ice making cell, the first through-hole may be contributed to reduce the deformation of the second region in the process of solidifying the water.

[0145] One portion of the second region may include a second through-hole providing a path through which the bubbles contained in the water in the ice making cell of the second region move or escape. When the second through-hole is defined as described above, the transparency of the solidified ice may be improved.

[0146] In one portion of the second region, a third through-hole may be defined to press the penetrating pusher. This is because it may be difficult for the non-penetrating type pusher to press the surface of the tray assembly so as to remove the ice when the degree of deformation resistance of the second region increases. The first, second, and third through-holes may overlap each other. The first, second, and third through-holes may be defined in one through-hole.

[0147] One portion of the second region may include a mounting part on which the ice separation heater is disposed. The induction of the ice in the ice making cell defined by the second region in the direction of the ice making cell defined by the first region may represent that the ice is first made in the second region. In this case, a time for which the ice is attached to the second region may be long, and the ice separation heater may be required to separate the ice from the second region. The thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell may be less than that of the other portion of the second region in which the ice separation heater is mounted. This is because the heat supplied by the ice separation heater increases in amount transferred to the ice making cell. The fixed end may be a portion of the wall defining the storage chamber or a bracket.

[0148] The relation between the coupling force of the transparent ice and the tray assembly is as follows.

[0149] To induce the ice to be made in the ice making cell defined by the second region in the direction of the ice making cell defined by the first region, it may be advantageous to increase in coupling force between the first and second regions arranged to contact each other. In the process of solidifying the water, when the pressure applied to the tray assembly while expanded is greater than the coupling

force between the first and second regions, the ice may be made in a direction in which the first and second regions are separated from each other. In the process of solidifying the water, when the pressure applied to the tray assembly while expanded is low, the coupling force between the first and second regions is low, it also has the advantage of inducing the ice to be made so that the ice is made in a direction of the region having the smallest degree of deformation resistance in the first and second regions.

[0150] There may be various examples of a method of increasing the coupling force between the first and second regions. For example, after the water supply is completed, the controller may change a movement position of the driver in the first direction to control one of the first and second regions so as to move in the first direction, and then, the movement position of the driver may be controlled to be additionally changed into the first direction so that the coupling force between the first and second regions increases. For another example, since the coupling force between the first and second regions increase, the degree of deformation resistances or the degree of restorations of the first and second regions may be different from each other with respect to the force applied from the driver so that the driver reduces the change of the shape of the ice making cell by the expanding the ice after the ice making process is started (or after the heater is turned on). For another example, the first region may include a first surface facing the second region. The second region may include a second surface facing the first region. The first and second surfaces may be disposed to contact each other. The first and second surfaces may be disposed to face each other. The first and second surfaces may be disposed to be separated from and coupled to each other. In this case, surface areas of the first surface and the second surface may be different from each other. In this configuration, the coupling force of the first and second regions may increase while reducing breakage of the portion at which the first and second regions contact each other. In addition, there is an advantage of reducing leakage of water supplied between the first and second regions.

[0151] The relationship between transparent ice and the degree of restoration is as follows.

[0152] The tray assembly may include a first portion that defines at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The second portion is configured to be deformed by the expansion of the ice made and then restored after the ice is removed. The second portion may include a horizontal extension part provided so that the degree of restoration with respect to the horizontal external force of the expanded ice increases. The second portion may include a vertical extension part provided so that the degree of restoration with respect to the vertical external force of the expanded ice increases. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0153] The second region may have different degree of restoration in a direction along the outer circumferential surface of the ice making cell. The first region may have different degree of deformation resistance in a direction along the outer circumferential surface of the ice making cell. The degree of restoration of one portion of the first region may be greater than that of the other portion of the first region. Also, the degree of deformation resistance of

one portion may be less than that of the other portion. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0154] The first and second regions defined to contact each other may have different degree of restoration in the direction along the outer circumferential surface of the ice making cell. Also, the first and second regions may have different degree of deformation resistances in the direction along the outer circumferential surface of the ice making cell. The degree of restoration of one of the first region may be greater than that of one of the second region. Also, the degree of deformation resistance of one of the first regions may be greater than that of one of the second region. Such a configuration may be assisted to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0155] In this case, as the water is solidified, a volume is expanded to apply a pressure to the tray assembly, which induces ice to be made in one direction of the first region in which the degree of deformation resistance decreases, or the degree of restoration increases. Here, the degree of restoration may be a degree of restoration after the external force is removed. The external force may a pressure applied to the tray assembly in the process of solidifying and expanding water in the ice making cell. The external force may be force in a vertical direction (Z-axis direction) of the pressure. The external force may be force acting in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region.

[0156] For example, in the thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell, one portion of the first region may be thinner than the other of the first region or thinner than one portion of the second region. One portion of the first region may be a portion at which the tray case is not surrounded. The other portion of the first region may be a portion that is surrounded by the tray case. One portion of the second region may be a portion that is surrounded by the tray case. One portion of the first region may be a portion of the first region that defines the lowermost end of the ice making cell. The first region may include a tray and a tray case locally surrounding the tray.

[0157] A minimum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the second region or less than that of one of the second region. A maximum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or less than that of the thickness of one portion of the second region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or may be less than that of one of the thickness of the second region. The uniformity of the thickness of one portion of the first region may be greater than that of the thickness of the other portion of the first region or greater than that of one of the thickness of the second region.

[0158] For another example, a shape of one portion of the first region may be different from that of the other portion of the first region or different from that of one portion of the

second region. A curvature of one portion of the first region may be different from that of the other portion of the first region or different from that of one portion of the second region. A curvature of one portion of the first region may be less than that of the other portion of the first region or less than that of one portion of the second region. One portion of the first region may include a flat surface. The other portion of the first region may include a curved surface. One portion of the second region may include a curved surface. One portion of the first region may include a shape that is recessed in a direction opposite to the direction in which the ice is expanded. One portion of the first region may include a shape recessed in a direction opposite to a direction in which the ice is made. In the ice making process, one portion of the first region may be modified in a direction in which the ice is expanded or a direction in which the ice is made. In the ice making process, in an amount of deformation from the center of the ice making cell toward the outer circumferential surface of the ice making cell, one portion of the first region is greater than the other portion of the first region. In the ice making process, in the amount of deformation from the center of the ice making cell toward the outer circumferential surface of the ice making cell, one portion of the first region is greater than one portion of the second region.

[0159] For another example, to induce ice to be made in a direction from the ice making cell defined by the second region to the ice making cell defined by the first region, one portion of the first region may include a first surface defining a portion of the ice making cell and a second surface extending from the first surface and supported by one surface of the other portion of the first region. The first region may be configured not to be directly supported by the other component except for the second surface. The other component may be a fixed end of the refrigerator.

[0160] One portion of the first region may have a pressing surface pressed by the non-penetrating type pusher. This is because when the degree of deformation resistance of the first region is low, or the degree of restoration is high, the difficulty in removing the ice by pressing the surface of the tray assembly may be reduced.

[0161] An ice making rate, at which ice is made inside the ice making cell, may affect the making of the transparent ice. The ice making rate may affect the transparency of the made ice. Factors affecting the ice making rate may be an amount of cold and/or heat, which are/is supplied to the ice making cell. The amount of cold and/or heat may affect the making of the transparent ice. The amount of cold and/or heat may affect the transparency of the ice.

[0162] In the process of making the transparent ice, the transparency of the ice may be lowered as the ice making rate is greater than a rate at which the bubbles in the ice making cell are moved or collected. On the other hand, if the ice making rate is less than the rate at which the bubbles are moved or collected, the transparency of the ice may increase. However, the more the ice making rate decreases, the more a time taken to make the transparent ice may increase. Also, the transparency of the ice may be uniform as the ice making rate is maintained in a uniform range.

[0163] To maintain the ice making rate uniformly within a predetermined range, an amount of cold and heat supplied to the ice making cell may be uniform. However, in actual use conditions of the refrigerator, a case in which the amount of cold is variable may occur, and thus, it is necessary to allow

a supply amount of heat to vary. For example, when a temperature of the storage chamber reaches a satisfaction region from a dissatisfaction region, when a defrosting operation is performed with respect to the cooler of the storage chamber, the door of the storage chamber may variously vary in state such as an opened state. Also, if an amount of water per unit height of the ice making cell is different, when the same cold and heat per unit height is supplied, the transparency per unit height may vary.

[0164] To solve this limitation, the controller may control the heater so that when a heat transfer amount between the cold within the storage chamber and the water of the ice making cell increases, the heating amount of transparent ice heater increases, and when the heat transfer amount between the cold within the storage chamber and the water of the ice making cell decreases, the heating amount of transparent ice heater decreases so as to maintain an ice making rate of the water within the ice making cell within a predetermined range that is less than an ice making rate when the ice making is performed in a state in which the heater is turned off.

[0165] The controller may control one or more of a cold supply amount of cooler and a heat supply amount of heater to vary according to a mass per unit height of water in the ice making cell. In this case, the transparent ice may be provided to correspond to a change in shape of the ice making cell.

[0166] The refrigerator may further include a sensor measuring information on the mass of water per unit height of the ice making cell, and the controller may control one of the cold supply amount of cooler and the heat supply amount of heater based on the information inputted from the sensor.

[0167] The refrigerator may include a storage part in which predetermined driving information of the cooler is recorded based on information on mass per unit height of the ice making cell, and the controller may control the cold supply amount of cooler to be changed based on the information.

[0168] The refrigerator may include a storage part in which predetermined driving information of the heater is recorded based on information on mass per unit height of the ice making cell, and the controller may control the heat supply amount of heater to be changed based on the information. For example, the controller may control at least one of the cold supply amount of cooler or the heat supply amount of heater to vary according to a predetermined time based on the information on the mass per unit height of the ice making cell. The time may be a time when the cooler is driven or a time when the heater is driven to make ice. For another example, the controller may control at least one of the cold supply amount of cooler or the heat supply amount of heater to vary according to a predetermined temperature based on the information on the mass per unit height of the ice making cell. The temperature may be a temperature of the ice making cell or a temperature of the tray assembly defining the ice making cell.

[0169] When the sensor measuring the mass of water per unit height of the ice making cell is malfunctioned, or when the water supplied to the ice making cell is insufficient or excessive, the shape of the ice making water is changed, and thus the transparency of the made ice may decrease. To solve this limitation, a water supply method in which an amount of water supplied to the ice making cell is precisely controlled is required. Also, the tray assembly may include a

structure in which leakage of the tray assembly is reduced to reduce the leakage of water in the ice making cell at the water supply position or the ice making position. Also, it is necessary to increase the coupling force between the first and second tray assemblies defining the ice making cell so as to reduce the change in shape of the ice making cell due to the expansion force of the ice during the ice making. Also, it is necessary to decrease in leakage in the precision water supply method and the tray assembly and increase in coupling force between the first and second tray assemblies so as to make ice having a shape that is close to the tray shape.

[0170] The degree of supercooling of the water inside the ice making cell may affect the making of the transparent ice. The degree of supercooling of the water may affect the transparency of the made ice.

[0171] To make the transparent ice, it may be desirable to design the degree of supercooling or lower the temperature inside the ice making cell and thereby to maintain a predetermined range. This is because the supercooled liquid has a characteristic in which the solidification rapidly occurs from a time point at which the supercooling is terminated. In this case, the transparency of the ice may decrease.

[0172] In the process of solidifying the liquid, the controller of the refrigerator may control the supercooling release part to operate so as to reduce a degree of supercooling of the liquid if the time required for reaching the specific temperature below the freezing point after the temperature of the liquid reaches the freezing point is less than a reference value. After reaching the freezing point, it is seen that the temperature of the liquid is cooled below the freezing point as the supercooling occurs, and no solidification occurs.

[0173] An example of the supercooling release part may include an electrical spark generating part. When the spark is supplied to the liquid, the degree of supercooling of the liquid may be reduced. Another example of the supercooling release part may include a driver applying external force so that the liquid moves. The driver may allow the container to move in at least one direction among X, Y, or Z axes or to rotate about at least one axis among X, Y, or Z axes. When kinetic energy is supplied to the liquid, the degree of supercooling of the liquid may be reduced. Further another example of the supercooling release part may include a part supplying the liquid to the container. After supplying the liquid having a first volume less than that of the container, when a predetermined time has elapsed or the temperature of the liquid reaches a certain temperature below the freezing point, the controller of the refrigerator may control an amount of liquid to additionally supply the liquid having a second volume greater than the first volume. When the liquid is divided and supplied to the container as described above, the liquid supplied first may be solidified to act as freezing nucleus, and thus, the degree of supercooling of the liquid to be supplied may be further reduced.

[0174] The more the degree of heat transfer of the container containing the liquid increase, the more the degree of supercooling of the liquid may increase. The more the degree of heat transfer of the container containing the liquid decrease, the more the degree of supercooling of the liquid may decrease.

[0175] The structure and method of heating the ice making cell in addition to the heat transfer of the tray assembly may affect the making of the transparent ice. As described above, the tray assembly may include a first region and a second

region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0176] The cold supplied to the ice making cell and the heat supplied to the ice making cell have opposite properties. To increase the ice making rate and/or improve the transparency of the ice, the design of the structure and control of the cooler and the heater, the relationship between the cooler and the tray assembly, and the relationship between the heater and the tray assembly may be very important.

[0177] For a constant amount of cold supplied by the cooler and a constant amount of heat supplied by the heater, it may be advantageous for the heater to be arranged to locally heat the ice making cell so as to increase the ice making rate of the refrigerator and/or to increase the transparency of the ice. As the heat transmitted from the heater to the ice making cell is transferred to an area other than the area on which the heater is disposed, the ice making rate may be improved. As the heater heats only a portion of the ice making cell, the heater may move or collect the bubbles to an area adjacent to the heater in the ice making cell, thereby increasing the transparency of the ice.

[0178] When the amount of heat supplied by the heater to the ice making cell is large, the bubbles in the water may be moved or collected in the portion to which the heat is supplied, and thus, the made ice may increase in transparency. However, if the heat is uniformly supplied to the outer circumferential surface of the ice making cell, the ice making rate of the ice may decrease. Therefore, as the heater locally heats a portion of the ice making cell, it is possible to increase the transparency of the made ice and minimize the decrease of the ice making rate.

[0179] The heater may be disposed to contact one side of the tray assembly. The heater may be disposed between the tray and the tray case. The heat transfer through the conduction may be advantageous for locally heating the ice making cell.

[0180] At least a portion of the other side at which the heater does not contact the tray may be sealed with a heat insulation material. Such a configuration may reduce that the heat supplied from the heater is transferred toward the storage chamber.

[0181] The tray assembly may be configured so that the heat transfer from the heater toward the center of the ice making cell is greater than that transfer from the heater in the circumference direction of the ice making cell.

[0182] The heat transfer of the tray toward the center of the ice making cell in the tray may be greater than the that transfer from the tray case to the storage chamber, or the thermal conductivity of the tray may be greater than that of the tray case. Such a configuration may induce the increase in heat transmitted from the heater to the ice making cell via the tray. In addition, it is possible to reduce the heat of the heater is transferred to the storage chamber via the tray case.

[0183] The heat transfer of the tray toward the center of the ice making cell in the tray may be less than that of the refrigerator case toward the storage chamber from the outside of the refrigerator case (for example, an inner case or an outer case), or the thermal conductivity of the tray may be less than that of the refrigerator case. This is because the more the heat or thermal conductivity of the tray increases, the more the supercooling of the water accommodated in the

tray may increase. The more the degree of supercooling of the water increase, the more the water may be rapidly solidified at the time point at which the supercooling is released. In this case, a limitation may occur in which the transparency of the ice is not uniform or the transparency decreases. In general, the case of the refrigerator may be made of a metal material including steel.

[0184] The heat transfer of the tray case in the direction from the storage chamber to the tray case may be greater than the that of the heat insulation wall in the direction from the outer space of the refrigerator to the storage chamber, or the thermal conductivity of the tray case may be greater than that of the heat insulation wall (for example, the insulation material disposed between the inner and outer cases of the refrigerator). Here, the heat insulation wall may represent a heat insulation wall that partitions the external space from the storage chamber. If the degree of heat transfer of the tray case is equal to or greater than that of the heat insulation wall, the rate at which the ice making cell is cooled may be excessively reduced.

[0185] The first region may be configured to have a different degree of heat transfer in a direction along the outer circumferential surface. The degree of heat transfer of one portion of the first region may be less than that of the other portion of the first region. Such a configuration may be assisted to reduce the heat transfer transferred through the tray assembly from the first region to the second region in the direction along the outer circumferential surface.

[0186] The first and second regions defined to contact each other may be configured to have a different degree of heat transfer in the direction along the outer circumferential surface. The degree of heat transfer of one portion of the first region may be configured to be less than the degree of heat transfer of one portion of the second region. Such a configuration may be assisted to reduce the heat transfer transferred through the tray assembly from the first region to the second region in the direction along the outer circumferential surface. In another aspect, it may be advantageous to reduce the heat transferred from the heater to one portion of the first region to be transferred to the ice making cell defined by the second region. As the heat transmitted to the second region is reduced, the heater may locally heat one portion of the first region. Thus, it may be possible to reduce the decrease in ice making rate by the heating of the heater. In another aspect, the bubbles may be moved or collected in the region in which the heater is locally heated, thereby improving the transparency of the ice. The heater may be a transparent ice heater.

[0187] For example, a length of the heat transfer path from the first region to the second region may be greater than that of the heat transfer path in the direction from the first region to the outer circumferential surface from the first region. For another example, in a thickness of the tray assembly in the direction of the outer circumferential surface of the ice making cell from the center of the ice making cell, one portion of the first region may be thinner than the other of the first region or thinner than one portion of the second region. One portion of the first region may be a portion at which the tray case is not surrounded. The other portion of the first region may be a portion that is surrounded by the tray case. One portion of the second region may be a portion that is surrounded by the tray case. One portion of the first region may be a portion of the first region that defines the

lowest end of the ice making cell. The first region may include a tray and a tray case locally surrounding the tray.

[0188] As described above, when the thickness of the first region is thin, the heat transfer in the direction of the center of the ice making cell may increase while reducing the heat transfer in the direction of the outer circumferential surface of the ice making cell. For this reason, the ice making cell defined by the first region may be locally heated.

[0189] A minimum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the second region or less than that of one of the second region. A maximum value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or less than that of the thickness of one portion of the second region. When the through-hole is defined in the region, the minimum value represents the minimum value in the remaining regions except for the portion in which the through-hole is defined. An average value of the thickness of one portion of the first region may be less than that of the thickness of the other portion of the first region or may be less than that of one of the thickness of the second region. The uniformity of the thickness of one portion of the first region may be greater than that of the thickness of the other portion of the first region or greater than that of one of the thickness of the second region.

[0190] For example, the tray assembly may include a first portion defining at least a portion of the ice making cell and a second portion extending from a predetermined point of the first portion. The first region may be defined in the first portion. The second region may be defined in an additional tray assembly that may contact the first portion. At least a portion of the second portion may extend in a direction away from the ice making cell defined by the second region. In this case, the heat transmitted from the heater to the first region may be reduced from being transferred to the second region.

[0191] The structure and method of cooling the ice making cell in addition to the degree of cold transfer of the tray assembly may affect the making of the transparent ice. As described above, the tray assembly may include a first region and a second region, which define an outer circumferential surface of the ice making cell. For example, each of the first and second regions may be a portion of one tray assembly. For another example, the first region may be a first tray assembly. The second region may be a second tray assembly.

[0192] For a constant amount of cold supplied by the cooler and a constant amount of heat supplied by the heater, it may be advantageous to configure the cooler so that a portion of the ice making cell is more intensively cooled to increase the ice making rate of the refrigerator and/or increase the transparency of the ice. The more the cold supplied to the ice making cell by the cooler increases, the more the ice making rate may increase. However, as the cold is uniformly supplied to the outer circumferential surface of the ice making cell, the transparency of the made ice may decrease. Therefore, as the cooler more intensively cools a portion of the ice making cell, the bubbles may be moved or collected to other regions of the ice making cell, thereby increasing the transparency of the made ice and minimizing the decrease in ice making rate.

[0193] The cooler may be configured so that the amount of cold supplied to the second region differs from that of cold supplied to the first region so as to allow the cooler to more intensively cool a portion of the ice making cell. The amount

of cold supplied to the second region by the cooler may be greater than that of cold supplied to the first region.

[0194] For example, the second region may be made of a metal material having a high cold transfer rate, and the first region may be made of a material having a cold rate less than that of the metal.

[0195] For another example, to increase the degree of cold transfer transmitted from the storage chamber to the center of the ice making cell through the tray assembly, the second region may vary in degree of cold transfer toward the central direction. The degree of cold transfer of one portion of the second region may be greater than that of the other portion of the second region. A through-hole may be defined in one portion of the second region. At least a portion of the heat absorbing surface of the cooler may be disposed in the through-hole. A passage through which the cold air supplied from the cooler passes may be disposed in the through-hole. The one portion may be a portion that is not surrounded by the tray case. The other portion may be a portion surrounded by the tray case. One portion of the second region may be a portion defining the uppermost portion of the ice making cell in the second region. The second region may include a tray and a tray case locally surrounding the tray. As described above, when a portion of the tray assembly has a high cold transfer rate, the supercooling may occur in the tray assembly having a high cold transfer rate. As described above, designs may be needed to reduce the degree of the supercooling.

[0196] Hereinafter, a specific embodiment of the refrigerator according to an embodiment will be described with reference to the drawings.

[0197] FIG. 1 is a front view of a refrigerator according to an embodiment.

[0198] Referring to FIG. 1, a refrigerator according to an embodiment may include a cabinet 14 including a storage chamber and a door that opens and closes the storage chamber. The storage chamber may include a refrigerating compartment 18 and a freezing compartment 32. The refrigerating compartment 18 is disposed at an upper side, and the freezing compartment 32 is disposed at a lower side. Each of the storage chamber may be opened and closed individually by each door. For another example, the freezing compartment may be disposed at the upper side and the refrigerating compartment may be disposed at the lower side. Alternatively, the freezing compartment may be disposed at one side of left and right sides, and the refrigerating compartment may be disposed at the other side.

[0199] The freezing compartment 32 may be divided into an upper space and a lower space, and a drawer 40 capable of being withdrawn from and inserted into the lower space may be provided in the lower space.

[0200] The door may include a plurality of doors 10, 20, 30 for opening and closing the refrigerating compartment 18 and the freezing compartment 32. The plurality of doors 10, 20, and 30 may include some or all of the doors 10 and 20 for opening and closing the storage chamber in a rotatable manner and the door 30 for opening and closing the storage chamber in a sliding manner. The freezing compartment 32 may be provided to be separated into two spaces even though the freezing compartment 32 is opened and closed by one door 30. In this embodiment, the freezing compartment 32 may be referred to as a first storage chamber, and the refrigerating compartment 18 may be referred to as a second storage chamber.

[0201] The freezing compartment 32 may be provided with an ice maker 200 capable of making ice. The ice maker 200 may be disposed, for example, in an upper space of the freezing compartment 32. An ice bin 600 in which the ice made by the ice maker 200 falls to be stored may be disposed below the ice maker 200. A user may take out the ice bin 600 from the freezing compartment 32 to use the ice stored in the ice bin 600. The ice bin 600 may be mounted on an upper side of a horizontal wall that partitions an upper space and a lower space of the freezing compartment 32 from each other. Although not shown, the cabinet 14 is provided with a duct supplying cold air to the ice maker 200 (not shown). The duct guides the cold air heat-exchanged with a refrigerant flowing through the evaporator to the ice maker 200. For example, the duct may be disposed behind the cabinet 14 to discharge the cold air toward a front side of the cabinet 14. The ice maker 200 may be disposed at a front side of the duct. Although not limited, a discharge hole of the duct may be provided in one or more of a rear wall and an upper wall of the freezing compartment 32.

[0202] Although the above-described ice maker 200 is provided in the freezing compartment 32, a space in which the ice maker 200 is disposed is not limited to the freezing compartment 32. For example, the ice maker 200 may be disposed in various spaces as long as the ice maker 200 receives the cold air. Therefore, hereinafter, the ice maker 200 will be described as being disposed in a storage chamber.

[0203] FIG. 2 is a perspective view of the ice maker according to an embodiment, and FIG. 3 is a front view of the ice maker of FIG. 2. FIG. 4 is a perspective view illustrating a state in which a bracket is removed from the ice maker of FIG. 3, and FIG. 5 is an exploded perspective view of the ice maker according to an embodiment.

[0204] Referring to FIGS. 2 to 5, each component of the ice maker 200 may be provided inside or outside the bracket 220, and thus, the ice maker 200 may constitute one assembly.

[0205] The ice maker 200 may include a first tray assembly and a second tray assembly. The first tray assembly may include a first tray 320, a first tray case, or all of the first tray 320 and a second tray case. The second tray assembly may include a second tray 380, a second tray case, or all of the second tray 380 and a second tray case. The bracket 220 may define at least a portion of a space that accommodates the first tray assembly and the second tray assembly.

[0206] The bracket 220 may be installed at, for example, the upper wall of the freezing compartment 32. The bracket 220 may be provided with a water supply part or liquid supply 240. The water supply part 240 may guide water supplied from the upper side to the lower side of the water supply part 240. A water supply pipe (not shown) to which water is supplied may be installed above the water supply part 240.

[0207] The water supplied to the water supply part 240 may move downward. The water supply part 240 may prevent the water discharged from the water supply pipe from dropping from a high position, thereby preventing the water from splashing. Since the water supply part 240 is disposed below the water supply pipe, the water may be guided downward without splashing up to the water supply part 240, and an amount of splashing water may be reduced even if the water moves downward due to the lowered height.

[0208] The ice maker 200 may include an ice making cell 320a (see FIG. 31) in which water is phase-changed into ice by the cold air. The first tray 320 may form at least a portion of the ice making cell 320a. The second tray 380 may include a second tray 380 forming the other portion of the ice making cell 320a. The second tray 380 may be disposed to be relatively movable with respect to the first tray 320. The second tray 380 may linearly rotate or rotate. Hereinafter, the rotation of the second tray 380 will be described as an example.

[0209] For example, in an ice making process, the second tray 380 may move with respect to the first tray 320 so that the first tray 320 and the second tray 380 contact each other. When the first tray 320 and the second tray 380 are in contact with each other, the complete ice making cell see 320a may be defined. On the other hand, the second tray 380 may move with respect to the first tray 320 during the ice making process after the ice making is completed, and the second tray 380 may be spaced apart from the first tray 320. In this embodiment, the first tray 320 and the second tray 380 may be arranged in a vertical direction in a state in which the ice making cell 320a is defined. Accordingly, the first tray 320 may be referred to as an upper tray, and the second tray 380 may be referred to as a lower tray.

[0210] A plurality of ice making cells 320a may be defined by the first tray 320 and the second tray 380. Hereinafter, in the drawing, three ice making cells 320a are provided as an example.

[0211] When water is cooled by cold air while water is supplied to the ice making cell 320a, ice having the same or similar shape as that of the ice making cell 320a may be made. In this embodiment, for example, the ice making cell 320a may be provided in a spherical shape or a shape similar to a spherical shape. The ice making cell 320a may have a rectangular parallelepiped shape or a polygonal shape.

[0212] For example, the first tray case may include the first tray supporter 340 and the first tray cover 320. The first tray supporter 340 and the first tray cover 320 may be integrally provided or coupled to each other with each other after being manufactured in separate configurations. For example, at least a portion of the first tray cover 300 may be disposed above the first tray 320. At least a portion of the first tray supporter 340 may be disposed under the first tray 320. The first tray cover 300 may be manufactured as a separate part from the bracket 220 and then may be coupled to the bracket 220 or integrally formed with the bracket 220. That is, the first tray case may include the bracket 220.

[0213] The ice maker 200 may further include a first heater case 280. An ice separation heater 290 (see FIG. 31) may be installed in the first heater case 280. The heater case 280 may be integrally formed with the first tray cover 300 or may be separately formed.

[0214] The ice separation heater 290 may be disposed at a position adjacent to the first tray 320. The ice separation heater 290 may be, for example, a wire type heater. For example, the ice separation ice heater 290 may be installed to contact the first tray 320 or may be disposed at a position spaced a predetermined distance from the first tray 320. In some cases, the ice separation heater 290 may supply heat to the first tray 320, and the heat supplied to the first tray 320 may be transferred to the ice making cell 320a. The first tray cover 300 may be provided to correspond to a shape of the ice making cell 320a of the first tray 320 and may be coupled to a lower side of the first tray 320.

[0215] The ice maker 200 may include a first pusher 260 separating the ice during an ice separation process. The first pusher 260 may receive power of the driver 480 to be described later. The first tray cover 300 may be provided with a guide slot 302 guiding movement of the first pusher 260. The guide slot 302 may be provided in a portion extending upward from the first tray cover 300. A guide connection part of the first pusher 260 to be described later may be inserted into the guide slot 302. Thus, the guide connection part may be guided along the guide slot 302.

[0216] The first pusher 260 may include at least one pushing bar 264. For example, the first pusher 260 may include a pushing bar 264 provided with the same number as the number of ice making cells 320a, but is not limited thereto. The pushing bar 264 may push out the ice disposed in the ice making cell 320a during the ice separation process. For example, the pushing bar 264 may be inserted into the ice making cell 320a through the first tray cover 300. Therefore, the first tray cover 300 may be provided with an opening 304 (or through-hole) through which a portion of the first pusher 260 passes.

[0217] The first pusher 260 may be coupled to a pusher link 500. In this case, the first pusher 260 may be coupled to the pusher link 500 so as to be rotatable. Therefore, when the pusher link 500 moves, the first pusher 260 may also move along the guide slot 302.

[0218] The second tray case may include, for example, a second tray cover 360 and a second tray supporter 400. The second tray cover 360 and the second tray supporter 400 may be integrally formed or coupled to each other with each other after being manufactured in separate configurations. For example, at least a portion of the second tray cover 360 may be disposed above the second tray 380. At least a portion of the second tray supporter 400 may be disposed below the second tray 380. The second tray supporter 400 may be disposed at a lower side of the second tray to support the second tray 380.

[0219] For example, at least a portion of the wall defining a second cell 381a the second tray 380 may be supported by the second tray supporter 400. A spring 402 may be connected to one side of the second tray supporter 400. The spring 402 may provide elastic force to the second tray supporter 400 to maintain a state in which the second tray 380 contacts the first tray 320.

[0220] The second tray 380 may include a circumferential wall 387 surrounding a portion of the first tray 320 in a state of contacting the first tray 320. The second tray cover 360 may cover at least a portion of the circumferential wall 387.

[0221] The ice maker 200 may further include a second heater case 420. A transparent ice heater 430 to be described later may be installed in the second heater case 420. The second heater case 420 may be integrally formed with the second tray supporter 400 or may be separately provided to be coupled to the second tray supporter 400.

[0222] The ice maker 200 may further include a driver 480 that provides driving force. The second tray 380 may relatively move with respect to the first tray 320 by receiving the driving force of the driver 480. The first pusher 260 may move by receiving the driving force of the driving force 480. A through-hole 282 may be defined in an extension part 281 extending downward in one side of the first tray cover 300. A through-hole 404 may be defined in the extension part 403 extending in one side of the second tray supporter 400.

[0223] The ice maker 200 may further include a shaft 440 (or a rotation shaft) that passes through the through-holes 282 and 404 together. A rotation arm 460 may be provided at each of both ends of the shaft 440. That is, the shaft 440 may be connected to the pair of rotation arms 460. The shaft 440 may rotate by receiving rotational force from the driver 480. One end of the rotation arm 460 may be connected to one end of the spring 402, and thus, a position of the rotation arm 460 may move to an initial position by restoring force when the spring 402 is tensioned. Alternatively, the rotation arm may be connected to the driver 480 to rotate by receiving rotational force from the driver 480. In this case, the shaft 440 may be connected to the rotation arm, which is not connected to the driver 480, of the pair of rotation arms 460 to transmit the rotational force.

[0224] The driver 480 may include a motor and a plurality of gears. A full ice detection lever 520 may be connected to the driver 480. The full ice detection lever 520 may also rotate by the rotational force provided by the driver 480.

[0225] The full ice detection lever 520 may have a 'C' shape as a whole. For example, the full ice detection lever 520 may include a first lever 521 and a pair of second levers 522 extending in a direction crossing the first lever 521 at both ends of the first lever 521. One of the pair of second levers 522 may be coupled to the driver 480, and the other may be coupled to the bracket 220 or the first tray cover 300. The full ice detection lever 520 may rotate to detect ice stored in the ice bin 600.

[0226] The driver 480 may further include a cam that rotates by the rotational power of the motor. The ice maker 200 may further include a sensor that senses the rotation of the cam. For example, the cam is provided with a magnet, and the sensor may be a hall sensor detecting magnetism of the magnet during the rotation of the cam. The sensor may output first and second signals that are different outputs according to whether the sensor senses a magnet. One of the first signal and the second signal may be a high signal, and the other may be a low signal. The controller 800 to be described later may determine a position of the second tray 380 (or the second tray assembly) based on the type and pattern of the signal outputted from the sensor. That is, since the second tray 380 and the cam rotate by the motor, the position of the second tray 380 may be indirectly determined based on a detection signal of the magnet provided in the cam. For example, a water supply position, an ice making position, and an ice separation position, which will be described later, may be distinguished and determined based on the signals outputted from the sensor. The ice making position, the water supply position, and the ice separation position may alternatively be referred to as first, second, and third positions.

[0227] The ice maker 200 may further include a second pusher 540. The second pusher 540 may be installed, for example, on the bracket 220. The second pusher 540 may include at least one pushing bar 544. For example, the second pusher 540 may include a pushing bar 544 provided with the same number as the number of ice making cells 320a, but is not limited thereto.

[0228] The pushing bar 544 may push the ice disposed in the ice making cell 320a. For example, the pushing bar 544 may pass through the second tray supporter 400 to contact the second tray 380 defining the ice making cell 320a and then press the contacting second tray 380. The first tray cover 300 may be rotatably coupled to the second tray

supporter 400 with respect to the second tray supporter 400 and then be disposed to change in angle about the shaft 440.

[0229] In this embodiment, the second tray 380 may be made of a non-metal material. For example, when the second tray 380 is pressed by the second pusher 540, the second tray 380 may be made of a flexible or soft material which is deformable. Although not limited, the second tray 380 may be made of, for example, a silicon material. Therefore, while the second tray 380 is deformed while the second tray 380 is pressed by the second pusher 540, pressing force of the second pusher 540 may be transmitted to ice. The ice and the second tray 380 may be separated from each other by the pressing force of the second pusher 540.

[0230] When the second tray 380 is made of the non-metal material and the flexible or soft material, the coupling force or attaching force between the ice and the second tray 380 may be reduced, and thus, the ice may be easily separated from the second tray 380. Also, if the second tray 380 is made of the non-metallic material and the flexible or soft material, after the shape of the second tray 380 is deformed by the second pusher 540, when the pressing force of the second pusher 540 is removed, the second tray 380 may be easily restored to its original shape.

[0231] For another example, the first tray 320 may be made of a metal material. In this case, since the coupling force or the attaching force between the first tray 320 and the ice is strong, the ice maker 200 according to this embodiment may include at least one of the ice separation heater 290 or the first pusher 260. For another example, the first tray 320 may be made of a non-metallic material. When the first tray 320 is made of the non-metallic material, the ice maker 200 may include only one of the ice separation heater 290 and the first pusher 260. Alternatively, the ice maker 200 may not include the ice separation heater 290 and the first pusher 260. Although not limited, the first tray 320 may be made of, for example, a silicon material. That is, the first tray 320 and the second tray 380 may be made of the same material.

[0232] When the first tray 320 and the second tray 380 are made of the same material, the first tray 320 and the second tray 380 may have different hardness to maintain sealing performance at the contact portion between the first tray 320 and the second tray 380.

[0233] In this embodiment, since the second tray 380 is pressed by the second pusher 540 to be deformed, the second tray 380 may have hardness less than that of the first tray 320 to facilitate the deformation of the second tray 380.

[0234] FIGS. 6 and 7 are perspective views of the bracket according to an embodiment.

[0235] Referring to FIGS. 6 and 7, the bracket 220 may be fixed to at least one surface of the storage chamber or to a cover member (to be described later) fixed to the storage chamber.

[0236] The bracket 220 may include a first wall 221 having a through-hole 221a defined therein. At least a portion of the first wall 221 may extend in a horizontal direction. The first wall 221 may include a first fixing wall 221b to be fixed to one surface of the storage chamber or the cover member. At least a portion of the first fixing wall 221b may extend in the horizontal direction. The first fixing wall 221b may also be referred to as a horizontal fixing wall. One or more fixing protrusions 221c may be provided on the first fixing wall 221b. A plurality of fixing protrusions 221c may

be provided on the first fixing wall **221b** to firmly fix the bracket **220**. The first wall **221** may further include a second fixing wall **221e** to be fixed to one surface of the storage chamber or the cover member. At least a portion of the second fixing wall **221e** may extend in a vertical direction. The second fixing wall **221e** may also be referred to as a vertical fixing wall. The second fixing wall **221e** may extend upward from the first fixing wall **221b**. The second fixing wall **221e** may include a fixing rib **221e1** and/or a hook **221e2**. In this embodiment, the first wall **221** may include at least one of the first fixing wall **221b** or the second fixing wall **221e** to fix the bracket **220**. The first wall **221** may be provided in a shape in which a plurality of walls are stepped in the vertical direction. In one example, a plurality of walls may be arranged with a height difference in the horizontal direction, and the plurality of walls may be connected by a vertical connection wall. The first wall **221** may further include a support wall **221d** supporting the first tray assembly. At least a portion of the support wall **221d** may extend in the horizontal direction. The support wall **221d** may be disposed at the same height as the first fixing wall **221b** or disposed at a different height. In FIG. 6, for example, the support wall **221d** is disposed at a position lower than that of the first fixing wall **221b**.

[0237] The bracket **220** may further include a second wall **222** having a through-hole **222a** through which cold air generated by a cooling part passes. The second wall **222** may extend from the first wall **221**. At least a portion of the second wall **222** may extend in the vertical direction. At least a portion of the through-hole **222a** may be disposed at a position higher than that of the support wall **221d**. In FIG. 6, for example, the lowermost end of the through-hole **222a** is disposed at a position higher than that of the support wall **221d**.

[0238] The bracket **220** may further include a third wall **223** on which the driver **480** is installed. The third wall **223** may extend from the first wall **221**. At least a portion of the third wall **223** may extend in the vertical direction. At least a portion of the third wall **223** may be disposed to face the second wall **222** while being spaced apart from the second wall **222**. At least a portion of the ice making cell (see **320a** in FIG. 49) may be disposed between the second wall **222** and the second wall **223**. The driver **480** may be installed on the third wall **223** between the second wall **222** and the third wall **223**. Alternatively, the driver **480** may be installed on the third wall **223** so that the third wall **223** is disposed between the second wall **222** and the driver **480**. In this case, a shaft hole **223a** through which a shaft of the motor constituting the driver **480** passes may be defined in the third wall **223**. FIG. 7 illustrates that the shaft hole **223a** is defined in the third wall **223**.

[0239] The bracket **220** may further include a fourth wall **224** to which the second pusher **540** is fixed. The fourth wall **224** may extend from the first wall **221**. The fourth wall **224** may connect the second wall **222** to the third wall **223**. The fourth wall **224** may be inclined at an angle with respect to the horizontal line and the vertical line. For example, the fourth wall **224** may be inclined in a direction away from the shaft hole **223a** from the upper side to the lower side. The fourth wall **224** may be provided with a mounting groove **224a** in which the second pusher **540** is mounted. The mounting groove **224a** may be provided with a coupling hole **224b** through which a coupling part coupled to the second pusher **540** passes.

[0240] The second tray **380** and the second pusher **540** may contact each other while the second tray assembly rotates while the second pusher **540** is fixed to the fourth wall **224**. Ice may be separated from the second tray **380** while the second pusher **540** presses the second tray **380**. When the second pusher **540** presses the second tray **380**, the ice also presses the second pusher **540** before the ice is separated from the second tray **380**. Force for pressing the second pusher **540** may be transmitted to the fourth wall **224**. Since the fourth wall **224** is provided in a thin plate shape, a strength reinforcement member **224c** may be provided on the fourth wall **224** to prevent the fourth wall **224** from being deformed or broken. For example, the strength reinforcement member **224c** may include ribs disposed in a lattice form. That is, the strength reinforcement member **224c** may include a first rib extending in the first direction and a second rib extending in a second direction crossing the first direction. In this embodiment, two or more of the first to fourth walls **221** to **224** may define a space in which the first and second tray assemblies are disposed.

[0241] FIG. 8 is a perspective view of the first tray when viewed from an upper side, and FIG. 9 is a perspective view of the first tray when viewed from a lower side. FIG. 10 is a cross-sectional view taken along line 10-10 of FIG. 8.

[0242] Referring to FIGS. 8 to 10, the first tray **320** may define a first cell **321a** that is a portion or a first portion of the ice making cell **320a**. The first tray **320** may include a first tray wall **321** defining a portion of the ice making cell **320a**.

[0243] For example, the first tray **320** may define a plurality of first cells **321a**. For example, the plurality of first cells **321a** may be arranged in a line. The plurality of first cells **321a** may be arranged in an X-axis direction in FIG. 9. For example, the first tray wall **321** may define the plurality of first cells **321a**.

[0244] The first tray wall **321** may include a plurality of first cell walls **3211** that respectively define the plurality of first cells **321a**, and a connection wall **3212** connecting the plurality of first cell walls **3211** to each other. The first tray wall **321** may be a wall extending in the vertical direction. The first tray **320** may include an opening **324**. The opening **324** may communicate with the first cell **321a**. The opening **324** may allow the cold air to be supplied to the first cell **321a**. The opening **324** may allow water for making ice to be supplied to the first cell **321a**. The opening **324** may provide a passage through which a portion of the first pusher **260** passes. For example, in the ice separation process, a portion of the first pusher **260** may be inserted into the ice making cell **320a** through the opening **324**. The first tray **320** may include a plurality of openings **324** corresponding to the plurality of first cells **321a**. One of the plurality of openings **324** **324a** may provide a passage of the cold air, a passage of the water, and a passage of the first pusher **260**. In the ice making process, the bubbles may escape through the opening **324**.

[0245] The first tray **320** may include a case accommodation part **321b**. For example, a portion of the first tray wall **321** may be recessed downward to provide the case accommodation part **321b**. At least a portion of the case accommodation part **321b** may be disposed to surround the opening **324**. A bottom surface of the case accommodation part **321b** may be disposed at a position lower than that of the opening **324**.

[0246] The first tray 320 may further include an auxiliary storage chamber 325 communicating with the ice making cell 320a. For example, the auxiliary storage chamber 325 may store water overflowed from the ice making cell 320a. The ice expanded in a process of phase-changing the supplied water may be disposed in the auxiliary storage chamber 325. That is, the expanded ice may pass through the opening 324 and be disposed in the auxiliary storage chamber 325. The auxiliary storage chamber 325 may be defined by a storage chamber wall 325a. The storage chamber wall 325a may extend upwardly around the opening 324. The storage chamber wall 325a may have a cylindrical shape or a polygonal shape. Substantially, the first pusher 260 may pass through the opening 324 after passing through the storage chamber wall 325a. The storage chamber wall 325a may define the auxiliary storage chamber 325 and also reduce deformation of the periphery of the opening 324 in the process in which the first pusher 260 passes through the opening 324 during the ice separation process. When the first tray 320 defines a plurality of first cells 321a, at least one 325b of the plurality of storage chamber walls 325a may support the water supply part 240. The storage chamber wall 325b supporting the water supply part 240 may have a polygonal shape. For example, the storage chamber wall 325b may include a round part rounded in a horizontal direction and a plurality of straight portions. For example, the storage chamber wall 325b may include a round wall 325b1, a pair of straight walls 325b2 and 325b3 extending side by side from both ends of the round wall 325b, and a connection wall 325b4 connecting the pair of straight walls 325b2 to each other. The connection wall 325b4 may be a rounded wall or a straight wall. An upper end of the connection wall 325b4 may be disposed at a position lower than that of an upper end of the remaining walls 325b1, 325b2, and 325b3. The connection wall 325b4 may support the water supply part 240. An opening 324a corresponding to the storage chamber wall 325b supporting the water supply part 240 may also be defined in the same shape as the storage chamber wall 325b.

[0247] The first tray 320 may further include a heater accommodation part 321c. The ice separation heater 290 may be accommodated in the heater accommodation part 321c. The ice separation heater 290 may contact a bottom surface of the heater accommodation part 321c. The heater accommodation part 321c may be provided on the first tray wall 321 as an example. The heater accommodation part 321c may be recessed downward from the case accommodation part 321b. The heater accommodation part 321c may be disposed to surround the periphery of the first cell 321a. For example, at least a portion of the heater accommodation part 321c may be rounded in the horizontal direction. The bottom surface of the heater accommodating portion 321c may be disposed at a position lower than that of the opening 324.

[0248] The first tray 320 may include a first contact surface 322c contacting the second tray 380. The bottom surface of the heater accommodating portion 321c may be disposed between the opening 324 and the first contact surface 322c. At least a portion of the heater accommodation part 321c may be disposed to overlap the ice making cell 320a (or the first cell 321a in the vertical direction).

[0249] The first tray 320 may further include a first extension wall 327 extending in the horizontal direction from the first tray wall 321. For example, the first extension

wall 327 may extend in the horizontal direction around an upper end of the first extension wall 327. One or more first coupling holes 327a may be provided in the first extension wall 327. Although not limited, the plurality of first coupling holes 327a may be arranged in one or more axes of the X axis and the Y axis. An upper end of the storage chamber wall 325b may be disposed at the same height or higher than a top surface of the first extension wall 327.

[0250] When the first tray 320 includes the plurality of first cells 321a, the length of the first tray 320 may be longer, but the width of the first tray 320 may be shorter than the length of the first tray 320 to prevent the volume of the first tray 320 from increasing.

[0251] FIG. 11 is a cross-sectional view taken along line 11-11 of FIG. 8.

[0252] Referring to FIGS. 9 and 11, the first tray 320 may further include a sensor accommodation part 321e in which the second temperature sensor 700 (or the tray temperature sensor) is accommodated. The second temperature sensor 700 may sense a temperature of water or ice of the ice making cell 320a. The second temperature sensor 700 may be disposed adjacent to the first tray 320 to sense the temperature of the first tray 320, thereby indirectly determining the water temperature or the ice temperature of the ice making cell 320a. In this embodiment, the water temperature or the ice temperature of the ice making cell 320a may be referred to as an internal temperature of the ice making cell 320a. The sensor accommodation part 321e may be recessed downward from the case accommodation part 321b. Here, a bottom surface of the sensor accommodation part 321e may be disposed at a position lower than that of the bottom surface of the heater accommodation part 321c to prevent the second temperature sensor 700 from interfering with the ice separation heater 290 in a state in which the second temperature sensor 700 is accommodated in the sensor accommodation part 321e.

[0253] The bottom surface of the sensor accommodating portion 321e may be disposed closer to the first contact surface 322c of the first tray 320 than the bottom surface of the heater accommodating portion 321c. The sensor accommodation part 321e may be disposed between two adjacent ice making cells 320a. For example, the sensor accommodation part 321e may be disposed between two adjacent first cells 321a. When the sensor accommodation part 321e is disposed between the two ice making cells 320a, the second temperature sensor 700 may be easily installed without increasing the volume of the second tray 250. Also, when the sensor accommodation part 321e is disposed between the two ice making cells 320a, the temperatures of at least two ice making cells 320a may be affected. Thus, the temperature sensor may be disposed so that the temperature sensed by the second temperature sensor maximally approaches an actual temperature inside the cell 320a.

[0254] The sensor accommodation part 321e may be disposed between the two adjacent first cells 321a among the three first cells 321a arranged in the X-axis direction.

[0255] FIG. 12 is a perspective view of a first tray cover, FIG. 13 is a bottom perspective view of the first tray cover, FIG. 14 is a plan view of the first tray cover, and FIG. 15 is a side view of a first tray case.

[0256] Referring to FIGS. 12 to 15, the first tray cover 300 may include an upper plate 301 contacting the first tray 320.

[0257] A bottom surface of the upper plate 301 may be coupled to contact an upper side of the first tray 320. For

example, the upper plate 301 may contact at least one of a top surface of the first portion 322 and a top surface of the second portion 323 of the first tray 320. A plate opening 304 (or through-hole) may be defined in the upper plate 301. The plate opening 304 may include a straight portion and a curved portion.

[0258] Water may be supplied from the water supply part 240 to the first tray 320 through the plate opening 304. Also, the extension part 264 of the first pusher 260 may pass through the plate opening 304 to separate ice from the first tray 320. Also, cold air may pass through the plate opening 304 to contact the first tray 320. A first case coupling part 301b extending upward may be disposed at a side of the straight portion of the plate opening 304 in the upper plate 301. A first case coupling part 301b may be coupled to the first heater case 280.

[0259] The first tray cover 300 may further include a circumferential wall 303 extending upward from an edge of the upper plate 301. The circumferential wall 303 may include two pairs of walls facing each other. For example, the pair of walls may be spaced apart from each other in the X-axis direction, and another pair of walls may be spaced apart from each other in the Y-axis direction.

[0260] The circumferential walls 303 spaced apart from each other in the Y-axis direction of FIG. 12 may include an extension wall 302e extending upward. The extension wall 302e may extend upward from a top surface of the circumferential wall 303.

[0261] The first tray cover 300 may be provided with a pair of guide slots 302 guiding movement of the first pusher 260. A portion of the guide slot 302 may be defined in the extension wall 302e, and the other portion may be defined in the circumferential wall 303 disposed below the extension wall 302e. A lower portion of the guide slot 302 may be defined in the circumferential wall 303.

[0262] The guide slot 302 may extend in the Z-axis direction of FIG. 12. The first pusher 260 may be inserted into the guide slot 302 to move. Also, the first pusher 260 may move up and down along the guide slot 302.

[0263] The guide slot 302 may include a first slot 302a extending perpendicular to the upper plate 301 and a second slot 302b that is bent at an angle from an upper end of the first slot 302a. Alternatively, the guide slot 302 may include only the first slot 302a extending in the vertical direction. The lower end 302d of the first slot 302a may be disposed lower than the upper end of the circumferential wall 303. Also, the upper end 302c of the first slot 302a may be disposed higher than the upper end of the circumferential wall 303. The portion bent from the first slot 302a to the second slot 302b may be disposed at a position higher than the circumferential wall 303. A length of the first slot 302a may be greater than that of the second slot 302b. The second slot 302b may be bent toward the horizontal extension part 305. When the first pusher 260 moves upward along the guide slot 302, the first pusher 260 rotates or is tilted at a predetermined angle in the portion moving along the second slot 302b.

[0264] When the first pusher 260 rotates, the pushing bar 264 of the first pusher 260 may rotate so that the pushing bar 264 is spaced apart vertically above the opening 324 of the first tray 320. When the first pusher 260 moves along the second slot 302b that is bent and extended, the end of the pushing bar 264 may be spaced apart so as not to contact with water supplied when water is supplied to the pushing

bar. Thus, the water may be cooled at the end of 264 to prevent the pushing bar 264 from being inserted into the opening 324 of the first tray 320.

[0265] The first tray cover 300 may include a plurality of coupling parts 301a coupling the first tray 320 to the first tray supporter 340 (see FIG. 16) to be described later. The plurality of coupling parts 301a may be disposed on the upper plate 301. The plurality of coupling parts 301a may be spaced apart from each other in the X-axis and/or Y-axis directions. The coupling part 301a may protrude upward from the top surface of the upper plate 301. For example, a portion of the plurality of coupling parts 301a may be connected to the circumferential wall 303.

[0266] The coupling part 301a may be coupled to a coupling member to fix the first tray 320. The coupling member coupled to the coupling part 301a may be, for example, a bolt. The coupling member may pass through the coupling hole 341a of the first tray supporter 340 and the first coupling hole 327a of the first tray 320 at the bottom surface of the first tray supporter 340 and then be coupled to the coupling part 301a.

[0267] A horizontal extension part 305 extending horizontally form the circumferential wall 303 may be disposed on one circumferential wall 303 of the circumferential walls 303 spaced apart from and facing each other in the Y-axis direction of FIG. 12. The horizontal extension part 305 may extend from the circumferential wall 303 in a direction away from the plate opening 304 so as to be supported by the support wall 221d of the bracket 220. A plurality of vertical coupling parts 303a may be provided on the other one of the circumferential walls 303 spaced apart from and facing each other in the Y-axis direction. The vertical coupling part 303a may be coupled to the first wall 221 of the bracket 220. The vertical coupling parts 303a may be arranged to be spaced apart from each other in the X-axis direction.

[0268] The upper plate 301 may be provided with a lower protrusion 306 protruding downward. The lower protrusion 306 may extend along the length of the upper plate 301 and may be disposed around the circumferential wall 303 of the other of the circumferential walls 303 spaced apart from each other in the Y-axis direction. Also, a step portion 306a may be disposed on the lower protrusion 306. The step portion 306a may be disposed between a pair of extension parts 281 described later. Thus, when the second tray 380 rotates, the second tray 380 and the first tray cover 300 may not interfere with each other.

[0269] The first tray cover 300 may further include a plurality of hooks 307 coupled to the first wall 221 of the bracket 220. For example, the hooks 307 may be provided on the horizontal protrusion 306. The plurality of hooks 307 may be spaced apart from each other in the X-axis direction. Also, the plurality of hooks 307 may be disposed between the pair of extension parts 281. Each of the hooks 307 may include a first portion 307a horizontally extending from the circumferential wall 303 in the opposite direction to the upper plate 301 and a second portion 307b bent from an end of the first portion 307a to extend vertically downward.

[0270] The first tray cover 300 may further include a pair of extension parts 281 to which the shaft 440 is coupled. For example, the pair of extension parts 281 may extend downward from the lower protrusion 306. The pair of extension parts 281 may be spaced apart from each other in the X-axis direction. Each of the extension parts 281 may include a through-hole 282 through which the shaft 440 passes.

[0271] The first tray cover 300 may further include an upper wire guide part 310 guiding a wire connected to the ice separation heater 290, which will be described later. The upper wire guide part 310 may, for example, extend upward from the upper plate 301. The upper wire guide part 310 may include a first guide 312 and a second guide 314, which are spaced apart from each other. For example, the first guide 312 and the second guide 314 may extend vertically upward from the upper plate 310.

[0272] The first guide 312 may include a first portion 312a extending from one side of the plate opening 304 in the Y-axis direction, a second portion 312b bent and extending from the first portion 312a, and a third portion 312c bent from the second portion 312b to extend in the X-axis direction. The third portion 312c may be connected to one circumferential wall 303. A first protrusion 313 may be disposed on an upper end of the second portion 312b to prevent the wire from being separated.

[0273] The second guide 314 may include a first extension part 314a disposed to face the second portion 312b of the first guide 312 and a second extension part 314b bent to extend from the first extension part 314a and disposed to face the third portion 312c. The second portion 312b of the first guide 312 and the first extension part 314a of the second guide 314 and also the third portion 312c of the first guide 312 and the second extension part 314b of the second guide 314 may be parallel to each other. A second protrusion 315 may be disposed on an upper end of the first extension part 314a to prevent the wire from being separated.

[0274] The wire guide slots 313a and 315a may be defined in the upper plate 310 to correspond to the first and second protrusions 313 and 315, and a portion of the wire may be the wire guide slots 313a and 315a to prevent the wire from being separated.

[0275] FIG. 16 is a plan view of a first tray supporter.

[0276] Referring to FIG. 16, the first tray supporter 340 may be coupled to the first tray cover 300 to support the first tray 320. In detail, the first tray supporter 340 includes a horizontal portion 341 contacting a bottom surface of the upper end of the first tray 320 and an insertion opening 342 through which a lower portion of the first tray 320 is inserted into a center of the horizontal portion 341. The horizontal portion 341 may have a size corresponding to the upper plate 301 of the first tray cover 300. Also, the horizontal portion 341 may include a plurality of coupling holes 341a engaged with the coupling parts 301a of the first tray cover 300. The plurality of coupling holes 341a may be spaced apart from each other in the X-axis and/or Y-axis direction of FIG. 16 to correspond to the coupling part 301a of the first tray cover 300.

[0277] When the first tray cover 300, the first tray 320, and the first tray supporter 340 are coupled to each other, the upper plate 301 of the first tray cover 300, the first extension wall 327 of the first tray 320, and the horizontal portion 341 of the first tray supporter 340 may sequentially contact each other. In detail, the bottom surface of the upper plate 301 of the first tray cover 300 and the top surface of the first extension wall 327 of the first tray 320 may contact each other, and the bottom surface of the first extension wall 327 of the first tray 320 and the top surface of the horizontal part 341 of the first tray supporter 340 may contact each other.

[0278] FIG. 17 is a perspective view of the second tray according to an embodiment, and FIG. 18 is a perspective

view of the second tray when viewed from a lower side. FIG. 19 is a bottom view of the second tray, and FIG. 20 is a plan view of the second tray.

[0279] Referring to FIGS. 17 to 20, the second tray 380 may define a second cell 381a which is another portion or a second portion of the ice making cell 320a. The second tray 380 may include a second tray wall 381 defining a portion of the ice making cell 320a. For example, the second tray 380 may define a plurality of second cells 381a. For example, the plurality of second cells 381a may be arranged in a line. Referring to FIG. 20, the plurality of second cells 381a may be arranged in the X-axis direction. For example, the second tray wall 381 may define the plurality of second cells 381a. The third tray wall 381 may include a plurality of second cell walls 3811 which respectively define the plurality of second cells 381a. The two adjacent second cell walls 3811 may be connected to each other.

[0280] The second tray 380 may include a circumferential wall 387 extending along a circumference of an upper end of the second tray wall 381. The circumferential wall 387 may be formed integrally with the second tray wall 381 and may extend from an upper end of the second tray wall 381. For another example, the circumferential wall 387 may be provided separately from the second tray wall 381 and disposed around the upper end of the second tray wall 381. In this case, the circumferential wall 387 may contact the second tray wall 381 or be spaced apart from the third tray wall 381. In any case, the circumferential wall 387 may surround at least a portion of the first tray 320. If the second tray 380 includes the circumferential wall 387, the second tray 380 may surround the first tray 320. When the second tray 380 and the circumferential wall 387 are provided separately from each other, the circumferential wall 387 may be integrally formed with the second tray case or may be coupled to the second tray case. For example, one second tray wall may define a plurality of second cells 381a, and one continuous circumferential wall 387 may surround the first tray 250.

[0281] The circumferential wall 387 may include a first extension wall 387b extending in the horizontal direction and a second extension wall 387c extending in the vertical direction. The first extension wall 387b may be provided with one or more second coupling holes 387a to be coupled to the second tray case. The plurality of second coupling holes 387a may be arranged in at least one axis of the X axis or the Y axis. The second tray 380 may include a second contact surface 382c contacting the first contact surface 322c of the first tray 320. The first contact surface 322c and the second contact surface 382c may be horizontal planes. Each of the first contact surface 322c and the second contact surface 382c may be provided in a ring shape. When the ice making cell 320a has a spherical shape, each of the first contact surface 322c and the second contact surface 382c may have a circular ring shape.

[0282] FIG. 21 is a cross-sectional view taken along line 21-21 of FIG. 17.

[0283] FIG. 21 illustrates a Y-Z cutting surface passing through the central line C1.

[0284] Referring to FIG. 21, the second tray 380 may include a first portion 382 that defines at least a portion of the ice making cell 320a. For example, the first portion 382 may be a portion or the whole of the second tray wall 381.

[0285] In this specification, the first portion 322 of the first tray 320 may be referred to as a third portion so as to be

distinguished from the first portion **382** of the second tray **380**. Also, the second portion **323** of the first tray **320** may be referred to as a fourth portion so as to be distinguished from the second portion **383** of the second tray **380**.

[0286] The first portion **382** may include a second cell surface **382b** (or an outer circumferential surface) defining the second cell **381a** of the ice making cell **320a**. The first portion **382** may be defined as an area between two dotted lines in FIG. 21. The uppermost end of the first portion **382** is the second contact surface **382c** contacting the first tray **320**.

[0287] The second tray **380** may further include a second portion **383**. The second portion **383** may reduce transfer of heat, which is transferred from the transparent ice heater **430** to the second tray **380**, to the ice making cell **320a** defined by the first tray **320**. That is, the second portion **383** serves to allow the heat conduction path to move in a direction away from the first cell **321a**. The second portion **383** may be a portion or the whole of the circumferential wall **387**. The second portion **383** may extend from a predetermined point of the first portion **382**. In the following description, for example, the second portion **383** is connected to the first portion **382**. The predetermined point of the first portion **382** may be one end of the first portion **382**. Alternatively, the predetermined point of the first portion **382** may be one point of the second contact surface **382c**. The second portion **383** may include the other end that does not contact one end contacting the predetermined point of the first portion **382**. The other end of the second portion **383** may be disposed farther from the first cell **321a** than one end of the second portion **383**.

[0288] At least a portion of the second portion **383** may extend in a direction away from the first cell **321a**. At least a portion of the second portion **383** may extend in a direction away from the second cell **381a**. At least a portion of the second portion **383** may extend upward from the second contact surface **382c**. At least a portion of the second portion **383** may extend horizontally in a direction away from the central line **C1**. A center of curvature of at least a portion of the second portion **383** may coincide with a center of rotation of the shaft **440** which is connected to the driver **480** to rotate.

[0289] The second portion **383** may include a first part **384a** extending from one point of the first portion **382**. The second portion **383** may further include a second part **384b** extending in the same direction as the extending direction with the first part **384a**. Alternatively, the second portion **383** may further include a third part **384b** extending in a direction different from the extending direction of the first part **384a**. Alternatively, the second portion **383** may further include a second part **384b** and a third part **384c** branched from the first part **384a**. For example, the first part **384a** may extend in the horizontal direction from the first portion **382**. A portion of the first part **384a** may be disposed at a position higher than that of the second contact surface **382c**. That is, the first part **384a** may include a horizontally extension part and a vertically extension part. The first part **384a** may further include a portion extending from the predetermined point in a vertical direction. A length of the second extension part **323b** in the Y-axis direction may be greater than that of the first extension part **323a**.

[0290] The extension direction of at least a portion of the first part **384a** may be the same as that of the second part **384b**. The extension directions of the second part **384b** and

the third part **384c** may be different from each other. The extension direction of the third part **384c** may be different from that of the first part **384a**. The third part **384a** may have a constant curvature based on the Y-Z cutting surface. That is, the same curvature radius of the third part **384a** may be constant in the longitudinal direction. The curvature of the second part **384b** may be zero. When the second part **384b** is not a straight line, the curvature of the second part **384b** may be less than that of the third part **384a**. The curvature radius of the second part **384b** may be greater than that of the third part **384a**.

[0291] At least a portion of the second portion **383** may be disposed at a position higher than or equal to that of the uppermost end of the ice making cell **320a**. In this case, since the heat conduction path defined by the second portion **383** is long, the heat transfer to the ice making cell **320a** may be reduced. A length of the second portion **383** may be greater than the radius of the ice making cell **320a**. The second portion **383** may extend up to a point higher than the center of rotation **C4** of the shaft **440**. For example, the second portion **383** may extend up to a point higher than the uppermost end of the shaft **440**.

[0292] The second portion **383** may include a first extension part **383a** extending from a first point of the first portion **382** and a second extension part **383b** extending from a second point of the first portion **382** so that transfer of the heat of the transparent ice heater **430** to the ice making cell **320a** defined by the first tray **320** is reduced. For example, the first extension part **383a** and the second extension part **383b** may extend in different directions with respect to the central line **C1**.

[0293] Referring to FIG. 21, the first extension part **383a** may be disposed at the left side with respect to the central line **C1**, and the second extension part **383b** may be disposed at the right side with respect to the central line **C1**. The first extension part **383a** and the second extension part **383b** may have different shapes based on the central line **C1**. The first extension part **383a** and the second extension part **383b** may be provided in an asymmetrical shape with respect to the central line **C1**. A length (horizontal length) of the second extension part **383b** in the Y-axis direction may be longer than the length (horizontal length) of the first extension part **383a**. The first extension part **383a** may be disposed closer to an edge part that is disposed at a side opposite to the portion of the second wall **222** or the third wall **223** of the bracket **220**, which is connected to the fourth wall **224**, than the second extension part **383a**. The second extension part **383b** may be disposed closer to the shaft **440** that provides a center of rotation of the second tray assembly than the first extension part **383a**.

[0294] In this embodiment, a length of the second extension part **383b** in the Y-axis direction may be greater than that of the first extension part **383a**. In this case, the heat conduction path may increase while reducing the width of the bracket **220** relative to the space in which the ice maker **200** is installed. Since the length of the second extension part **383b** in the Y-axis direction is greater than that of the first extension part **383a**, the second tray assembly including the second tray **380** contacting the first tray **320** may increase in radius of rotation. When the rotation radius of the second tray assembly increases centrifugal force of the second tray assembly may increase. Thus, in the ice separation process, separating force for separating the ice from the second tray assembly may increase to improve ice separation perfor-

mance. The center of curvature of at least a portion of the second extension part **383b** may be a center of curvature of the shaft **440** which is connected to the driver **480** to rotate.

[0295] A distance between an upper portion of the first extension part **383a** and an upper portion of the second extension part **383b** may be greater than that between a lower portion of the first extension part **383a** and a lower portion of the second extension part **383b** with respect to the Y-Z cutting surface passing through the central line C1. For example, a distance between the first extension part **383a** and the second extension part **383b** may increase upward.

[0296] Each of the first extension part **383a** and the third extension part **383b** may include first to third parts **384a**, **384b**, and **384c**.

[0297] In another aspect, the third part **384c** may also be described as including the first extension part **383a** and the second extension part **383b** extending in different directions with respect to the central line C1.

[0298] The first portion **382** may have a variable radius in the Y-axis direction. The first portion **382** may include a first region **382d** (see region A in FIG. 21) and a second region **382e**. The curvature of at least a portion of the first region **382d** may be different from that of at least a portion of the second region **382e**. The first region **382d** may include the lowermost end of the ice making cell **320a**. The second region **382e** may have a diameter greater than that of the first region **382d**. The first region **382d** and the second region **382e** may be divided vertically.

[0299] The transparent ice heater **430** may contact the first region **382d**. The first region **382d** may include a heater contact surface **382g** contacting the transparent ice heater **430**. The heater contact surface **382g** may be, for example, a horizontal plane. The heater contact surface **382g** may be disposed at a position higher than that of the lowermost end of the first portion **382**.

[0300] The second region **382e** may include the second contact surface **382c**. The first region **382d** may have a shape recessed in a direction opposite to a direction in which ice is expanded in the ice making cell **320a**. A distance from the center of the ice making cell **320a** to the second region **382e** may be less than that from the center of the ice making cell **320a** to the portion at which the shape recessed in the first area **382d** is disposed. For example, the first region **382d** may include a pressing part **382f** that is pressed by the second pusher **540** during the ice separation process. When pressing force of the second pusher **540** is applied to the pressing part **382f**, the pressing part **382f** is deformed, and thus, ice is separated from the first portion **382**. When the pressing force applied to the pressing part **382f** is removed, the pressing part **382f** may return to its original shape. The central line C1 may pass through the first region **382d**. For example, the central line C1 may pass through the pressing part **382f**. The heater contact surface **382g** may be disposed to surround the pressing unit **382f**. The heater contact surface **382g** may be disposed at a position higher than that of the lowermost end of the pressing part **382f**. At least a portion of the heater contact surface **382g** may be disposed to surround the central line C1. Accordingly, at least a portion of the transparent ice heater **430** contacting the heater contact surface **382g** may be disposed to surround the central line C1. Therefore, the transparent ice heater **430** may be prevented from interfering with the second pusher **540** while the second pusher **540** presses the pressing unit **382f**. A distance from the center of the ice making cell **320a**

to the pressing part **382f** may be different from that from the center of the ice making cell **320a** to the second region **382e**.

[0301] FIG. 22 is a perspective view of the second tray cover, and FIG. 23 is a plan view of the second tray cover.

[0302] Referring to FIGS. 22 and 23, the second tray cover **360** includes an opening **362** (or through-hole) into which a portion of the second tray **380** is inserted. For example, when the second tray **380** is inserted below the second tray cover **360**, a portion of the second tray **380** may protrude upward from the second tray cover **360** through the opening **362**.

[0303] The second tray cover **360** may include a vertical wall **361** and a curved wall **363** surrounding the opening **362**. In detail, the vertical wall **361** may define three surfaces of the second tray cover **360**, and the curved wall **363** may define the other surface of the second tray cover **360**. The vertical wall **361** may be a wall extending vertically upward, and the curved wall **363** may be a wall rounded away from the opening **362** upward. The vertical walls **361** and the curved walls **363** may be provided with a plurality of coupling parts **361a**, **361c**, and **363a** to be coupled to the second tray **380** and the second tray case **400**. The vertical wall **361** and the curved wall **363** may further include a plurality of coupling grooves **361b**, **361d**, and **363b** corresponding to the plurality of coupling parts **361a**, **361c**, and **363a**. A coupling member may be inserted into the plurality of coupling parts **361a**, **361c**, and **363a** to pass through the second tray **380** and then be coupled to the coupling parts **401a**, **401b**, and **401c** of the second tray supporter **400**. Here, the coupling part may protrude upward from the vertical wall **361** and the curved wall **363** through the plurality of coupling grooves **361b**, **361d**, and **363b** to prevent an interference with other components.

[0304] A plurality of first coupling parts **361a** may be provided on the wall facing the curved wall **363** of the vertical wall **361**. In detail, the plurality of first coupling parts **361a** may be spaced apart from each other in the X-axis direction of FIG. 22. In addition, a first coupling groove **361b** corresponding to each of the first coupling parts **361a** may be provided. For example, the first coupling groove **361b** may be defined by recessing the vertical wall **361**, and the first coupling part **361a** may be provided in the recessed portion of the first coupling groove **361b**.

[0305] In addition, the vertical wall **361** may further include a plurality of second coupling parts **361c**. The plurality of second coupling parts **361c** may be provided on the vertical walls **361** that are spaced apart from each other in the X-axis direction. In detail, the plurality of second coupling parts **361c** may be disposed closer to the first coupling parts **361a** than the third coupling parts **363a**, which will be described later. This is done for preventing the interference with the extension **403** of the second tray supporter **400** when being coupled to a second tray supporter **400** that will be described later. For example, the vertical wall **361** in which the plurality of second coupling parts **361c** are disposed may further include a second coupling groove **361d** defined by spacing portions except for the second coupling parts **361c** apart from each other. The curved wall **363** may be provided with a plurality of third coupling parts **363a** to be coupled to the second tray **380** and the second tray supporter **400**. For example, the plurality of third coupling parts **363a** may be spaced apart from each other in the X-axis direction of FIG. 22. The curved wall **363** may be provided with a third coupling groove **363b** corre-

sponding to each of the third coupling parts **363a**. For example, the third coupling groove **363b** may be defined by vertically recessing the curved wall **363**, and the third coupling part **363a** may be provided in the recessed portion of the third coupling groove **363b**.

[0306] FIG. 24 is a perspective view illustrating an upper portion of a second tray supporter, and FIG. 25 is a perspective view illustrating a lower portion of the second tray supporter. FIG. 26 is a cross-sectional view taken along line 26-26 of FIG. 24.

[0307] Referring to FIGS. 24 to 26, the second tray supporter **400** may include a support body **407** on which a lower portion of the second tray **380** is seated. The support body **407** may include an accommodation space **406a** in which a portion of the second tray **380** is accommodated. The accommodation space **406a** may be defined corresponding to the first portion **382** of the second tray **380**, and a plurality of accommodation spaces **406a** may be provided.

[0308] The support body **407** may include a lower opening **406b** (or a through-hole) through which a portion of the second pusher **540** passes. For example, three lower openings **406b** may be provided in the support body **407** to correspond to the three accommodation spaces **406a**. Also, a portion of the lower portion of the second tray **380** may be exposed by the lower opening **406b**. At least a portion of the second tray **380** may be disposed in the lower opening **406b**.

[0309] A top surface **407a** of the support body **407** may extend in the horizontal direction. The second tray supporter **400** may include a top surface **407a** of the support body **407** and a stepped lower plate **401**. The lower plate **401** may be disposed at a position higher than that of the top surface **407a** of the support body **407**.

[0310] The lower plate **401** may include a plurality of coupling parts **401a**, **401b**, and **401c** to be coupled to the second tray cover **360**. The second tray **380** may be inserted and coupled between the second tray cover **360** and the second tray supporter **400**. For example, the second tray **380** may be disposed below the second tray cover **360**, and the second tray **380** may be accommodated above the second tray supporter **400**. Also, the first extension wall **387b** of the second tray **380** may be coupled to the coupling parts **361a**, **361b**, and **361c** of the second tray cover **360** and the coupling parts **400a**, **401b**, and **401c** of the second tray supporter **400**. The plurality of first coupling parts **401a** may be spaced apart from each other in the X-axis direction. Also, the first coupling part **401a** and the second and third coupling parts **401b** and **401c** may be spaced apart from each other in the Y-axis direction. The third coupling part **401c** may be disposed farther from the first coupling part **401a** than the second coupling part **401b**.

[0311] The second tray supporter **400** may further include a vertical extension wall **405** extending vertically downward from an edge of the lower plate **401**. One surface of the vertical extension wall **405** may be provided with a pair of extension parts **403** coupled to the shaft **440** to allow the second tray **380** to rotate.

[0312] The pair of extension parts **403** may be spaced apart from each other in the X-axis direction. Also, each of the extension parts **403** may further include a through-hole **404**. The shaft **440** may pass through the through-hole **404**, and the extension part **281** of the first tray cover **300** may be disposed inside the pair of extension parts **403**. Also, the

through-hole **404** may further include a central portion **404a** and an extension hole **404b** extending symmetrically to the central portion **404a**.

[0313] The second tray supporter **400** may further include a spring coupling part **402a** to which a spring **402** is coupled. The spring coupling part **402a** may provide a ring to be hooked with a lower end of the spring **402**. Also, one of the walls spaced apart from and facing each other in the X-axis direction of the vertical extension wall **405** is provided with a guide hole **408** guiding the transparent ice heater **430** to be described later or the wire connected to the transparent ice heater **430**.

[0314] The second tray supporter **400** may further include a link connection part **405a** to which the pusher link **500** is coupled. For example, the link connection part **405a** may protrude from the vertical extension wall **405** in the X-axis direction. The link connection part **405a** may be disposed on an area between the center line CL1 and the through-hole **404** with respect to FIG. 26. Also, the lower plate **401** may further include a plurality of second heater coupling parts **409** coupled to the second heater case **420**. The plurality of second heater coupling parts **409** may be arranged to be spaced apart from each other in the X-axis direction and/or the Y-axis direction.

[0315] Referring to FIG. 26, the second tray supporter **400** may include a first portion **411** supporting the second tray **380** defining at least a portion of the ice making cell **320a**. In FIG. 26, the first portion **411** may be an area between two dotted lines. For example, the support body **407** may define the first portion **411**. The second tray supporter **400** may further include a second portion **413** extending from a predetermined point of the first portion **411**.

[0316] The second portion **413** may reduce transfer of heat, which is transfer from the transparent ice heater **430** to the second tray supporter **400**, to the ice making cell **320a** defined by the first tray **320**. At least a portion of the second portion **413** may extend in a direction away from the first cell **321a** defined by the first tray **320**. The direction away from the ice making cell **320a** may be a horizontal direction passing through a center of the ice making cell. The direction away from the ice making cell **320a** may be a downward direction with respect to a horizontal line passing through the center of the ice making cell.

[0317] The second portion **413** may include a first part **414a** extending in the horizontal direction from the predetermined point and a second part **414b** extending in the same direction as the first part **414a**. The second portion **413** may include a first part **414a** extending in the horizontal direction from the predetermined point, and a third part **414c** extending in a direction different from that of the first part **414a**. The second portion **413** may include a first part **414a** extending in the horizontal direction from the predetermined point, and a second part **414b** and a third part **414c**, which are branched from the first part **414a**.

[0318] A top surface **407a** of the support body **407** may provide, for example, the first part **414a**. The first part **414a** may further include a fourth part **414d** extending in the vertical line direction. The lower plate **401** may provide, for example, the fourth part **414d**. The vertical extension wall **405** may provide, for example, the third part **414c**. A length of the third part **414c** may be greater than that of the second part **414b**. The second part **414b** may extend in the same direction as the first part **414a**. The third part **414c** may extend in a direction different from that of the first part **414a**.

The second portion **413** may be disposed at the same height as the lowermost end of the first cell **321a** or extend up to a lower point.

[0319] The second portion **413** may include a first extension part **413a** and a second extension part **413b** which are disposed opposite to each other with respect to the center line CL1 corresponding to the center line C1 of the ice making cell **320a**. Referring to FIG. 26, the first extension part **413a** may be disposed at a left side with respect to the center line CL1, and the second extension part **413b** may be disposed at a right side with respect to the center line CL1.

[0320] The first extension part **413a** and the second extension part **413b** may have different shapes with respect to the center line CL1. The first extension part **413a** and the second extension part **413b** may have shapes that are asymmetrical to each other with respect to the center line CL1. A length of the second extension part **413b** may be greater than that of the first extension part **413a** in the horizontal direction. That is, a length of the thermal conductivity of the second extension **413b** is greater than that of the first extension part **413a**.

[0321] The first extension part **413a** may be disposed closer to an edge part that is disposed at a side opposite to the portion of the second wall **222** or the third wall **223** of the bracket **220**, which is connected to the fourth wall **224**, than the second extension part **413b**. The second extension part **413b** may be disposed closer to the shaft **440** that provides a center of rotation of the second tray assembly than the first extension part **413a**.

[0322] In this embodiment, since the length of the second extension part **413b** in the Y-axis direction is greater than that of the first extension part **413a**, the second tray assembly including the second tray **380** contacting the first tray **320** may increase in radius of rotation. A center of curvature of at least a portion of the second extension part **413a** may coincide with a center of rotation of the shaft **440** which is connected to the driver **480** to rotate. The first extension part **413a** may include a portion **414e** extending upwardly with respect to the horizontal line. The portion **414e** may surround, for example, a portion of the second tray **380**.

[0323] In another aspect, the second tray supporter **400** may include a first region **415a** including the lower opening **406b** and a second region **415b** having a shape corresponding to the ice making cell **320a** to support the second tray **380**. For example, the first region **415a** and the second region **415b** may be divided vertically. In FIG. 26, for example, the first region **415a** and the second region **415b** are divided by a dashed-dotted line that extends in the horizontal direction. The first region **415a** may support the second tray **380**.

[0324] The controller controls the ice maker to allow the second pusher **540** to move from a first point outside the ice making cell **320a** to a second point inside the second tray supporter **400** via the lower opening **406b**.

[0325] A deformation resistance degree of the second tray supporter **400** may be greater than that of the second tray **380**. A restoration degree of the second tray supporter **400** may be less than that of the second tray **380**.

[0326] In another aspect, the second tray supporter **400** includes a first region **415a** including a lower opening **406b** and a second region **415b** disposed farther from the transparent ice heater **430** than the first region **415a**.

[0327] The transparent ice heater **430** will be described in detail.

[0328] The controller **800** according to this embodiment may control the transparent ice heater **430** so that heat is supplied to the ice making cell **320a** in at least partial section while cold air is supplied to the ice making cell **320a** to make the transparent ice.

[0329] An ice making rate may be delayed so that bubbles dissolved in water within the ice making cell **320a** may move from a portion at which ice is made toward liquid water by the heat of the transparent ice heater **430**, thereby making transparent ice in the ice maker **200**. That is, the bubbles dissolved in water may be induced to escape to the outside of the ice making cell **320a** or to be collected into a predetermined position in the ice making cell **320a**.

[0330] When a cold air supply part **900** to be described later supplies cold air to the ice making cell **320a**, if the ice making rate is high, the bubbles dissolved in the water inside the ice making cell **320a** may be frozen without moving from the portion at which the ice is made to the liquid water, and thus, transparency of the ice may be reduced.

[0331] On the contrary, when the cold air supply part **900** supplies the cold air to the ice making cell **320a**, if the ice making rate is low, the above limitation may be solved to increase in transparency of the ice. However, there is a limitation in which a making time increases.

[0332] Accordingly, the transparent ice heater **430** may be disposed at one side of the ice making cell **320a** so that the heater locally supplies heat to the ice making cell **320a**, thereby increasing in transparency of the made ice while reducing the ice making time.

[0333] When the transparent ice heater **430** is disposed on one side of the ice making cell **320a**, the transparent ice heater **430** may be made of a material having thermal conductivity less than that of the metal to prevent heat of the transparent ice heater **430** from being easily transferred to the other side of the ice making cell **320a**.

[0334] Alternatively, at least one of the first tray **320** and the second tray **380** may be made of a resin including plastic so that the ice attached to the trays **320** and **380** is separated in the ice making process.

[0335] At least one of the first tray **320** or the second tray **380** may be made of a flexible or soft material so that the tray deformed by the pushers **260** and **540** is easily restored to its original shape in the ice separation process.

[0336] The transparent ice heater **430** may be disposed at a position adjacent to the second tray **380**. The transparent ice heater **430** may be, for example, a wire type heater. For example, the transparent ice heater **430** may be installed to contact the second tray **380** or may be disposed at a position spaced a predetermined distance from the second tray **380**. For another example, the second heater case **420** may not be separately provided, but the transparent heater **430** may be installed on the second tray supporter **400**. In some cases, the transparent ice heater **430** may supply heat to the second tray **380**, and the heat supplied to the second tray **380** may be transferred to the ice making cell **320a**.

[0337] FIG. 27 is a view of the first pusher according to an embodiment, wherein FIG. 27, view (a) is a perspective view of the first pusher, and FIG. 27, view (b) is a side view of the first pusher.

[0338] Referring to FIG. 27, the first pusher **260** may include a pushing bar **264**. The pushing bar **264** may include a first edge **264a** on which a pressing surface pressing ice or a tray in the ice separation process is disposed and a second

edge **264b** disposed at a side opposite to the first edge **264a**. For example, the pressing surface may be flat or curved surface.

[0339] The pushing bar **264** may extend in the vertical direction and may be provided in a straight line shape or a curved shape in which at least a portion of the pushing bar **264** is rounded. A diameter of the pushing bar **264** is less than that of the opening **324** of the first tray **320**. Accordingly, the pushing bar **264** may be inserted into the ice making cell **320a** through the opening **324**. Thus, the first pusher **260** may be referred to as a through pusher passing through the ice making cell **320a**.

[0340] When the ice maker includes a plurality of ice making cells **320a**, the first pusher **260** may include a plurality of pushing bars **264**. Two adjacent pushing bars **264** may be connected to each other by the connection part **263**. The connection part **263** may connect upper ends of the pushing bars **264** to each other. Thus, the second edge **264a** and the connection part **263** may be prevented from interfering with the first tray **320** while the pushing bar **264** is inserted into the ice making cell **320a**.

[0341] The first pusher **260** may include a guide connection part **265** passing through the guide slot **302**. For example, the guide connection part **265** may be provided at each of both sides of the first pusher **260**. A vertical cross-section of the guide connection part **265** may have a circular, oval, or polygonal shape. The guide connection part **265** may be disposed in the guide slot **302**. The guide connection part **265** may move in a longitudinal direction along the guide slot **302** in a state of being disposed in the guide slot **302**. For example, the guide connection part **265** may move in the vertical direction. Although the guide slot **302** has been described as being provided in the first tray cover **300**, it may be alternatively provided in the wall defining the bracket **220** or the storage chamber.

[0342] The guide connection part **265** may further include a link connection part **266** to be coupled to the pusher link **500**. The link connection part **266** may be disposed at a position lower than that of the second edge **264b**. The link connection part **266** may be provided in a cylindrical shape so that the link connection part **266** rotates in the state in which the link connection part **266** is coupled to the pusher link **500**.

[0343] FIG. 28 is a view illustrating a state in which the first pusher is connected to the second tray assembly by a pusher link, and FIG. 29 is a perspective view illustrating the rotation arm of the present invention.

[0344] Referring to FIGS. 28 and 29, the pusher link **500** may connect the first pusher **500** to the second tray assembly. For example, the pusher link **500** may be connected to the first pusher **260** and the second tray case.

[0345] The pusher link **500** may include a link body **502**. The link body **502** may have a rounded shape. As the link body **502** is provided in a round shape, the pusher link **500** may allow the first pusher **260** to rotate and also to vertically move while the second tray assembly rotates.

[0346] The pusher link **500** may include a first connection part **504** provided at one end of the link body **502** and a second connection part **506** provided at the other end of the link body **502**. The first connection part **504** may include a first coupling hole **504a** to which the link connection part **266** is coupled. The link connection part **266** may be connected to the first connection part **504** after passing through the guide slot **302**. The second connection part **506**

may be coupled to the second tray supporter **400**. The second connection part **506** may include a second coupling hole **506a** to which the link connection part **405a** provided on the second tray supporter **400** is coupled. The second connection part **504** may be connected to the second tray supporter **400** at a position spaced apart from the rotation center **C4** of the shaft **440** or the rotation center **C4** of the second tray assembly. Therefore, according to this embodiment, the pusher link **500** connected to the second tray assembly rotates together by the rotation of the second tray assembly. While the pusher link **500** rotates, the first pusher **260** connected to the pusher link **500** moves vertically along the guide slot **302**. The pusher link **502** may serve to convert rotational force of the second tray assembly into vertical movement force of the first pusher **260**. Accordingly, the first pusher **260** may also be referred to as a movable pusher.

[0347] The rotation arm **460** may pass through the through-hole **404** of the extension part **403**. For example, the rotation arm **460** may be connected to the shaft **440** in the through-hole **404**.

[0348] A hook hole **562** hooked with the spring **402** may be formed in one side of the rotation arm **460**. The other side of the rotary arm **560** may be provided with a coupling part coupled to the extension part **403** of the second tray supporter **400**.

[0349] The rotation arm **460** may be connected to a first point of the second tray assembly, and a spring **402** connected to the rotation arm **460** may be connected to a second point of the second tray assembly.

[0350] The coupling part may include a cylindrical first protrusion **463**. The first protrusion **463** may be coupled to a central portion **404a** of the through-hole **404**. The shaft **440** may be coupled to the first protrusion **463**. The coupling part may include a plurality or a pair of second protrusions **464** protruding in a radial direction of the first protrusion **463**. The second protrusion **464** may be disposed in an extension hole **404b** of the through-hole **404**.

[0351] The extension hole **404b** in the circumferential direction based on a rotation center **C4** of the shaft **440** may be greater than a length of the second protrusion **464** so that the second tray supporter **400** and the rotation arm **460** relatively rotate with each other in a predetermined angle range. Thus, in the state in which the second protrusion **464** is disposed in the extension hole **404b**, the second tray supporter **400** and the rotation arm **460** may relatively rotate with respect to each other in a range of a difference in the length of the second protrusion **464** in the circumferential direction and the length of the extension hole **404**.

[0352] According to this structure, in the state in which the first tray **320** and the second tray **380** are in contact with each other at the ice making position, the second tray **380** may be limited in reverse rotation, but the rotation arm **460** may relatively rotate with respect to the second tray supporter **400**. Thus, in the state in which the first tray **320** and the second tray **380** are in contact with each other at the ice making position so that the second tray **380** is stopped, the driver **480** may additionally rotate, and the rotation arm **460** may rotate at a predetermined angle by the additional rotation of the driver **480**. That is, the rotation arm **460** may additionally rotate in the reverse direction. Since the spring **402** is connected to the rotation arm **460**, elastic force of the spring **402** increases by the additional rotation of the rotation arm **402** in the reverse direction.

[0353] Since the increasing elastic force acts as the second tray 380, adhesion between the second tray 380 and the first tray 320 is improved. When the adhesion between the second tray 380 and the first tray 320 is improved, water leakage between the first tray 320 and the second tray 380 is prevented. Thus, according to this embodiment, the rotation angle of the driver 480 is greater than the rotation angle of the second tray assembly. For example, at the ice making position, the driver 480 may additionally rotate in the reverse direction by, for example, 15 degrees.

[0354] At the ice making position, since the rotation arm 460 stops at the additionally rotating position of the rotation arm 460, when the driver 480 rotates to separate ice, the rotational force of the rotation arm 460 is not transmitted to the second tray supporter 400 until the rotation arm 460 rotates at a predetermined angle in the forward direction.

[0355] When the rotation arm 460 rotates at the predetermined angle in the forward direction, the second protrusion 464 may be in contact with one end of the extension hole 404b, and thus, the rotation arm 460 and the second tray supporter 400 may rotate together.

[0356] The power of the driver 480 is transmitted to the second tray supporter 400 by the rotation arm 460 until moving to the ice separation position.

[0357] In order to move from the ice separation position to the water supply position, the driver 480 rotates in the reverse direction. When the driver 480 rotates in the reverse direction, the second tray supporter 400 is in a state of being capable of rotating in the reverse direction, and the second tray supporter 400 rotates in the reverse direction by the power of the driver 480 and/or the elastic force of the spring 402 connected to the rotation arm 460.

[0358] FIG. 30 is a perspective view of the second pusher according to an embodiment.

[0359] Referring to FIG. 30, the second pusher 540 according to this embodiment may include a pushing bar 544. The pushing bar 544 may include a first edge 544a on which a pressing surface pressing the second tray 380 is disposed and a second edge 544b disposed at a side opposite to the first edge 544a.

[0360] The pushing bar 544 may have a curved shape to increase in time taken to press the second tray 380 without interfering with the second tray 380 that rotates in the ice separation process. The first edge 544a may be a plane and include a vertical surface or an inclined surface. The second edge 544b may be coupled to the fourth wall 224 of the bracket 220, or the second edge 544b may be coupled to the fourth wall 224 of the bracket 220 by the coupling plate 542. The coupling plate 542 may be seated in the mounting groove 224a defined in the fourth wall 224 of the bracket 220.

[0361] When the ice maker 200 includes the plurality of ice making cells 320a, the second pusher 540 may include a plurality of pushing bars 544. The plurality of pushing bars 544 may be connected to the coupling plate 542 while being spaced apart from each other in the horizontal direction. The plurality of pushing bars 544 may be integrally formed with the coupling plate 542 or coupled to the coupling plate 542. The first edge 544a may be disposed to be inclined with respect to the center line C1 of the ice making cell 320a. The first edge 544a may be inclined in a direction away from the center line C1 of the ice making cell 320a from an upper end toward a lower end. An angle of the inclined surface defined

by the first edge 544a with respect to the vertical line may be less than that of the inclined surface defined by the second edge 544b.

[0362] The direction in which the pushing bar 544 extends from the center of the first edge 544a toward the center of the second edge 544a may include at least two directions. For example, the pushing bar 544 may include a first portion extending in a first direction and a second portion extending in a direction different from the second portion. At least a portion of the line connecting the center of the second edge 544a to the center of the first edge 544a along the pushing bar 544 may be curved. The first edge 544a and the second edge 544b may have different heights. The first edge 544a may be disposed to be inclined with respect to the second edge 544b.

[0363] FIG. 31 is a cross-sectional view taken along line 31-31 of FIG. 2.

[0364] Referring to FIG. 31, the ice maker 200 may include a first tray assembly 201 and a second tray assembly 211, which are connected to each other.

[0365] The second tray assembly 211 may include a first portion 212 defining at least a portion of the ice making cell 320a and a second portion 213 extending from a predetermined point of the first portion 212. The second portion 213 may reduce transfer of heat from the transparent ice heater 430 to the ice making cell 320a defined by the first tray assembly 201. The first portion 212 may be an area disposed between two dotted lines in FIG. 31.

[0366] The predetermined point of the first portion 212 may be an end of the first portion 212 or a point at which the first tray assembly 201 and the second tray assembly 211 meet each other. At least a portion of the first portion 212 may extend in a direction away from the ice making cell 320a defined by the first tray assembly 201. At least two portions of the second portion 213 may be branched to reduce heat transfer in the direction extending to the second portion 213. A portion of the second portion 213 may extend in the horizontal direction passing through the center of the ice making cell 320a. A portion of the second portion 213 may extend in an upward direction with respect to a horizontal line passing through the center of the ice making chamber 320a.

[0367] The second portion 213 includes a first part 213c extending in the horizontal direction passing through the center of the ice making cell 320a, a second part 213d extending upward with respect to the horizontal line passing through the center of the ice making cell 320a, a third part extending downward.

[0368] The first portion 212 may have different heat transfer in a direction along the outer circumferential surface of the ice making cell 320a to reduce transfer of heat, which is transferred from the transparent ice heater 430 to the second tray assembly 211, to the ice making cell 320a defined by the first tray assembly 201. The transparent ice heater 430 may be disposed to heat both sides with respect to the lowermost end of the first portion 212.

[0369] The first portion 212 may include a first region 214a and a second region 214b. In FIG. 31, the first region 214a and the second region 214b are divided by a dashed-dotted line that extends in the horizontal direction. The second region 214b may be a region defined above the first region 214a. The heat transfer rate of the second region 214b may be greater than that of the first region 214a.

[0370] The first region 214a may include a portion at which the transparent ice heater 430 is disposed. That is, the transparent ice heater 430 may be disposed in the first region 214a. The lowermost end 214a1 of the ice making cell 320a in the first region 214a may have a heat transfer rate less than that of the other portion of the first region 214a. The second region 214b may include a portion in which the first tray assembly 201 and the second tray assembly 211 contact each other. The first region 214a may provide a portion of the ice making cell 320a. The second region 214b may provide the other portion of the ice making cell 320a. The second region 214b may be disposed farther from the transparent ice heater 430 than the first region 214a.

[0371] A portion of the first region 214a may have the heat transfer degree less than that of the other part of the first region 214a to reduce transfer of heat, which is transferred from the transparent ice heater 430 to the first region 214a, to the ice making cell 320a defined by the second region 214b. To make ice in the direction from the ice making cell 320a defined by the first region 214a to the ice making cell 320a defined by the second region 214b, a portion of the first region 214a may have a deformation resistance degree less than that of the other portion of the first region 214a and a restoration degree greater than that of the other portion of the first region 214a.

[0372] A portion of the first region 214a may be thinner than the other portion of the first region 214a in the thickness direction from the center of the ice making cell 320a to the outer circumferential surface direction of the ice making cell 320a. For example, the first region 214a may include a second tray case surrounding at least a portion of the second tray 380 and at least a portion of the second tray 380.

[0373] An average cross-sectional area or average thickness of the first tray assembly 201 may be greater than that of the second tray assembly 211 with respect to the Y-Z cutting surface. A maximum cross-sectional area or maximum thickness of the first tray assembly 201 may be greater than that of the second tray assembly 211 with respect to the Y-Z cutting surface. A minimum cross-sectional area or minimum thickness of the first tray assembly 201 may be greater than that of the second tray assembly 211 with respect to the Y-Z cutting surface. Based on the Y-Z cut plane, the uniformity of the minimum cross-sectional area or the minimum thickness of the first tray assembly 201 may be greater than the uniformity of the minimum cross-sectional area or the minimum thickness of the second tray assembly 211.

[0374] The rotation center C4 may be eccentric with respect to a line bisecting the length in the Y-axis direction of the bracket 220. Also, the ice making cell 320a may be eccentric with respect to a line bisecting a length in the Y-axis direction of the bracket 200. The rotation center C4 may be disposed closer to the second pusher 540 than to the ice making cell 320a.

[0375] The second portion 213 may include a first extension part 213a and a second extension part 213b, which are disposed at sides opposite to each other with respect to the central line C1. The first extension part 213a may be disposed at a left side of the center line C1 in FIG. 31, and the second extension part 213b may be disposed at a right side of the center line C1.

[0376] The water supply part 240 may be disposed close to the first extension part 213a. The first tray assembly 301 may include a pair of guide slots 302, and the water supply

part 240 may be disposed in a region between the pair of guide slots 302. A length of the guide slot 320 may be greater than the sum of a radius of the ice making cell 320a and a height of the auxiliary storage chamber 325.

[0377] FIG. 32 is a control block diagram of the refrigerator according to an embodiment of the present invention.

[0378] Referring to FIG. 32, the refrigerator according to this embodiment may include a cooler supplying the cold to the freezing compartment 32 (or the ice making cell).

[0379] In FIG. 32, for example, the cooler includes a cold air supply part 900. The cold air supply part 900 may supply cold air to the freezing compartment 32 using a refrigerant cycle. For example, the cold air supply part 900 may include a compressor compressing the refrigerant. A temperature of the cold air supplied to the freezing compartment 32 may vary according to the output (or frequency) of the compressor. Alternatively, the cold air supply part 900 may include a fan blowing air to an evaporator. An amount of cold air supplied to the freezing compartment 32 may vary according to the output (or rotation rate) of the fan. Alternatively, the cold air supply part 900 may include a refrigerant valve controlling an amount of refrigerant flowing through the refrigerant cycle. An amount of refrigerant flowing through the refrigerant cycle may vary by adjusting an opening degree by the refrigerant valve, and thus, the temperature of the cold air supplied to the freezing compartment 32 may vary. Therefore, in this embodiment, the cold air supply part 900 may include one or more of the compressor, the fan, and the refrigerant valve. The cold air supply part 900 may further include the evaporator exchanging heat between the refrigerant and the air. The cold air heat-exchanged with the evaporator may be supplied to the ice maker 200.

[0380] The refrigerator according to this embodiment may further include a controller 800 that controls the cold air supply part 900. Also, the refrigerator may further include a water supply valve 242 controlling an amount of water supplied through the water supply part 240.

[0381] The controller 800 may control a portion or all of the ice separation heater 290, the transparent ice heater 430, the driver 480, the cold air supply part 900, and the water supply valve 242.

[0382] In this embodiment, when the ice maker 200 includes both the ice separation heater 290 and the transparent ice heater 430, an output of the ice separation heater 290 and an output of the transparent ice heater 430 may be different from each other. When the outputs of the ice separation heater 290 and the transparent ice heater 430 are different from each other, an output terminal of the ice separation heater 290 and an output terminal of the transparent ice heater 430 may be provided in different shapes, incorrect connection of the two output terminals may be prevented. Although not limited, the output of the ice separation heater 290 may be set larger than that of the transparent ice heater 430. Accordingly, ice may be quickly separated from the first tray 320 by the ice separation heater 290. In this embodiment, when the ice separation heater 290 is not provided, the transparent ice heater 430 may be disposed at a position adjacent to the second tray 380 described above or be disposed at a position adjacent to the first tray 320.

[0383] The refrigerator may further include a first temperature sensor 33 (or an internal temperature sensor) that senses a temperature of the freezing compartment 32. The controller 800 may control the cold air supply part 900 based

on the temperature sensed by the first temperature sensor 33. Also, the controller 800 may determine whether ice making is completed based on the temperature sensed by the second temperature sensor 700.

[0384] FIG. 33 is a flowchart for explaining a process of making ice in the ice maker according to an embodiment. FIG. 34 is a view for explaining a height reference depending on a relative position of the transparent heater with respect to the ice making cell, and FIG. 35 is a view for explaining an output of the transparent heater per unit height of water within the ice making cell. FIG. 36 is a cross-sectional view illustrating a position relationship between the first tray assembly and the second tray assembly at a water supply position, and FIG. 37 is a view illustrating a state in which supply of water is completed in FIG. 36.

[0385] FIG. 38 is a cross-sectional view illustrating a position relationship between the first tray assembly and a second tray assembly at the ice making position, and FIG. 39 is a view illustrating a state in which the pressing part of the second tray is deformed in a state in which ice making is completed. FIG. 40 is a cross-sectional view illustrating a position relationship between the first tray assembly and the second tray assembly in the ice separation process, and FIG. 41 is a cross-sectional view illustrating a position relationship between the first tray assembly and the second tray assembly at the ice separation position.

[0386] Referring to FIGS. 33 to 41, to make ice in the ice maker 200, the controller 800 moves the second tray assembly 211 to a water supply position (S1). In this specification, a direction in which the second tray assembly 211 moves from the ice making position of FIG. 38 to the ice separation position of FIG. 41 may be referred to as forward movement (or forward rotation). On the other hand, the direction from the ice separation position of FIG. 41 to the water supply position of FIG. 36 may be referred to as reverse movement (or reverse rotation).

[0387] The movement to the water supply position of the second tray assembly 211 is detected by a sensor, and when it is detected that the second tray assembly 211 moves to the water supply position, the controller 800 stops the driver 480. At least a portion of the second tray 380 may be spaced apart from the first tray 320 at the water supply position of the second tray assembly 211.

[0388] At the water supply position of the second tray assembly 211, the first tray assembly 201 and the second tray assembly 211 define a first angle $\theta 1$ with respect to the rotation center C4. That is, the first contact surface 322c of the first tray 320 and the second contact surface 382c of the second tray 380 define a first angle therebetween.

[0389] The water supply starts when the second tray 380 moves to the water supply position (S2). For the water supply, the controller 800 turns on the water supply valve 242, and when it is determined that a predetermined amount of water is supplied, the controller 800 may turn off the water supply valve 242. For example, in the process of supplying water, when a pulse is outputted from a flow sensor (not shown), and the outputted pulse reaches a reference pulse, it may be determined that a predetermined amount of water is supplied. For the water supply, the controller 800 turns on the water supply valve 242, and when it is determined that a predetermined amount of water is supplied, the controller 800 may turn off the water supply valve 242. For example, in the process of supplying water, when a pulse is outputted from a flow sensor (not shown),

and the outputted pulse reaches a reference pulse, it may be determined that a predetermined amount of water is supplied. Accordingly, leakage of the water, which supplied to the ice making cell 320a, between the first tray assembly 201 and the second tray assembly 211 while the second tray 380 moves from the water supply position to the ice making position may be reduced. Also, it is possible to reduce a phenomenon in which water expanded in the ice making process leaks between the first tray assembly 201 and the second tray assembly 211 and is frozen.

[0390] After the water supply is completed, the controller 800 controls the driver 480 to allow the second tray assembly 211 to move to the ice making position (S3). For example, the controller 800 may control the driver 480 to allow the second tray assembly 211 to move from the water supply position in the reverse direction. When the second tray assembly 211 move in the reverse direction, the second contact surface 382c of the second tray 380 comes close to the first contact surface 322c of the first tray 320. Then, water between the second contact surface 382c of the second tray 380 and the first contact surface 322c of the first tray 320 is divided into each of the plurality of second cells 381a and then is distributed. When the second contact surface 382c of the second tray 380 and the first contact surface 322c of the first tray 320 contact each other, water is filled in the first cell 321a. As described above, when the second contact surface 382c of the second tray 380 contacts the first contact surface 322c of the first tray 320, the leakage of water in the ice making cell 320a may be reduced. The movement to the ice making position of the second tray assembly 211 is detected by a sensor, and when it is detected that the second tray assembly 211 moves to the ice making position, the controller 800 stops the driver 480.

[0391] In the state in which the second tray assembly 211 moves to the ice making position, ice making is started (S4).

[0392] At the ice making position of the second tray assembly 211, the second portion 383 of the second tray 380 may face the second portion 323 of the first tray 320. At least a portion of each of the second portion 383 of the second tray 380 and the second portion 323 of the first tray 320 may extend in a horizontal direction passing through the center of the ice making cell 320a. At least a portion of each of the second portion 383 of the second tray 380 and the second portion 323 of the first tray 320 is disposed at the same height or higher than the uppermost end of the ice making cell 320a. At least a portion of each of the second portion 383 of the second tray 380 and the second portion 323 of the first tray 320 may be lower than the uppermost end of the auxiliary storage chamber 325. At the ice making position of the second tray assembly 211, the second portion 383 of the second tray 380 may be spaced apart from the second portion 323 of the first tray 320. The space may extend to a portion having a height equal to or greater than the uppermost end of the ice making cell 320a defined by the first portion 322 of the first tray 320. The space may extend to a point lower than the uppermost end of the auxiliary storage chamber 325.

[0393] The ice separation heater 290 provides heat to reduce freezing of water in the space between the second portion 383 of the second tray 380 and the second portion 323 of the first tray 320.

[0394] As described above, the second portion 383 of the second tray 380 serves as a leakage prevention part. It is advantageous that a length of the leakage prevention part is

provided as long as possible. This is because as the length of the leak prevention part increases, an amount of water leaking between the first and second tray assemblies is reduced. A length of the leakage prevention part defined by the second portion 383 may be greater than a distance from the center of the ice making cell 320a to the outer circumferential surface of the ice making cell 320a.

[0395] A second surface facing the first portion 322 of the first tray 320 at the first portion 382 of the second tray 380 may have a surface area greater than that of the first surface facing the first portion 382 of the second tray 380 at the first portion 322 of the first tray 320. Due to a difference in surface area, coupling force between the first tray assembly 201 and the second tray assembly 211 may increase.

[0396] The ice making may be started when the second tray 380 reaches the ice making position. Alternatively, when the second tray 380 reaches the ice making position, and the water supply time elapses, the ice making may be started. When ice making is started, the controller 800 may control the cold air supply part 900 to supply cool air to the ice making cell 320a.

[0397] After the ice making is started, the controller 800 may control the transparent ice heater 430 to be turned on in at least partial sections of the cold air supply part 900 supplying the cold air to the ice making cell 320a. When the transparent ice heater 430 is turned on, since the heat of the transparent ice heater 430 is transferred to the ice making cell 320a, the ice making rate of the ice making cell 320a may be delayed. According to this embodiment, the ice making rate may be delayed so that the bubbles dissolved in the water inside the ice making cell 320a move from the portion at which ice is made toward the liquid water by the heat of the transparent ice heater 430 to make the transparent ice in the ice maker 200.

[0398] In the ice making process, the controller 800 may determine whether the turn-on condition of the transparent ice heater 430 is satisfied (S5). In this embodiment, the transparent ice heater 430 is not turned on immediately after the ice making is started, and the transparent ice heater 430 may be turned on only when the turn-on condition of the transparent ice heater 430 is satisfied (S6).

[0399] Generally, the water supplied to the ice making cell 320a may be water having normal temperature or water having a temperature lower than the normal temperature. The temperature of the water supplied is higher than a freezing point of water. Thus, after the water supply, the temperature of the water is lowered by the cold air, and when the temperature of the water reaches the freezing point of the water, the water is changed into ice.

[0400] In this embodiment, the transparent ice heater 430 may not be turned on until the water is phase-changed into ice. If the transparent ice heater 430 is turned on before the temperature of the water supplied to the ice making cell 320a reaches the freezing point, the speed at which the temperature of the water reaches the freezing point by the heat of the transparent ice heater 430 is slow. As a result, the starting of the ice making may be delayed. The transparency of the ice may vary depending on the presence of the air bubbles in the portion at which ice is made after the ice making is started. If heat is supplied to the ice making cell 320a before the ice is made, the transparent ice heater 430 may operate regardless of the transparency of the ice. Thus, according to this embodiment, after the turn-on condition of the transparent ice heater 430 is satisfied, when the trans-

parent ice heater 430 is turned on, power consumption due to the unnecessary operation of the transparent ice heater 430 may be prevented. Alternatively, even if the transparent ice heater 430 is turned on immediately after the start of ice making, since the transparency is not affected, it is also possible to turn on the transparent ice heater 430 after the start of the ice making.

[0401] In this embodiment, the controller 800 may determine that the turn-on condition of the transparent ice heater 430 is satisfied when a predetermined time elapses from the set specific time point. The specific time point may be set to at least one of the time points before the transparent ice heater 430 is turned on. For example, the specific time point may be set to a time point at which the cold air supply part 900 starts to supply cooling power for the ice making, a time point at which the second tray assembly 211 reaches the ice making position, a time point at which the water supply is completed, and the like. In this embodiment, the controller 800 determines that the turn-on condition of the transparent ice heater 430 is satisfied when a temperature sensed by the second temperature sensor 700 reaches a turn-on reference temperature. For example, the turn-on reference temperature may be a temperature for determining that water starts to freeze at the uppermost side (side of the opening 324) of the ice making cell 320a.

[0402] When a portion of the water is frozen in the ice making cell 320a, the temperature of the ice in the ice making cell 320a is below zero. The temperature of the first tray 320 may be higher than the temperature of the ice in the ice making cell 320a. Alternatively, although water is present in the ice making cell 320a, after the ice starts to be made in the ice making cell 320a, the temperature sensed by the second temperature sensor 700 may be below zero. Thus, to determine that making of ice is started in the ice making cell 320a on the basis of the temperature detected by the second temperature sensor 700, the turn-on reference temperature may be set to the below-zero temperature. That is, when the temperature sensed by the second temperature sensor 700 reaches the turn-on reference temperature, since the turn-on reference temperature is below zero, the ice temperature of the ice making cell 320a is below zero, i.e., lower than the below reference temperature. Therefore, it may be indirectly determined that ice is made in the ice making cell 320a. As described above, when the transparent ice heater 430 is turned on, the heat of the transparent ice heater 430 is transferred into the ice making cell 320a.

[0403] In this embodiment, when the second tray 380 is disposed below the first tray 320, the transparent ice heater 430 is disposed to supply the heat to the second tray 380, the ice may be made from an upper side of the ice making cell 320a.

[0404] In this embodiment, since ice is made from the upper side in the ice making cell 320a, the bubbles move downward from the portion at which the ice is made in the ice making cell 320a toward the liquid water. Since density of water is greater than that of ice, water or bubbles may be convex in the ice making cell 320a, and the bubbles may move to the transparent ice heater 430. In this embodiment, the mass (or volume) per unit height of water in the ice making cell 320a may be the same or different according to the shape of the ice making cell 320a. For example, when the ice making cell 320a is a rectangular parallelepiped, the mass (or volume) per unit height of water in the ice making cell 320a is the same. On the other hand, when the ice

making cell 320a has a shape such as a sphere, an inverted triangle, a crescent moon, etc., the mass (or volume) per unit height of water is different.

[0405] When the cooling power of the cold air supply part 900 is constant, if the heating amount of the transparent ice heater 430 is the same, since the mass per unit height of water in the ice making cell 320a is different, an ice making rate per unit height may be different. For example, if the mass per unit height of water is small, the ice making rate is high, whereas if the mass per unit height of water is high, the ice making rate is slow. As a result, the ice making rate per unit height of water is not constant, and thus, the transparency of the ice may vary according to the unit height. In particular, when ice is made at a high rate, the bubbles may not move from the ice to the water, and the ice may contain the bubbles to lower the transparency. That is, the more the variation in ice making rate per unit height of water decreases, the more the variation in transparency per unit height of made ice may decrease.

[0406] Therefore, in this embodiment, the control part 800 may control the cooling power and/or the heating amount so that the cooling power of the cold air supply part 900 and/or the heating amount of the transparent ice heater 430 is variable according to the mass per unit height of the water of the ice making cell 320a.

[0407] In this specification, the variable of the cooling power of the cold air supply part 900 may include one or more of a variable output of the compressor, a variable output of the fan, and a variable opening degree of the refrigerant valve. Also, in this specification, the variation in the heating amount of the transparent ice heater 430 may represent varying the output of the transparent ice heater 430 or varying the duty of the transparent ice heater 430. In this case, the duty of the transparent ice heater 430 represents a ratio of the turn-on time and a sum of the turn-on time and the turn-off time of the transparent ice heater 430 in one cycle, or a ratio of the turn-off time and a sum of the turn-on time and the turn-off time of the transparent ice heater 430 in one cycle.

[0408] In this specification, a reference of the unit height of water in the ice making cell 320a may vary according to a relative position of the ice making cell 320a and the transparent ice heater 430. For example, as shown in FIG. 34, view (a), the transparent ice heater 430 at the bottom surface of the ice making cell 320a may be disposed to have the same height. In this case, a line connecting the transparent ice heater 430 is a horizontal line, and a line extending in a direction perpendicular to the horizontal line serves as a reference for the unit height of the water of the ice making cell 320a.

[0409] In the case of FIG. 34, view (a), ice is made from the uppermost side of the ice making cell 320a and then is grown. On the other hand, as shown in FIG. 34, view (b), the transparent ice heater 430 at the bottom surface of the ice making cell 320a may be disposed to have different heights. In this case, since heat is supplied to the ice making cell 320a at different heights of the ice making cell 320a, ice is made with a pattern different from that of FIG. 34, view (a). For example, in FIG. 34, view (b), ice may be made at a position spaced apart from the uppermost end to the left side of the ice making cell 320a, and the ice may be grown to a right lower side at which the transparent ice heater 430 is disposed.

[0410] Accordingly, in FIG. 34, view (b), a line (reference line) perpendicular to the line connecting two points of the transparent ice heater 430 serves as a reference for the unit height of water of the ice making cell 320a. The reference line of FIG. 34, view (b) is inclined at a predetermined angle from the vertical line.

[0411] FIG. 35 illustrates a unit height division of water and an output amount of transparent ice heater per unit height when the transparent ice heater is disposed as shown in FIG. 34, view (a).

[0412] Hereinafter, an example of controlling an output of the transparent ice heater so that the ice making rate is constant for each unit height of water will be described.

[0413] Referring to FIG. 35, when the ice making cell 320a is formed, for example, in a spherical shape, the mass per unit height of water in the ice making cell 320a increases from the upper side to the lower side to reach the maximum and then decreases again. For example, the water (or the ice making cell itself) in the spherical ice making cell 320a having a diameter of about 50 mm is divided into nine sections (section A to section I) by 6 mm height (unit height). Here, it is noted that there is no limitation on the size of the unit height and the number of divided sections.

[0414] When the water in the ice making cell 320a is divided into unit heights, the height of each section to be divided is equal to the section A to the section H, and the section I is lower than the remaining sections. Alternatively, the unit heights of all divided sections may be the same depending on the diameter of the ice making cell 320a and the number of divided sections. Among the many sections, the section E is a section in which the mass of unit height of water is maximum. For example, in the section in which the mass per unit height of water is maximum, when the ice making cell 320a has spherical shape, a diameter of the ice making cell 320a, a horizontal cross-sectional area of the ice making cell 320a, or a circumference of the ice may be maximum.

[0415] As described above, when assuming that the cooling power of the cold air supply part 900 is constant, and the output of the transparent ice heater 430 is constant, the ice making rate in section E is the lowest, the ice making rate in the sections A and I is the fastest.

[0416] In this case, since the ice making rate varies for the height, the transparency of the ice may vary for the height. In a specific section, the ice making rate may be too fast to contain bubbles, thereby lowering the transparency. Therefore, in this embodiment, the output of the transparent ice heater 430 may be controlled so that the ice making rate for each unit height is the same or similar while the bubbles move from the portion at which ice is made to the water in the ice making process.

[0417] Specifically, since the mass of the section E is the largest, the output W5 of the transparent ice heater 430 in the section E may be set to a minimum value. Since the volume of the section D is less than that of the section E, the volume of the ice may be reduced as the volume decreases, and thus it is necessary to delay the ice making rate. Thus, an output W6 of the transparent ice heater 430 in the section D may be set to a value greater than an output W5 of the transparent ice heater 430 in the section E.

[0418] Since the volume in the section C is less than that in the section D by the same reason, an output W3 of the transparent ice heater 430 in the section C may be set to a value greater than the output W4 of the transparent ice heater

430 in the section D. Since the volume in the section B is less than that in the section C, an output **W2** of the transparent ice heater **430** in the section B may be set to a value greater than the output **W3** of the transparent ice heater **430** in the section C. Since the volume in the section A is less than that in the section B, an output **W1** of the transparent ice heater **430** in the section A may be set to a value greater than the output **W2** of the transparent ice heater **430** in the section B.

[0419] For the same reason, since the mass per unit height decreases toward the lower side in the section E, the output of the transparent ice heater **430** may increase as the lower side in the section E (see **W6**, **W7**, **W8**, and **W9**). Thus, according to an output variation pattern of the transparent ice heater **430**, the output of the transparent ice heater **430** is gradually reduced from the first section to the intermediate section after the transparent ice heater **430** is initially turned on.

[0420] The output of the transparent ice heater **430** may be minimum in the intermediate section in which the mass of unit height of water is minimum. The output of the transparent ice heater **430** may again increase step by step from the next section of the intermediate section.

[0421] The output of the transparent ice heater **430** in two adjacent sections may be set to be the same according to the type or mass of the made ice. For example, the output of section C and section D may be the same. That is, the output of the transparent ice heater **430** may be the same in at least two sections.

[0422] Alternatively, the output of the transparent ice heater **430** may be set to the minimum in sections other than the section in which the mass per unit height is the smallest. For example, the output of the transparent ice heater **430** in the section D or the section F may be minimum. The output of the transparent ice heater **430** in the section E may be equal to or greater than the minimum output.

[0423] In summary, in this embodiment, the output of the transparent ice heater **430** may have a maximum initial output. In the ice making process, the output of the transparent ice heater **430** may be reduced to the minimum output of the transparent ice heater **430**.

[0424] The output of the transparent ice heater **430** may be gradually reduced in each section, or the output may be maintained in at least two sections. The output of the transparent ice heater **430** may increase from the minimum output to the end output. The end output may be the same as or different from the initial output. In addition, the output of the transparent ice heater **430** may incrementally increase in each section from the minimum output to the end output, or the output may be maintained in at least two sections.

[0425] Alternatively, the output of the transparent ice heater **430** may be an end output in a section before the last section among a plurality of sections. In this case, the output of the transparent ice heater **430** may be maintained as an end output in the last section. That is, after the output of the transparent ice heater **430** becomes the end output, the end output may be maintained until the last section.

[0426] As the ice making is performed, an amount of ice existing in the ice making cell **320a** may decrease. Thus, when the transparent ice heater **430** continues to increase until the output reaches the last section, the heat supplied to the ice making cell **320a** may be reduced. As a result, excessive water may exist in the ice making cell **320a** even after the end of the last section. Therefore, the output of the

transparent ice heater **430** may be maintained as the end output in at least two sections including the last section.

[0427] The transparency of the ice may be uniform for each unit height, and the bubbles may be collected in the lowermost section by the output control of the transparent ice heater **430**. Thus, when viewed on the ice as a whole, the bubbles may be collected in the localized portion, and the remaining portion may become totally transparent.

[0428] As described above, even if the ice making cell **320a** does not have the spherical shape, the transparent ice may be made when the output of the transparent ice heater **430** varies according to the mass for each unit height of water in the ice making cell **320a**.

[0429] The heating amount of the transparent ice heater **430** when the mass for each unit height of water is large may be less than that of the transparent ice heater **430** when the mass for each unit height of water is small. For example, while maintaining the same cooling power of the cold air supply part **900**, the heating amount of the transparent ice heater **430** may vary so as to be inversely proportional to the mass per unit height of water. Also, it is possible to make the transparent ice by varying the cooling power of the cold air supply part **900** according to the mass per unit height of water. For example, when the mass per unit height of water is large, the cold force of the cold air supply part **900** may increase, and when the mass per unit height is small, the cold force of the cold air supply part **900** may decrease. For example, while maintaining a constant heating amount of the transparent ice heater **430**, the cooling power of the cold air supply part **900** may vary to be proportional to the mass per unit height of water.

[0430] Referring to the variable cooling power pattern of the cold air supply part **900** in the case of making the spherical ice, the cooling power of the cold air supply part **900** from the initial section to the intermediate section during the ice making process may increase.

[0431] The cooling power of the cold air supply part **900** may be maximum in the intermediate section in which the mass per unit height of water is maximum. The cooling power of the cold air supply part **900** may be reduced again from the next section of the intermediate section. Alternatively, the transparent ice may be made by varying the cooling power of the cold air supply part **900** and the heating amount of the transparent ice heater **430** according to the mass for each unit height of water. For example, the heating power of the transparent ice heater **430** may vary so that the cooling power of the cold air supply part **900** may be proportional to the mass per unit height of water. The heating power of the transparent ice heater **430** may be inversely proportional to the mass per unit height of water.

[0432] According to this embodiment, when one or more of the cooling power of the cold air supply part **900** and the heating amount of the transparent ice heater **430** are controlled according to the mass per unit height of water, the ice making rate per unit height of water may be substantially the same or may be maintained within a predetermined range.

[0433] As illustrated in FIG. 39, a convex portion **382f** may be deformed in a direction away from the center of the ice making cell **320a** by being pressed by the ice. The lower portion of the ice may have the spherical shape by the deformation of the convex portion **382f**.

[0434] The controller **800** may determine whether the ice making is completed based on the temperature sensed by the second temperature sensor **700** (**S8**). When it is determined

that the ice making is completed, the controller **800** may turn off the transparent ice heater **430** (S9). For example, when the temperature sensed by the second temperature sensor **700** reaches a first reference temperature, the controller **800** may determine that the ice making is completed to turn off the transparent ice heater **430**.

[0435] In this case, since a distance between the second temperature sensor **700** and each ice making cell **320a** is different, in order to determine that the ice making is completed in all the ice making cells **320a**, the controller **800** may perform the ice separation after a certain amount of time, at which it is determined that ice making is completed, has passed or when the temperature sensed by the second temperature sensor **700** reaches a second reference temperature lower than the first reference temperature.

[0436] When the ice making is completed, the controller **800** operates one or more of the ice separation heater **290** and the transparent ice heater **430** (S10).

[0437] When at least one of the ice separation heater **290** or the transparent ice heater **430** is turned on, heat of the heater is transferred to at least one of the first tray **320** or the second tray **380** so that the ice may be separated from the surfaces (inner surfaces) of one or more of the first tray **320** and the second tray **380**. Also, the heat of the heaters **290** and **430** is transferred to the contact surface of the first tray **320** and the second tray **380**, and thus, the first contact surface **322c** of the first tray **320** and the second contact surface **382c** of the second tray **380** may be in a state capable of being separated from each other.

[0438] When at least one of the ice separation heater **290** and the transparent ice heater **430** operate for a predetermined time, or when the temperature sensed by the second temperature sensor **700** is equal to or higher than an off reference temperature, the controller **800** is turned off the heaters **290** and **430**, which are turned on (S10). Although not limited, the turn-off reference temperature may be set to above zero temperature.

[0439] The controller **800** operates the driver **480** to allow the second tray assembly **211** to move in the forward direction (S11).

[0440] As illustrated in FIG. 40, when the second tray **380** move in the forward direction, the second tray **380** is spaced apart from the first tray **320**. The moving force of the second tray **380** is transmitted to the first pusher **260** by the pusher link **500**. Then, the first pusher **260** descends along the guide slot **302**, and the extension part **264** passes through the opening **324** to press the ice in the ice making cell **320a**. In this embodiment, ice may be separated from the first tray **320** before the extension part **264** presses the ice in the ice making process. That is, ice may be separated from the surface of the first tray **320** by the heater that is turned on. In this case, the ice may move together with the second tray **380** while the ice is supported by the second tray **380**. For another example, even when the heat of the heater is applied to the first tray **320**, the ice may not be separated from the surface of the first tray **320**. Therefore, when the second tray assembly **211** moves in the forward direction, there is possibility that the ice is separated from the second tray **380** in a state in which the ice contacts the first tray **320**.

[0441] In this state, in the process of moving the second tray **380**, the extension part **264** passing through the opening **324** may press the ice contacting the first tray **320**, and thus,

the ice may be separated from the tray **320**. The ice separated from the first tray **320** may be supported by the second tray **380** again.

[0442] When the ice moves together with the second tray **380** while the ice is supported by the second tray **380**, the ice may be separated from the tray **250** by its own weight even if no external force is applied to the second tray **380**.

[0443] While the second tray **380** moves, even if the ice does not fall from the second tray **380** by its own weight, when the second pusher **540** contacts the second tray **540** as illustrated in FIGS. 40 and 41 to press the second tray **380**, the ice may be separated from the second tray **380** to fall downward.

[0444] For example, as illustrated in FIG. 40, while the second tray assembly **311** moves in the forward direction, the second tray **380** may contact the extension part **544** of the second pusher **540**. As illustrated in FIG. 40, when the second tray **380** contacts the second pusher **540**, the first tray assembly **201** and the second tray assembly **211** form a second angle $\theta 2$ therebetween with respect to the rotation center C4. That is, the first contact surface **322c** of the first tray **320** and the second contact surface **382c** of the second tray **380** form a second angle therebetween. The second angle may be greater than the first angle and may be close to about 90 degrees.

[0445] When the second tray assembly **211** continuously moves in the forward direction, the extension part **544** may press the second tray **380** to deform the second tray **380** and the extension part **544**. Thus, the pressing force of the extension part **544** may be transferred to the ice so that the ice is separated from the surface of the second tray **380**. The ice separated from the surface of the second tray **380** may drop downward and be stored in the ice bin **600**.

[0446] In this embodiment, as shown in FIG. 41, the position at which the second tray **380** is pressed by the second pusher **540** and deformed may be referred to as an ice separation position. As illustrated in FIG. 41, at the ice separation position of the second tray assembly **211**, the first tray assembly **201** and the second tray assembly **211** may form a third angle $\theta 3$ based on the rotation center C4. That is, the first contact surface **322c** of the first tray **320** and the second contact surface **382c** of the second tray **380** form the third angle $\theta 3$. The third angle $\theta 3$ is greater than the second angle $\theta 2$. For example, the third angle $\theta 3$ is greater than about 90 degrees and less than about 180 degrees.

[0447] At the ice separation position, a distance between a first edge **544a** of the second pusher **540** and a second contact surface **382c** of the second tray **380** may be less than that between the first edge **544a** of the second pusher **540** and the lower opening **406b** of the second tray supporter **400** so that the pressing force of the second pusher **540** increases.

[0448] An attachment degree between the first tray **320** and the ice is greater than that between the second tray **380** and the ice. Thus, a minimum distance between the first edge **264a** of the first pusher **260** and the first contact surface **322c** of the first tray **320** at the ice separation position may be greater than a minimum distance between the second edge **544a** of the second pusher **540** and the second contact surface **382c** of the second tray **380**.

[0449] At the ice separation position, a distance between the first edge **264a** of the first pusher **260** and the line passing through the first contact surface **322c** of the first tray **320** may be greater than 0 and may be less than about $\frac{1}{2}$ of a radius of the ice making cell **320a**. Accordingly, since the

first edge **264a** of the first pusher **260** moves to a position close to the first contact surface **322c** of the first tray **320**, the ice is easily separated from the first tray **320**.

[0450] Whether the ice bin **600** is full may be detected while the second tray assembly **211** moves from the ice making position to the ice separation position. For example, the full ice detection lever **520** rotates together with the second tray assembly **211**, and the rotation of the full ice detection lever **520** is interrupted by ice while the full ice detection lever **520** rotates. In this case, it may be determined that the ice bin **600** is in a full ice state. On the other hand, if the rotation of the full ice detection lever **520** is not interfered with the ice while the full ice detection lever **520** rotates, it may be determined that the ice bin **600** is not in the ice state.

[0451] After the ice is separated from the second tray **380**, the controller **800** controls the driver **480** to allow the second tray assembly **211** to move in the reverse direction (S11). Then, the second tray assembly **211** moves from the ice separation position to the water supply position. When the second tray assembly **211** moves to the water supply position of FIG. 36, the controller **800** stops the driver **480** (S1).

[0452] When the second tray **380** is spaced apart from the extension part **544** while the second tray assembly **211** moves in the reverse direction, the deformed second tray **380** may be restored to its original shape.

[0453] In the reverse movement of the second tray assembly **211**, the moving force of the second tray **380** is transmitted to the first pusher **260** by the pusher link **500**, and thus, the first pusher **260** ascends, and the extension part **264** is removed from the ice making cell **320a**.

[0454] FIG. 42 is a view illustrating an operation of the pusher link when the second tray assembly moves from the ice making position to the ice separation position. FIG. 42, view (a) illustrates the ice making position, FIG. 42, view (b) illustrates the water supply position, FIG. 42, view (c) illustrates the position at which the second tray contacts the second pusher, and FIG. 42, view (d) illustrates the ice separation position.

[0455] Referring to FIG. 42, the pushing bar **264** of the first pusher **260** may include the first edge **264a** and the second edge **264b** as described above. The first pusher **260** may move by receiving power from the driver **480**.

[0456] The controller **800** may control the first edge **264a** so as to be disposed at a different position from the ice making position so that a phenomenon in which water supplied into the ice making cell **320a** at the water supply position is attached to the first pusher **260** and then frozen in the ice making process is reduced.

[0457] In this specification, the control of the position by the controller **800** may be understood as controlling the position by controlling the driver **480**.

[0458] The controller **800** may control the position so that the first edge **264a** is disposed at different positions at the water supply position, the ice making position, and the ice separation position.

[0459] The controller **800** control the first edge **264a** to allow the first edge **264a** to move in the first direction in the process of moving from the ice separation position to the water supply position and to allow the first edge **264a** to additionally move in the first direction in the process of moving from the water supply position to the ice making position. Alternatively, the controller **800** controls the first edge **264a** to allow the first edge **264a** to move in the first

direction in the process of moving from the ice separation position to the water supply position and allow the first edge to move in a second direction different from the first direction in the process of moving from the water supply position to the ice making position.

[0460] For example, the first edge **264a** may move in the first direction by the first slot **302a** of the guide slot **302**, and the second edge **264a** may rotate in a second direction or move in a second direction inclined with the first direction by the second slot **302b**. The first edge **264a** may be disposed at a first point outside the ice making cell **320a** at the ice making position and may be controlled to be disposed at a second point of the ice making cell **320a** during the ice separation process.

[0461] FIG. 42 is a view for explaining a method for controlling the refrigerator when a heat transfer amount between cold air and water vary in the ice making process.

[0462] Referring to FIGS. 32 and 42, cooling power of the cold air supply part **900** may be determined corresponding to the target temperature of the freezing compartment **32**. The cold air generated by the cold air supply part **900** may be supplied to the freezing chamber **32**. The water of the ice making cell **320a** may be phase-changed into ice by heat transfer between the cold water supplied to the freezing chamber **32** and the water of the ice making cell **320a**.

[0463] In this embodiment, a heating amount of the transparent ice heater **430** for each unit height of water may be determined in consideration of predetermined cooling power of the cold air supply part **900**.

[0464] In this embodiment, the heating amount of the transparent ice heater **430** determined in consideration of the predetermined cooling power of the cold air supply part **900** is referred to as a reference heating amount. The magnitude of the reference heating amount per unit height of water is different. However, when the amount of heat transfer between the cold of the freezing compartment **32** and the water in the ice making cell **320a** is variable, if the heating amount of the transparent ice heater **430** is not adjusted to reflect this, the transparency of ice for each unit height varies.

[0465] In this embodiment, the case in which the heat transfer amount between the cold and the water increase may be a case in which the cooling power of the cold air supply part **900** increases or a case in which the air having a temperature lower than the temperature of the cold air in the freezing compartment **32** is supplied to the freezing compartment **32**.

[0466] On the other hand, the case in which the heat transfer amount between the cold and the water decrease may be a case in which the cooling power of the cold air supply part **900** decreases or a case in which the air having a temperature higher than the temperature of the cold air in the freezing compartment **32** is supplied to the freezing compartment **32**.

[0467] For example, a target temperature of the freezing compartment **32** is lowered, an operation mode of the freezing compartment **32** is changed from a normal mode to a rapid cooling mode, an output of at least one of the compressor or the fan increases, or an opening degree increases, the cooling power of the cold air supply part **900** may increase.

[0468] On the other hand, the target temperature of the freezer compartment **32** increases, the operation mode of the freezing compartment **32** is changed from the rapid cooling

mode to the normal mode, the output of at least one of the compressor or the fan decreases, or the opening degree of the refrigerant valve decreases, the cooling power of the cold air supply part 900 may decrease.

[0469] When the cooling power of the cold air supply part 900 increases, the temperature of the cold air around the ice maker 200 is lowered to increase in ice making rate. On the other hand, if the cooling power of the cold air supply part 900 decreases, the temperature of the cold air around the ice maker 200 increases, the ice making rate decreases, and also, the ice making time increases.

[0470] Therefore, in this embodiment, when the amount of heat transfer of cold and water increases so that the ice making rate is maintained within a predetermined range lower than the ice making rate when the ice making is performed with the transparent ice heater 430 that is turned off, the heating amount of transparent ice heater 430 may be controlled to increase.

[0471] On the other hand, when the amount of heat transfer between the cold and the water decreases, the heating amount of transparent ice heater 430 may be controlled to decrease.

[0472] In this embodiment, when the ice making rate is maintained within the predetermined range, the ice making rate is less than the rate at which the bubbles move in the portion at which the ice is made, and no bubbles exist in the portion at which the ice is made.

[0473] When the cooling power of the cold air supply part 900 increases, the heating amount of transparent ice heater 430 may increase. On the other hand, when the cooling power of the cold air supply part 900 decreases, the heating amount of transparent ice heater 430 may decrease.

[0474] Hereinafter, the case in which the target temperature of the freezing compartment 32 varies will be described with an example.

[0475] The controller 800 may control the output of the transparent ice heater 430 so that the ice making rate may be maintained within the predetermined range regardless of the target temperature of the freezing compartment 32.

[0476] For example, the ice making may be started (S4), and a change in heat transfer amount of cold and water may be detected (S31). For example, it may be sensed that the target temperature of the freezing compartment 32 is changed through an input part (not shown).

[0477] The controller 800 may determine whether the heat transfer amount of cold and water increases (S32). For example, the controller 800 may determine whether the target temperature increases.

[0478] As the result of the determination in the process (S32), when the target temperature increases, the controller 800 may decrease the reference heating amount of transparent ice heater 430 that is predetermined in each of the current section and the remaining sections. The variable control of the heating amount of the transparent ice heater 430 may be normally performed until the ice making is completed (S35). On the other hand, if the target temperature decreases, the controller 800 may increase the reference heating amount of transparent ice heater 430 that is predetermined in each of the current section and the remaining sections. The variable control of the heating amount of the transparent ice heater 430 may be normally performed until the ice making is completed (S35).

[0479] In this embodiment, the reference heating amount that increases or decreases may be predetermined and then

stored in a memory. According to this embodiment, the reference heating amount for each section of the transparent ice heater increases or decreases in response to the change in the heat transfer amount of cold and water, and thus, the ice making rate may be maintained within the predetermined range, thereby realizing the uniform transparency for each unit height of the ice.

[0480] FIG. 44 is an exploded perspective view of the driver according to an embodiment of the present invention, and FIG. 45 is a plan view illustrating an internal configuration of the driver. FIG. 46 is a view illustrating a cam and an operation lever of the driver, and FIG. 47 is a view illustrating a position relationship between a hall sensor and a magnet depending on rotation of the cam.

[0481] View (a) of FIG. 47 illustrates a state in which the hall sensor and the magnet are aligned at the first position of a magnet lever, and view (b) of FIG. 47 illustrates a state in which the hall sensor and the magnet are not aligned at the first position of the magnet lever.

[0482] Referring to FIGS. 44 to 47, the driver 480 may include an operation lever 4840 that is organically interlocked by a motor 4822, a cam 4830 rotating by the motor 4822, and a cam surface for the detection lever of the cam 4830. The driver 480 may further include a lever coupling part 4850 that rotates (swings) the full ice detection lever 520 in the left and right direction while rotating by the operation lever 4840. The driver 480 may include a magnet lever 4860, which is organically interlocked along the cam surface for the magnet of the cam 4830, the motor 4822, the cam 4830, the operation lever 4840, and the lever coupling part 4850, and a case in which the magnet lever 4860 is embedded.

[0483] The case may include a first case 4811 in which the motor 4822, the cam 4830, the operation lever 4840, the lever coupling part 4850, and the magnet lever 4860 are embedded, and a second case 4815 that covers the first case 4811. The motor 4822 generates power for rotating the cam 4830.

[0484] The driver 480 may further include a control panel 4821 coupled to an inner side of the first case 4811. The motor 4822 may be connected to the control panel 4821.

[0485] A hall sensor 4823 may be provided on the control panel 4821. The hall sensor 4824 may output a first signal and a second signal according to a position relative to the magnet lever 4860.

[0486] As illustrated in FIG. 46, the cam 4830 may include a coupling part 4831 to which the rotation arm 460 is coupled. The coupling part 4831 serves as a rotation shaft of the cam 4830.

[0487] The cam 4830 may include a gear 4832 to transmit power to the motor 4822. The gear 4832 may be formed on an outer circumferential surface of the cam 4830. The cam 4830 may include a cam surface 4833 for the detection lever and a cam surface 4834 for the magnet. That is, the cam 4830 forms a path through which the levers 4840 and 4860 move. A cam groove 4833a for the detection lever, which rotates the full ice detection lever 520 by lowering the operation lever 4840 is formed in the cam surface 4833 for the detection lever.

[0488] A cam groove 4834a for the magnet, which lowers the magnet lever 4860 so that the magnet lever 4860 and the hall sensor 423 are separated from each other is formed in the cam surface 4834 for the magnet.

[0489] A reduction gear 4870 that reduces rotational force of the motor 4822 to transmit the rotational force to the cam 4830 may be provided between the cam 4830 and the motor 4822. The reduction gear 4870 may include a first reduction gear 4871 connected to the motor 4822 to transmit power, a second reduction gear 4872 engaged with the first reduction gear 4871, and a third reduction gear 4873 connecting the second reduction gear 4872 to the cam 4830 to transmit the power.

[0490] One end of the operation lever 4840 is fitted and coupled to the rotation shaft of the third reduction gear 4873 so as to be freely rotatable, and a gear 4882 formed at the other end of the operation lever 4840 is connected to the lever coupling part 4850 so as to transmit the power. That is, when the operation lever 4840 move, the lever coupling part 4850 rotates.

[0491] The lever coupling part 4850 has one end rotatably connected to the operation lever 4840 inside the case and the other end protruding to the outside of the case so as to be coupled to the full ice detection lever 520.

[0492] The magnet lever 4860 may include a central portion rotatably provided on the case, an end that is organically interlocked along the cam surface 4834 for the magnet of the cam 4830, and a magnet 4861 that is aligned with the hall sensor 4824 or spaced apart from the hall sensor 4823.

[0493] As illustrated in view (a) of FIG. 47, when the magnet 4881 is aligned with the hall sensor 4824, any one of the first signal and the second signal may be output from the hall sensor 4824.

[0494] As illustrated in view (b) of FIG. 47, when the magnet 4881 is out of the position facing the hall sensor 4824, the other signal of the first signal and the second signal is output from the hall sensor 4824.

[0495] A blocking member 4880 that selectively blocks the cam groove 4833a for the detection lever so that the operation lever 4840 moving along the cam surface 4833 for the detection lever is not inserted into the cam groove 4833a for the detection lever when the full ice detection lever 500 returns to its original position may be provided on the rotation shaft of the cam 4830.

[0496] That is, the blocking member 4880 may include a coupling part 4881 rotatably coupled to the rotation shaft of the cam 4830 and a hook groove 4882 formed in one side of the coupling part 4881 and coupled to the protrusion 4813 formed on the bottom surface of the case to restrict a rotation angle of the coupling part 4881.

[0497] Also, the blocking member 4880 may further include a support protrusion 4883 that is provided outside the coupling part 4881 to restrict an operation of the operation lever 4840 so that the operation lever 4840 is not inserted into the cam groove 4833a for the detection lever while being supported on or separated from the operation lever 4840 when the cam gear rotates in the forward or reverse direction.

[0498] Also, the driver 480 may further include an elastic member that provides elastic force so that the lever coupling part 4850 rotates in one direction. One end of the elastic member may be connected to the lever coupling part 4850, and the other end may be fixed to the case.

[0499] A protrusion 4833b may be provided between the cam surface 4833 for the detection lever of the cam 4830 and the cam groove 4833a.

[0500] Since the rotation arm 460 is connected to the cam 4830, the rotation angle of the cam 4830 in the process of moving from the ice making position to the ice separation position or the process of moving from the ice separation position to the ice making position may be the same as that of the second tray assembly.

[0501] However, as described above, due to the relatively rotatable structure of the rotation arm 460 and the second tray supporter 400, in the state in which the second tray assembly moves to the ice making position, the cam 4830 may additionally rotate in a state in which the second tray assembly is stopped.

[0502] Referring to FIG. 38, the ice making position may be a position at which at least a portion of the ice making cell formed by the second tray 380 reaches a reference line passing through the rotation center C4 (rotation center of the driver) of the shaft 440. Referring to FIG. 36, the water supply position may be a position before at least a portion of the ice making cell formed by the second tray 380 reaches the reference line passing through the rotation center C4 of the shaft 440.

[0503] It is assumed that the rotation angle of the cam 4830 is 0 at the ice making position. The cam 4830 may further rotate in the reverse direction due to a difference in length between the second protrusion 463 of the rotation arm 460 and the extension hole 404b of the extension part 403. That is, at the ice making position of the second tray assembly, the cam 4830 may additionally rotate in the reverse direction.

[0504] At the ice making position, the rotation angle of the cam 4830 when the cam 4830 rotates in the reverse direction may be referred to as a negative (−) rotation angle.

[0505] At the ice making position, the rotation angle of the cam 4830 when the cam 4830 rotates in the forward direction toward the water supply position or the ice separation position may be referred to as a positive (+) rotation angle. Hereinafter, in the case of the positive (+) rotation angle, the positive (+) value will be omitted.

[0506] At the ice making position, the cam 4830 may rotate to the water supply position at a first rotation angle. The first rotation angle may be greater than 0 degrees and less than 20 degrees. Preferably, the first rotation angle may be greater than 5 degrees and less than 15 degrees.

[0507] Since the water dropping into the second tray 380 is evenly spread into the plurality of ice making cell 320a by the setting of the water supply position according to the present invention, the overflowing of the water dropping into the second tray 380 may be prevented.

[0508] At the ice making position, the cam 4830 may rotate to the ice making position at a second rotation angle. A rotation angle of the second may be greater than 90 degrees and less than 180 degrees. Preferably, the second rotation angle may be greater than 90 degrees and less than 150 degrees. More preferably, the second rotation angle may be greater than 90 degrees and less than 150 degrees.

[0509] When the second rotation angle is greater than 90 degrees, ice may be easily separated from the second tray 380 while the second tray 380 is pressed by the second pusher 540. As a result, the separated ice may smoothly drop down without being caught on the end of the second tray 380.

[0510] At the ice separation position, the cam 4830 may additionally rotate at a third angle. The cam 4830 may additionally rotate in the forward direction at the third

rotation angle in the state in which the second tray assembly moves to the ice separation position by an assembly tolerance of the cam **4830** and the rotation arm **460**, a difference in rotation angle of the pair of rotation arms due to the cam **4830** being coupled to one of the pair of rotation arms **460**, and the like. When the cam **4830** further rotates in the forward direction, pressing force applied by the second pusher **540** to press the second tray **380** may increase.

[0511] At the ice separation position, the cam **4830** may rotate in the reverse direction, and after the second tray assembly moves to the water supply position, the cam **4830** may further rotate in the reverse direction. The reverse direction may be a direction opposite to the direction of gravity. In consideration of the inertia of the tray assembly and the motor, if the cam further rotates in the direction opposite to the direction of gravity, it is advantageous in controlling the water supply position.

[0512] At the ice making position, the cam **4830** may rotate at a fourth rotation angle in the reverse direction. The fourth rotation angle may be set in a range of 0 degrees and negative (−) 30 degrees. Preferably, the fourth rotation angle may be set in a range of negative (−) 5 degrees and negative (−) 25 degrees. More preferably, the fourth rotation angle may be set in a range of negative (−) 10 degrees and negative (−) 20 degrees.

1. An ice maker comprising:
 - a first tray assembly configured to form a first portion of a cell;
 - a second tray assembly configured to form a second portion of the cell, the first and second portions configured to form a space in which liquid is phase-changed to ice; and
 - a driver configured to move the second tray assembly, wherein the driver includes:
 - a cam, and
 - a plurality of levers,
 wherein the cam includes a plurality of cam surfaces for the plurality of levers.
2. The ice maker of claim 1, wherein the cam further includes a rotation shaft and a gear formed on one surface of the cam.
3. The ice maker of claim 1, wherein the plurality of cam surfaces includes a first cam surface and a second cam surface,
 - the plurality of levers includes a first lever that moves along the first cam surface and a second lever that moves along the second cam surface.
4. The ice maker of claim 3, wherein the first lever is connected to a full ice taction lever.
5. The ice maker of claim 3, wherein the first cam surface includes a cam groove.
6. The ice maker of claim 5, wherein the driver further comprises a blocking member that selectively blocks the cam groove so that the first lever moving along the first cam surface is not inserted into the cam groove.
7. The ice maker of claim 6, wherein the blocking member includes a coupling part rotatably coupled to a rotation shaft of the cam.
8. The ice maker of claim 3, wherein the second cam surface includes a cam groove.

9. An ice maker comprising:
 - a first tray assembly configured to form a first portion of a cell;
 - a second tray assembly configured to form a second portion of the cell, the first and second portions configured to form a space in which liquid is phase-changed to ice; and
 - a driver configured to move the second tray assembly, wherein the driver includes:
 - a motor, and
 - a cam rotated by the motor,
 wherein after the ice is separated from the cell, the motor drives the cam in a first cam direction until the second tray assembly moves to a liquid supply position.
10. The ice maker of claim 9, wherein the motor additionally drives the cam in the first cam direction until the second tray assembly moves to an ice making position.
11. The refrigerator of claim 10, wherein the motor additionally drives the cam in the first cam direction after the second tray assembly moves to the ice making position.
12. The ice maker of claim 9, wherein after an ice making is completed, the motor drives the cam in a second direction until the second tray assembly moves to an ice separation position.
13. The ice maker of claim 12, wherein the motor additionally drives the cam in the second direction after the second tray assembly moves to the ice separation position.
14. An ice maker comprising:
 - a first tray assembly configured to form a first portion of a cell;
 - a second tray assembly configured to form a second portion of the cell, the first and second portions configured to form a space in which liquid is phase-changed to ice; and
 - a driver configured to move the second tray assembly, wherein the driver includes:
 - a motor,
 - a cam rotated by the motor and having a cam surface with a cam groove,
 - a lever moves along the cam surface, and
 - a blocking member that selectively blocks the cam groove.
15. The ice maker of claim 14, wherein the lever is connected to a full ice taction lever.
16. The ice maker of claim 14, wherein the blocking member includes a coupling part rotatably coupled to a rotation shaft of the cam.
17. The ice maker of claim 16, wherein the blocking member includes a support protrusion provided outside the coupling part to restrict an operation of the lever.
18. A refrigerator comprising:
 - a cabinet including a storage chamber;
 - a door that opens and closes the storage chamber; and
 - an ice maker of claim 1.
19. A refrigerator comprising:
 - a cabinet including a storage chamber;
 - a door that opens and closes the storage chamber; and
 - an ice maker of claim 9.
20. A refrigerator comprising:
 - a cabinet including a storage chamber;
 - a door that opens and closes the storage chamber; and
 - an ice maker of claim 14.